

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J70 MLB

LAST_MODIFIED= Tue Apr 15 17:23:51 2014

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
A	0002721096	PRODUCTION RELEASED	2014-04-15

Page	Contents	Sync	Date
1	Table of Contents	J16_MLB_IG	00/27/2013
2	BOM Configuration	J70_TONY	11/06/2013
3	DEBUG LEDS	J70_GAREN	10/22/2013
4	Holes/PD parts	J16_MLB_IG	00/27/2013
5	CPU GFX/NCTF/RSVD	J70_TONY	00/11/2013
6	CPU Misc/JTAG/CFG/RSVD	J70_TONY	00/24/2013
7	CPU DDR3/LPDDR3 Interfaces	J70_DINI	00/23/2013
8	CPU/PCH POWER	J70_TONY	10/16/2013
9	CPU/PCH GROUNDS	J41_MLB	00/27/2013
10	CPU Decoupling	J70_TONY	11/14/2013
11	PCH Decoupling	J41_MLB	00/27/2013
12	PCH Audio/JTAG/SATA/CLK	J70_GAREN	01/30/2014
13	PCH PM/PCI/GFX	J70_NICK	11/20/2013
14	PCH PCIe/USB/LPC/SPI/SMBus	J70_TONY	10/15/2013
15	PCH GPIO/MISC/LPIO	J70_DINI	01/21/2014
16	CPU/PCH Merged XDP	J70_TONY	11/07/2013
17	Chipset Support	J70_NICK	02/25/2014
18	Project Chipset Support	J70_NICK	11/20/2013
19	DDR3 Signal Aliases	MASTER	MASTER
20	DDR3 VREF MARGINING	J70_GAREN	10/24/2013
21	LPDDR3 DRAM Channel A (0-31)	J41_MLB	00/03/2013
22	LPDDR3 DRAM Channel A (32-63)	J41_MLB	00/03/2013
23	LPDDR3 DRAM Channel B (0-31)	J41_MLB	00/03/2013
24	LPDDR3 DRAM Channel B (32-63)	J41_MLB	00/03/2013
25	LPDDR3 DRAM Termination	J41_MLB	00/03/2013
26	Thunderbolt Host (1 of 2)	J70_NICK	01/27/2014
27	Thunderbolt Host (2 of 2)	J70_NICK	01/12/2014
28	Thunderbolt Connector A	J70_NICK	10/16/2013
29	Thunderbolt Connector B	J70_NICK	10/16/2013
30	DDC Crossbar	J70_TONY	00/13/2013
31	AIRPORT/BT	J70_DINI	10/08/2013
32	SATA/SSD Connectors	J70_DINI	10/14/2013
33	HDD Connector	J16_MLB_IG	00/27/2013
34	ETHERNET PHY (CAESAR IV)	J70_GAREN	00/19/2013
35	Ethernet Support & Connector	J16_MLB_IG	05/01/2013
36	SD READER CONNECTOR	J70_GAREN	00/23/2013
37	Camera Controller	J70_GAREN	02/05/2014
38	Camera Controller Support	J70_GAREN	10/24/2013
39	Internal DP Support	J16_MLB_IG	00/27/2013
40	Internal DP MUXing	J70_TONY	00/05/2013
41	EXTERNAL USB PORTS A & B	J70_GAREN	10/22/2013
42	EXTERNAL USB PORTS C & D	J70_GAREN	10/22/2013
43	SMC	J70_NICK	00/24/2013
44	SMC Support	J70_NICK	01/15/2014
45	SPI and Debug Connector	J70_TONY	00/24/2013
46	SMBus Connections	J70_NICK	02/14/2014
47	I and V Sense	J70_NICK	11/05/2013
48	Temperature Sensors	J70_NICK	11/05/2013

Page	Contents	Sync	Date
49	System Fan	J16_MLB_IG	08/27/2013
50	AUDIO:CODEC ANALOG	J70_GAREN	12/17/2013
51	AUDIO:CODEC, DIGITAL	J70_GAREN	01/30/2014
52	AUDIO: SPEAKER AMP, LEFT CHANNEL	J70_GAREN	03/07/2014
53	AUDIO: SPEAKER AMP, RIGHT CHANEL	J70_GAREN	03/07/2014
54	AUDIO: JACK TRANSLATORS	J70_GAREN	03/11/2014
55	AUDIO: Speaker ID	J70_GAREN	03/07/2014
56	Power Connectors / VReg G3Hot	J16_MLB_IG	08/27/2013
57	CPU VR12.6 VCC Regulator IC	J70_ROSSANA	09/17/2013
58	CPU VR12.5 VCC Power Stage	J70_GAREN	03/07/2014
59	VReg VDDQ S3 / 1.8V S3	J70_ROSSANA	09/12/2013
60	VREG 1V05 S0 / 1V5 S0	J16_MLB_IG	08/27/2013
61	VReg 3.3V S5/5V S4	J70_GAREN	03/07/2014
62	LCD Backlight Driver (LP8561)	J70_GAREN	03/07/2014
63	FET-Controlled S0 and S4	J16_MLB_IG	08/27/2013
64	PM Regulator Enables	J16_MLB_IG	08/27/2013
65	PM Power Good	J70_TONY	10/10/2013
66	Power Aliases	J16_MLB_IG	08/27/2013
67	Unused Signal Aliases	J16_MLB_IG	08/27/2013
68	Functional / ICT Test	J16_MLB_IG	08/27/2013
69	J70 RULE DEFINITIONS	J70_NICK	09/12/2013
70	DDR3 Constraints	J70_NICK	09/30/2013
71	CPU CONSTRAINTS	J70_NICK	09/12/2013
72	PCH PCIe/DMI Constaints	J70_DINI	10/28/2013
73	SATA/FDI/XDP Constraints	J70_TONY	10/23/2013
74	PCH and BR Constraints	J70_NICK	10/15/2013
75	USB/Ethernet/SD Constraints	J70_DINI	10/28/2013
76	SMBus/Sensor Constraints	J70_NICK	10/15/2013
77	VReg Constraints	J70_GAREN	03/07/2014
78	CPU VReg Constraints	J70_GAREN	03/07/2014
79	Platform VReg Constraints	J70_GAREN	03/07/2014
80	TBT/DP Constraints	J70_NICK	10/22/2013
81	BLC Constraints	J70_GAREN	03/07/2014

Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-1407	1	SCH, MLB, J70	SCH	CRITICAL	
820-4668	1	PCBF, MLB, J70	PCB	CRITICAL	
685-1351	1	PCBA, MLB, J70, COMMON PARTS	CMNPPTS		MLB_CMNPPTS

Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-1058	PCBA, MLB, DEV, J70	DEVELOPMENT, J70_DEVEL
939-00046	PCBA, MLB, DEV, J70, CPU_INT	MLB_CMNPTS, ALTERNATE, CPU: SOCKET, DDR3: HYNIX_4GB_29nm, SSD: Y
639-6212	PCBA, MLB, J70, HY_4GB_38nm, HDD	MLB_CMNPTS, CPU: ULT, DDR3: HYNIX_4GB_38nm, SSD: N
639-6213	PCBA, MLB, J70, EL_4GB_38nm, HDD	MLB_CMNPTS, CPU: ULT, DDR3: ELPIDA_4GB_38nm, SSD: N
639-6214	PCBA, MLB, J70, HY_4GB_38nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: HYNIX_4GB_38nm, SSD: Y
639-6215	PCBA, MLB, J70, EL_4GB_38nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: ELPIDA_4GB_38nm, SSD: Y
639-6216	PCBA, MLB, J70, HY_8GB_38nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: HYNIX_8GB_38nm, SSD: Y
639-6217	PCBA, MLB, J70, EL_8GB_38nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: ELPIDA_8GB_38nm, SSD: Y
639-6218	PCBA, MLB, J70, HY_4GB_29nm, HDD	MLB_CMNPTS, CPU: ULT, DDR3: HYNIX_4GB_29nm, SSD: N
639-6219	PCBA, MLB, J70, EL_4GB_25nm, HDD	MLB_CMNPTS, CPU: ULT, DDR3: ELPIDA_4GB_25nm, SSD: N
639-6220	PCBA, MLB, J70, HY_4GB_29nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: HYNIX_4GB_29nm, SSD: Y
639-6221	PCBA, MLB, J70, EL_4GB_25nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: ELPIDA_4GB_25nm, SSD: Y
639-6222	PCBA, MLB, J70, HY_8GB_29nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: HYNIX_8GB_29nm, SSD: Y
639-6223	PCBA, MLB, J70, EL_8GB_25nm, SSD	MLB_CMNPTS, ALTERNATE, CPU: ULT, DDR3: ELPIDA_8GB_25nm, SSD: Y
685-1351	PCBA, MLB, J70, COMMON PARTS	J70_COMMON

BOM Groups

BOM GROUP	BOM OPTIONS
J70_COMMON	COMMON, ALTERNATE, J70_PROGPARTS, SMCREG:DIV, XDP, TEMPSNSDEV, SMBUS0:ISOL, SMBUS1:ISOL, AUDIO:I2S, RTCRST:Y
J70_PROGPARTS	SMC:PROG, BOOTROM:PROG, CAMROM:PROG, TBTROM:PROG, ENETROM:PROG
J70_DEVEL	XDP_CONN, LPCPLUS, TEMPSNSDEV

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00011	1	IC, CPU, HSW, SR1GT, PRQ, 1.4, 15W, 2+3, 3H, BGA	U0500	CRITICAL	CPU:ULT
998-7866	1	INTERPOSER, BGA1160, SINGLE SIDE	U0500	CRITICAL	CPU: SOCKET

ASIC Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S1273	1	IC, TBT, RR-4C, A8, PRQ, C10, FCDBGA, 288, T&R	U2800	CRITICAL	
343S0616	1	IC, BCM57766A, CIV+, A8, 8x8	U3900	CRITICAL	

CPU DRAM SPD Straps

BOM GROUP	BOM OPTIONS
DDR3:HYNIX_4GB_38nm	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:HYNIX_4GB_38nm
DDR3:HYNIX_8GB_38nm	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:L, DRAM_TYPE:HYNIX_8GB_38nm
DDR3:ELPIDA_4GB_38nm	RAMCFG0:H, RAMCFG1:H, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:ELPIDA_4GB_38nm
DDR3:ELPIDA_8GB_38nm	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:L, DRAM_TYPE:ELPIDA_8GB_38nm
DDR3:HYNIX_4GB_29nm	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:H, DRAM_TYPE:HYNIX_4GB_29nm
DDR3:HYNIX_8GB_29nm	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:H, DRAM_TYPE:HYNIX_8GB_29nm
DDR3:ELPIDA_4GB_25nm	RAMCFG0:H, RAMCFG1:H, RAMCFG2:L, RAMCFG3:H, DRAM_TYPE:ELPIDA_4GB_25nm
DDR3:ELPIDA_8GB_25nm	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:H, DRAM_TYPE:ELPIDA_8GB_25nm
DDR3:SAMSUNG_4GB	RAMCFG0:L, RAMCFG1:H, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:SAMSUNG_4GB
DDR3:SAMSUNG_8GB	RAMCFG0:L, RAMCFG1:H, RAMCFG2:H, RAMCFG3:L, DRAM_TYPE:SAMSUNG_8GB

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0677	4	IC, SDRAM, 38nm, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=HYNIX_4Gb_38nm
333S0681	4	IC, SDRAM, 38nm, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=HYNIX_8Gb_38nm
333S0678	4	IC, SDRAM, 38nm, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=ELPIDA_4Gb_38nm
333S0666	4	IC, SDRAM, 38nm, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=ELPIDA_8Gb_38nm
333S0787	4	IC, SDRAM, 29nm, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=HYNIX_4Gb_29nm
333S0785	4	IC, SDRAM, 29nm, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=HYNIX_8Gb_29nm
333S0793	4	IC, SDRAM, 25nm, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=ELPIDA_4Gb_25nm
333S0791	4	IC, SDRAM, 25nm, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=ELPIDA_8Gb_25nm
333S0676	4	IC, SDRAM, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=SAMSUNG_8Gb
333S0680	4	IC, SDRAM, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE=SAMSUNG_8Gb

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S00053	1	IC,EFI,V0174,370	U5210	CRITICAL	BOOTROM:PROG
33S00807	1	IC,64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S00050	1	IC,SMC-B3,EXTERNAL,V2,23F70,PVT,370	U5000	CRITICAL	SMC:PROG
33SC1159	1	IC,SMC12-A3,40MHZ/50MSPS,SCPL FW,15780A	U5000	CRITICAL	SMC:BLANK
341S3778	1	IC,CAMERA,FLASH,V7230,316	U4202	CRITICAL	CAMROM:PROG
33S00852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341S00048	1	IC,EPROM,T29,REDWOOD RDG,V30-B,370	U2890	CRITICAL	TBTROM:PROG
33S00915	1	IC,FLASH,SPI,4MBIT,50MHZ	U2890	CRITICAL	TBTROM:BLANK
341S3912	1	IC,ENET SPI ROM,V0000YX,V1,15,116/1160/117	U3990	CRITICAL	ENETROM:PROG
33S1025	1	IC,SERIAL FLASH,2MBIT,2.7V,REV F	U3990	CRITICAL	ENETROM:BLANK

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0155	377S0104		ALL	USB3 diodes
377S0124	377S0057		ALL	TVS
376S1218	376S1081		ALL	P/NCh dual FET
155S0578	155S0367		ALL	1200HM EMI BEAD
128S0368	128S0365		ALL	150uF AL POLY
138S0681	138S0638		ALL	Taiyo 10uF 805 alt
197S0479	197S0478		ALL	12 MHZ Cam. Xtal
197S0486	197S0478		ALL	12 MHZ Cam. Xtal
107S0251	107S0249		ALL	Sense resistor
197S0481	197S0480		ALL	25MHZ Xtal
197S0343	197S0480		ALL	25MHz Xtal
138S0860	138S0775		ALL	Single-source 1uF 402
138S0859	138S0788		ALL	Single-source 10uF
378S0391	378S0390		ALL	Debug LEDs
341S00016	341S3912		ALL	ENET ROM,ADEST0,V1.15
138S0747	138S0773		ALL	1uF, X6S, 402
197S0542	197S0544		ALL	24 MHz PCH Xtal
197S0545	197S0544		ALL	24 Mhz PCH Xtal
376S1217	376S0855		ALL	Dual N-Ch FET
376S1129	376S0855		ALL	Dual N-Ch FET
376S0572	376S0659		ALL	Single P-Ch FET
376S0972	376S1017		ALL	Single N-Ch FET

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1089	376S1128		ALL	Dual N-Ch FET
155S0660	155S0513		ALL	220HM EMI BEAD
155S0694	155S0387		ALL	4700HM EMI BEAD
127S0164	127S0162		ALL	1UF 25V TANT
376S1217	376S0855	SSD:Y	ALL	Dual N-Ch FET
376S1129	376S0855	SSD:Y	ALL	Dual N-Ch FET
197S0369	197S0392		ALL	32 KHz PCH Xtal
197S0399	197S0392		ALL	32 KHz PCH Xtal
311S0649	311S0541		ALL	Single AND Gate
107S0375	107S0151		ALL	DDR Sense Res
107S0372	107S0181		ALL	CPU VR Sense Res

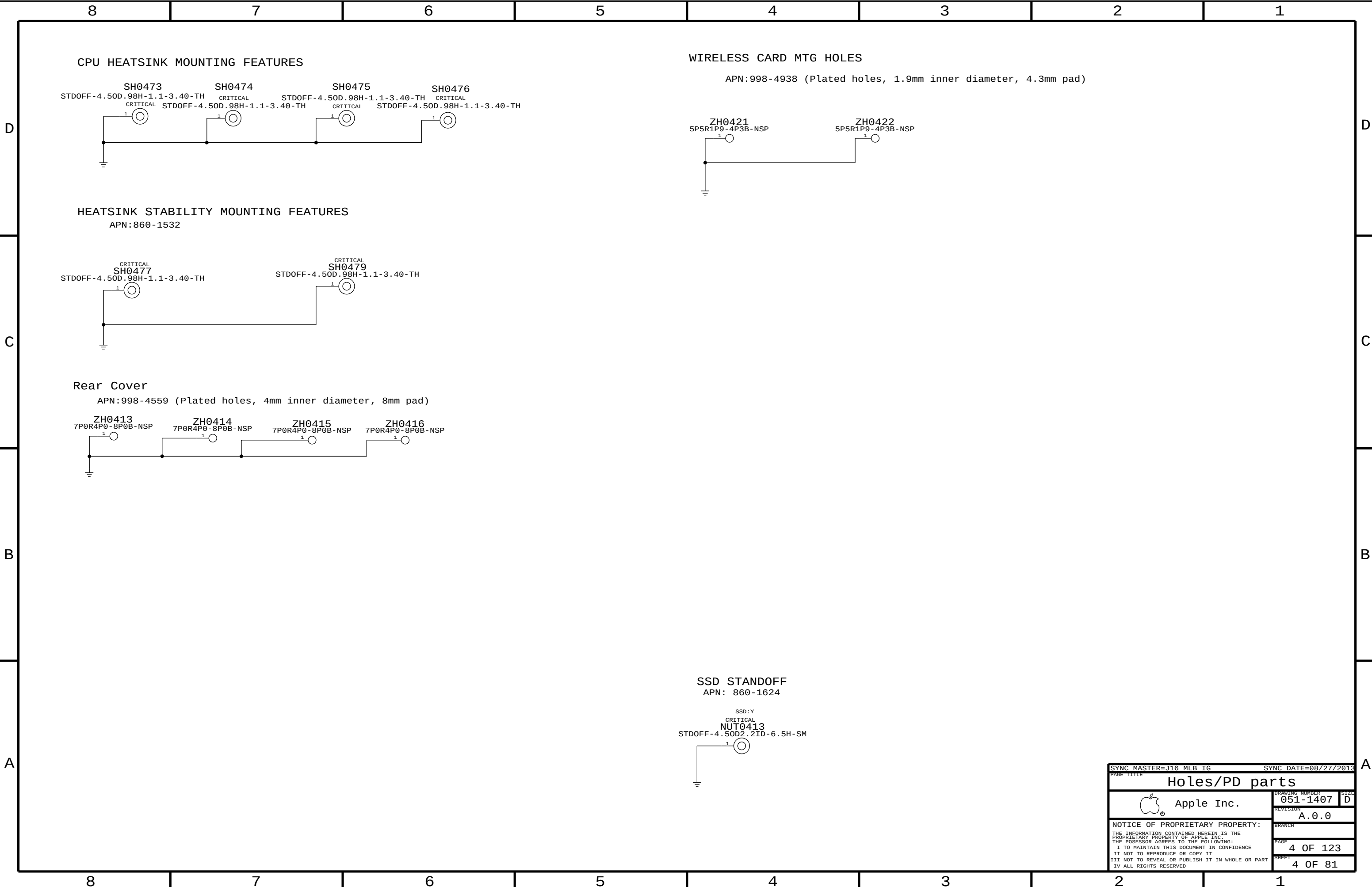
CPU DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
N/A	0	1
ELPIDA	1	1

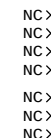
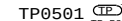
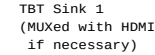
SIZE	CFG 2	DIE REV	CFG 3
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8GB	1	B	1

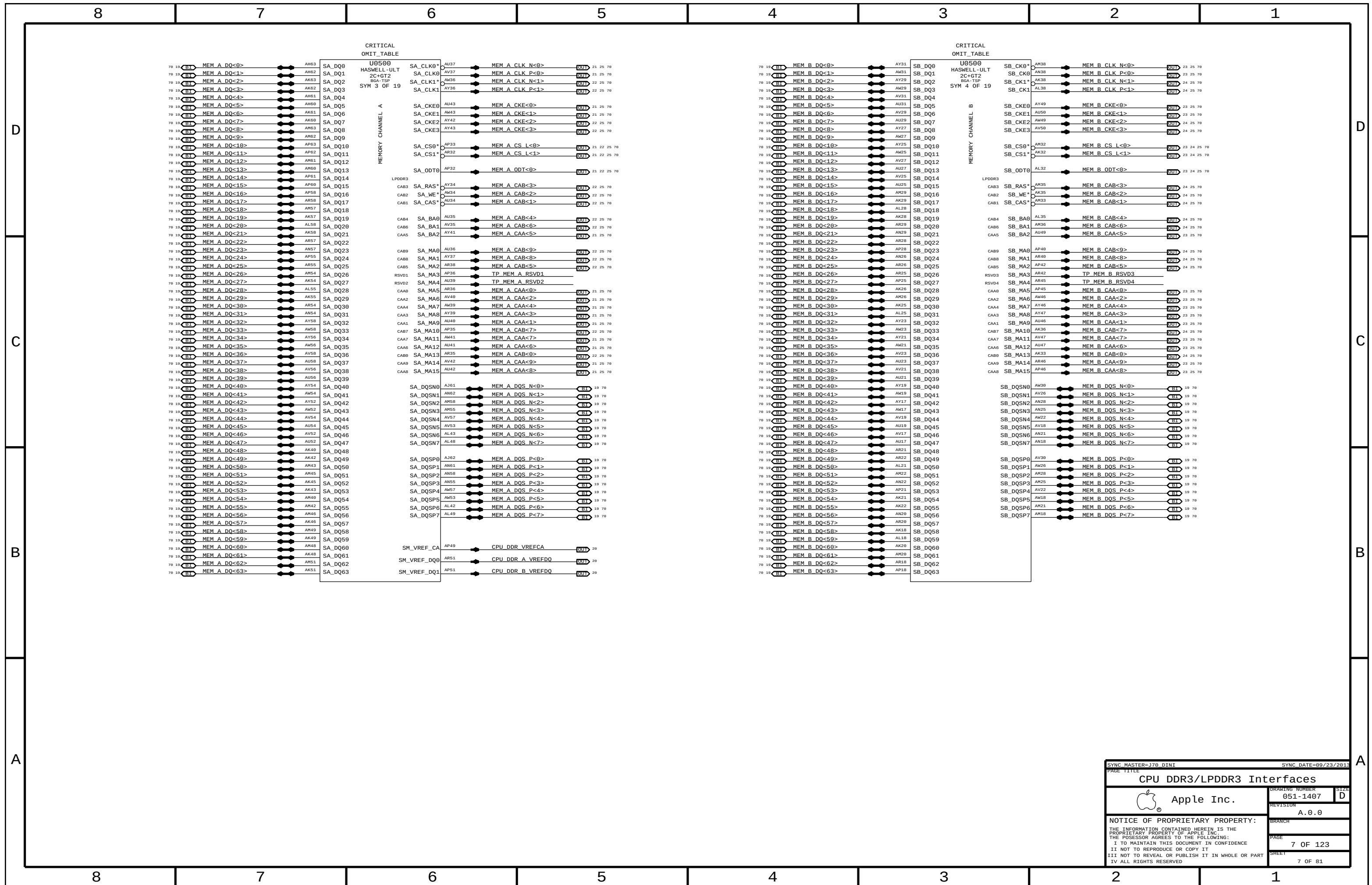
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		PAGE	2 OF 123
		SHEET	2 OF 81

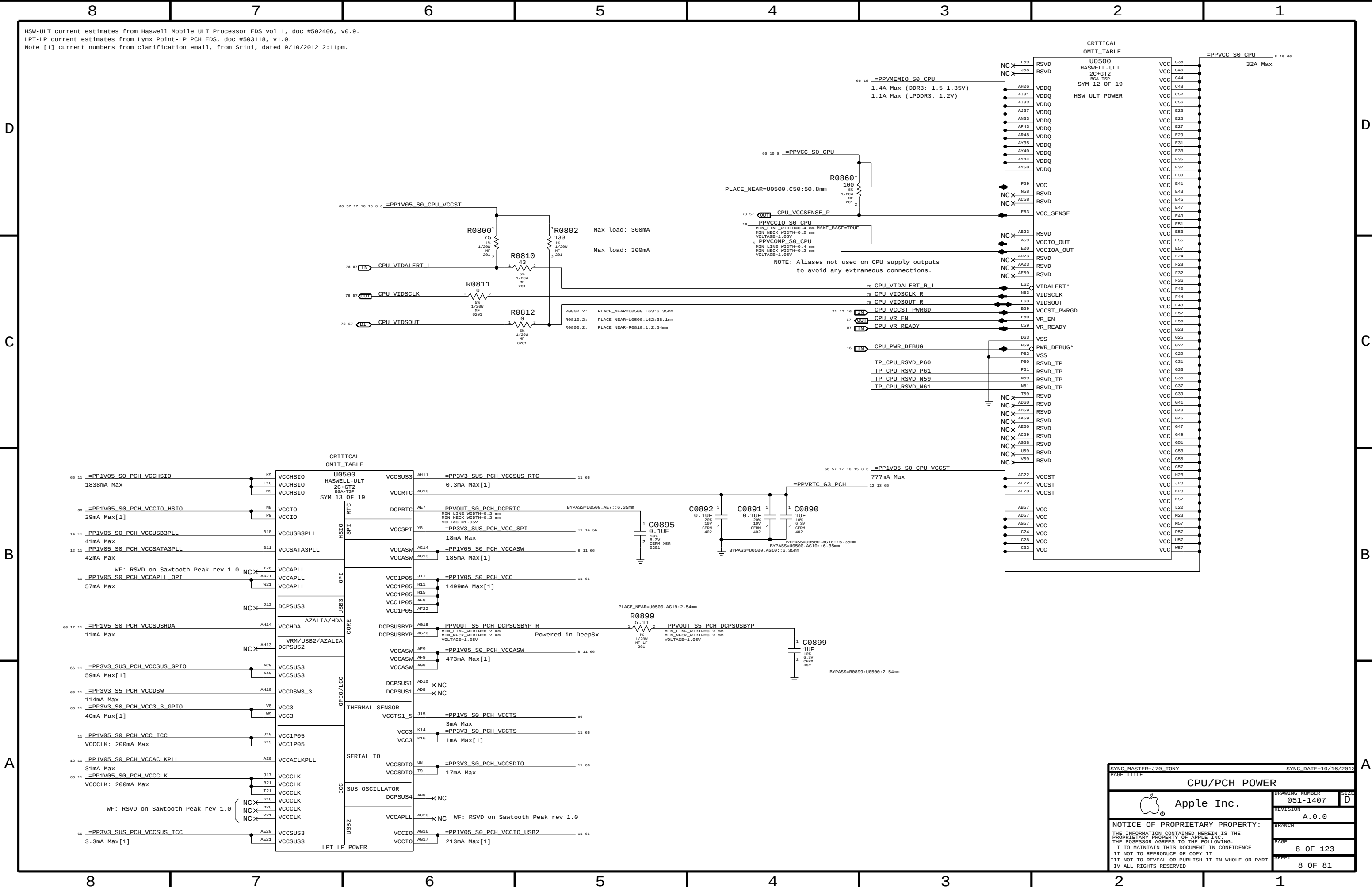


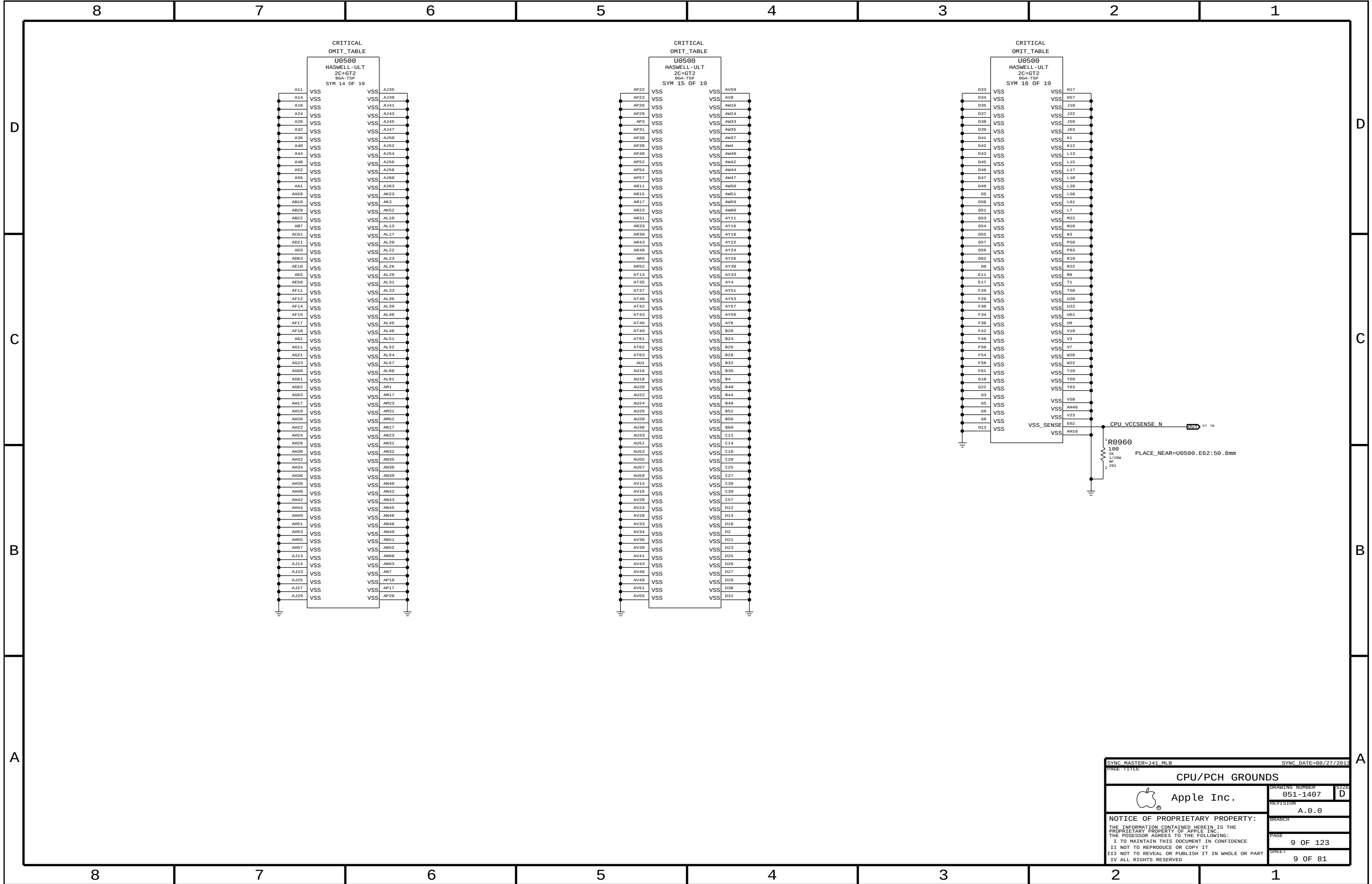


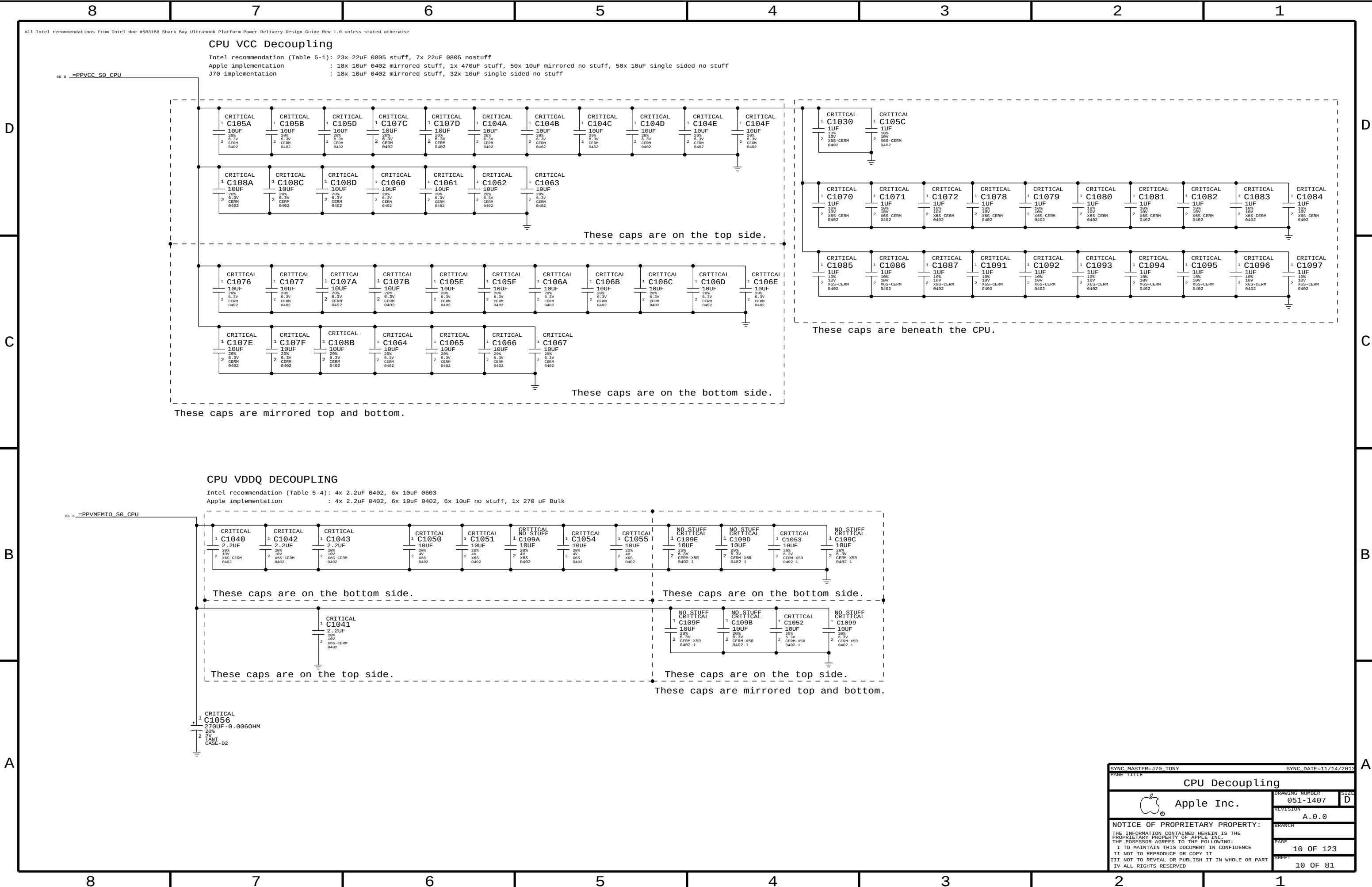
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		PAGE	4 OF 123
		SHEET	4 OF 81

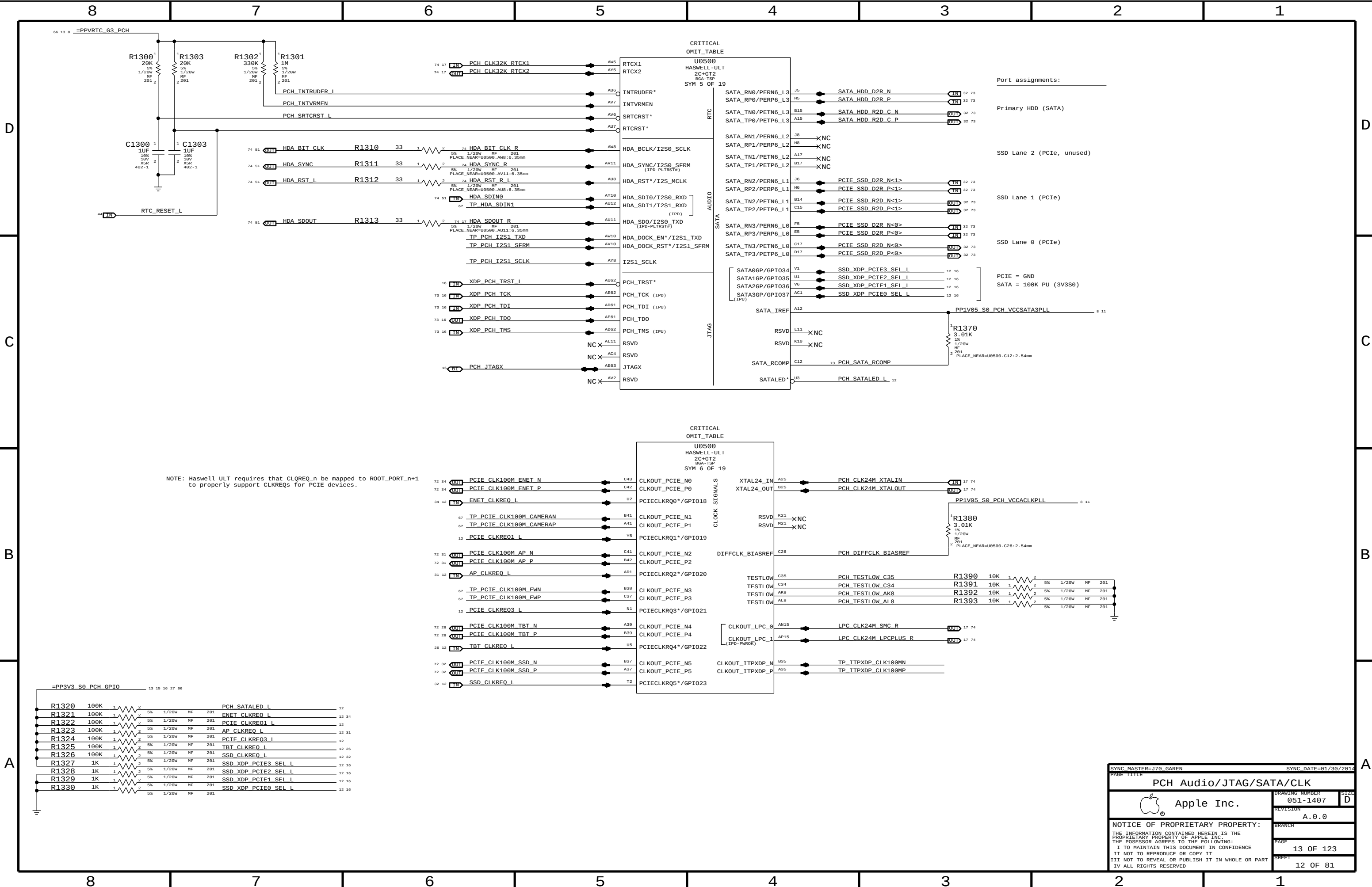






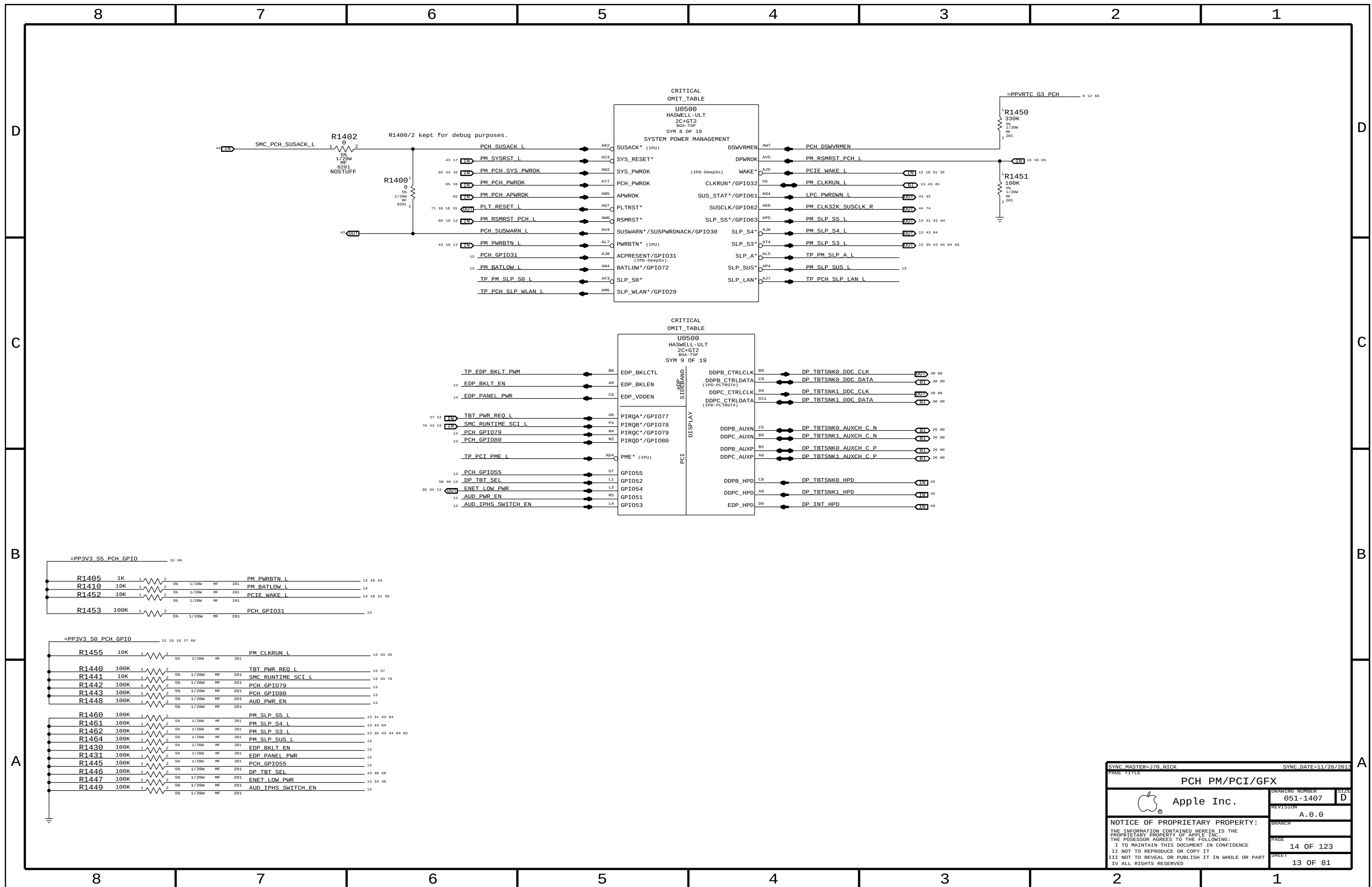


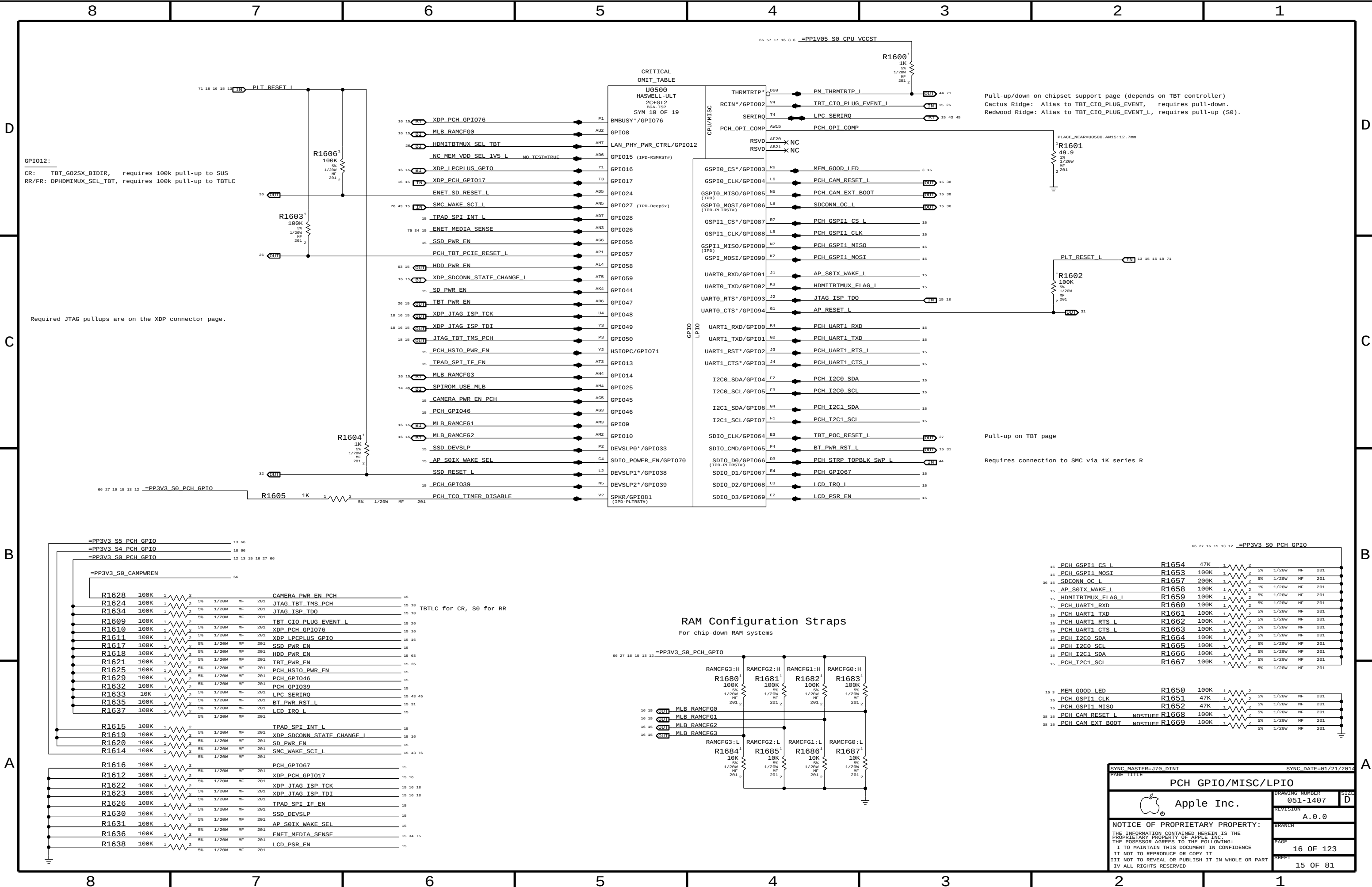


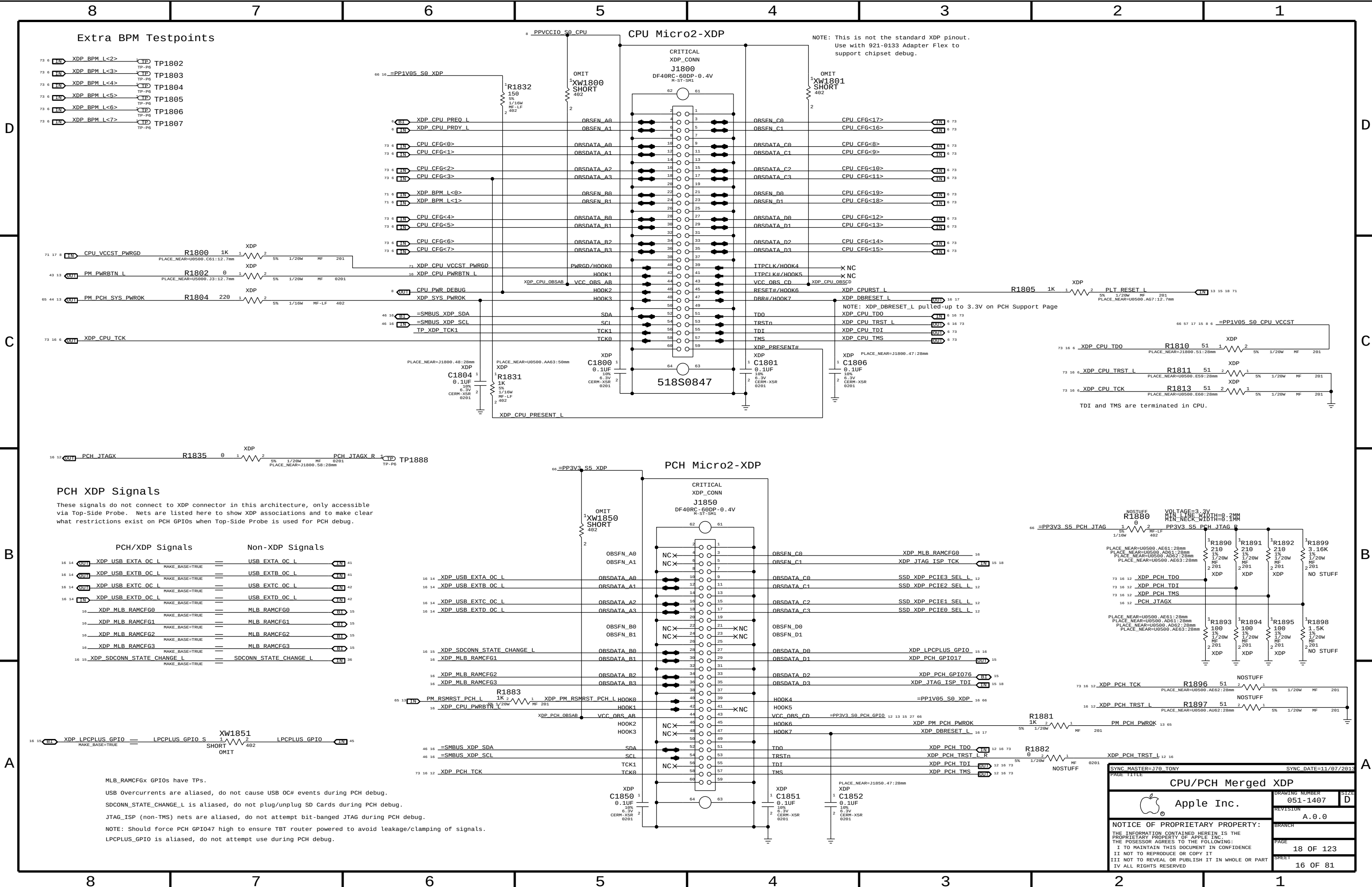


NOTE: Haswell ULT requires that CLQREQ_n be mapped to ROOT_PORT_n+1 to properly support CLKREQs for PCIe devices.

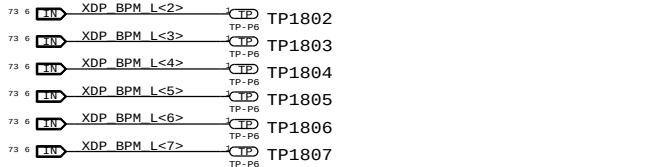
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		PAGE	13 OF 123
		SHEET	12 OF 81







Extra BPM Testpoints



CPU Micro2-XDP

NOTE: This is not the standard XDP pinout.
Use with 921-0133 Adapter Flex to
support chipset debug.

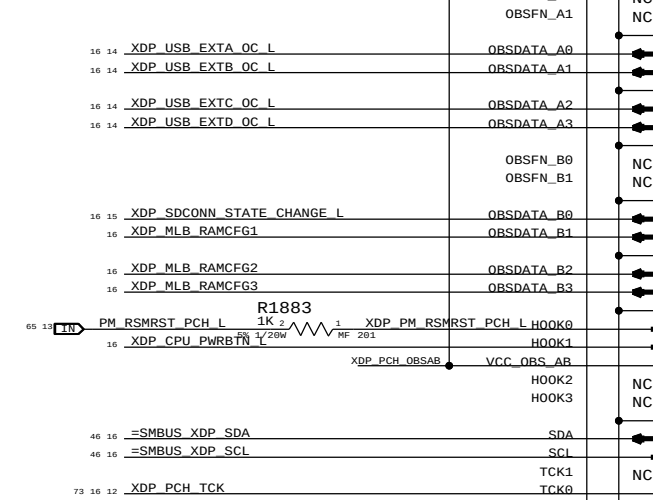
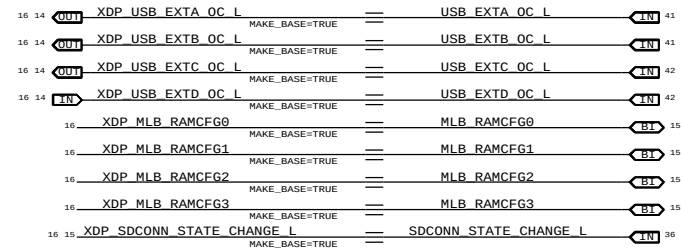
PCH Micro2-XDP

PCH XDP Signals

These signals do not connect to XDP connector in this architecture, only accessible via Top-Side Probe. Nets are listed here to show XDP associations and to make clear what restrictions exist on PCH GPIOs when Top-Side Probe is used for PCH debug.

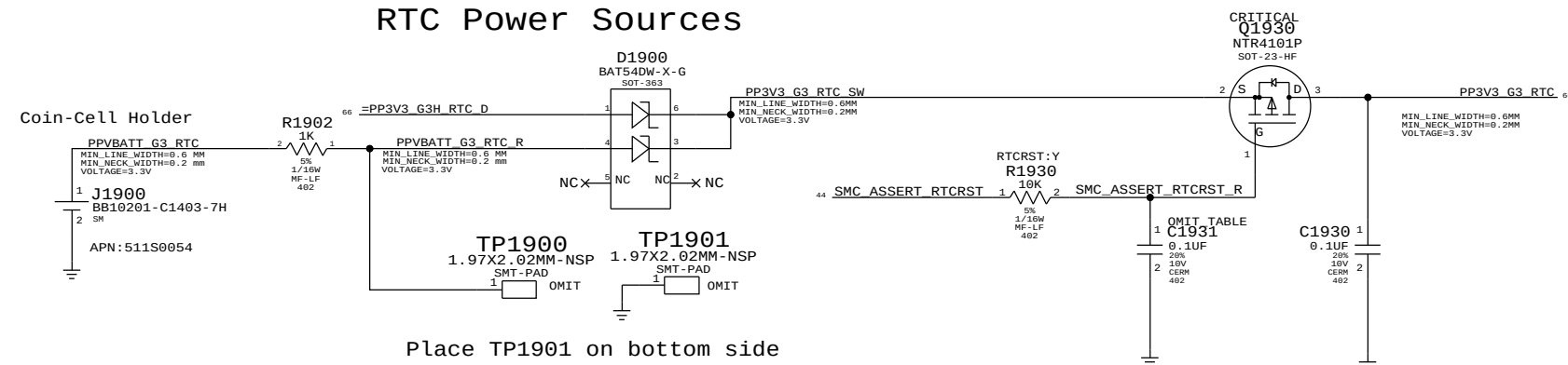
PCH/XDP Signals

Non-XDP Signals



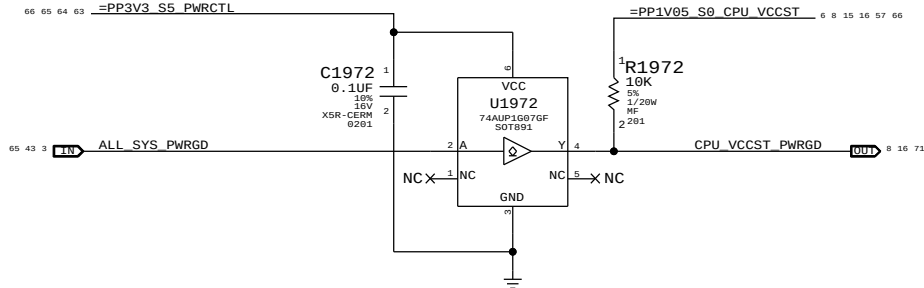
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RTC Power Sources

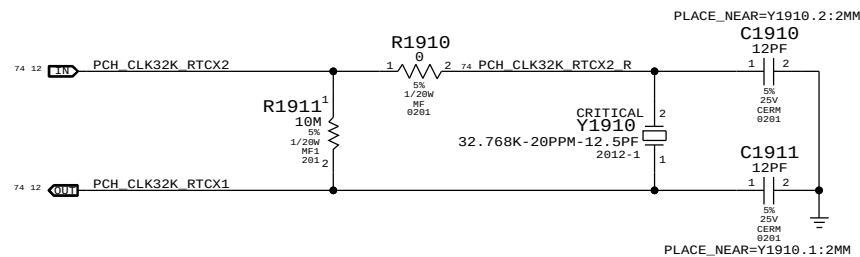


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S1059	1	CAP, 0.1 UF, 402	C1931	RTCRST:Y
116S0090	1	RES, 10K OHM, 402	C1931	RTCRST:N

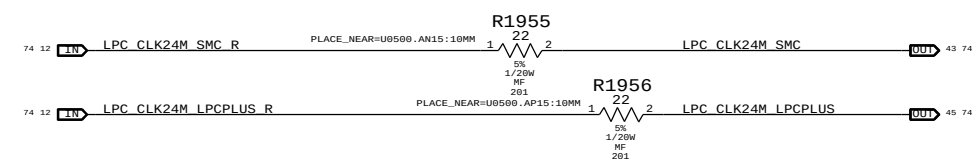
ALL_SYS_PWRGD/CPU_VCCST_PWRGD Level-Shifter



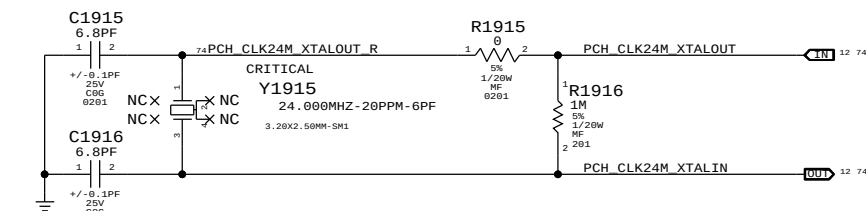
PCH RTC Crystal



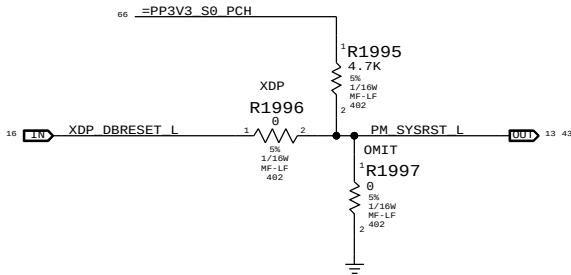
Clock series termination



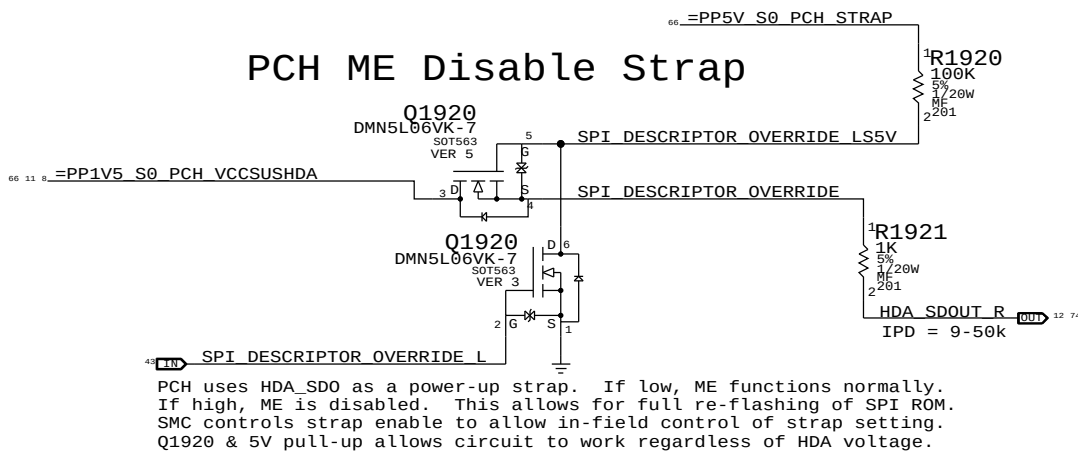
PCH 24MHz Crystal



PCH Reset Button



PCH ME Disable Strap



D

C



A

8



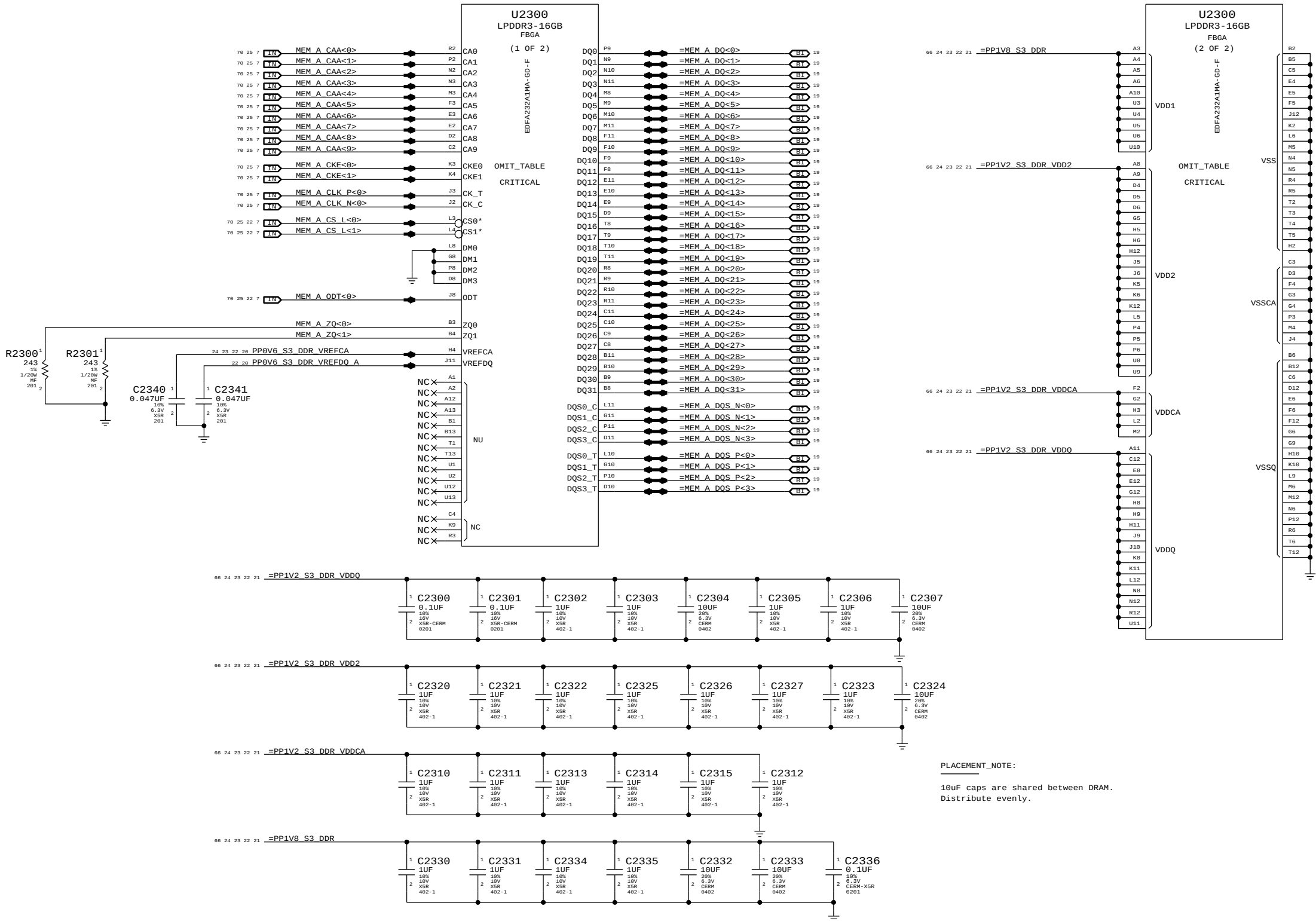
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4



DRAWING NUMBER		SIZE
051-1407		D
REVISION		
A.0.0		
BRANCH		
PAGE		
20 OF 123		
SHEET		
18 OF 81		

LPDDR3 CHANNEL A (0-31)

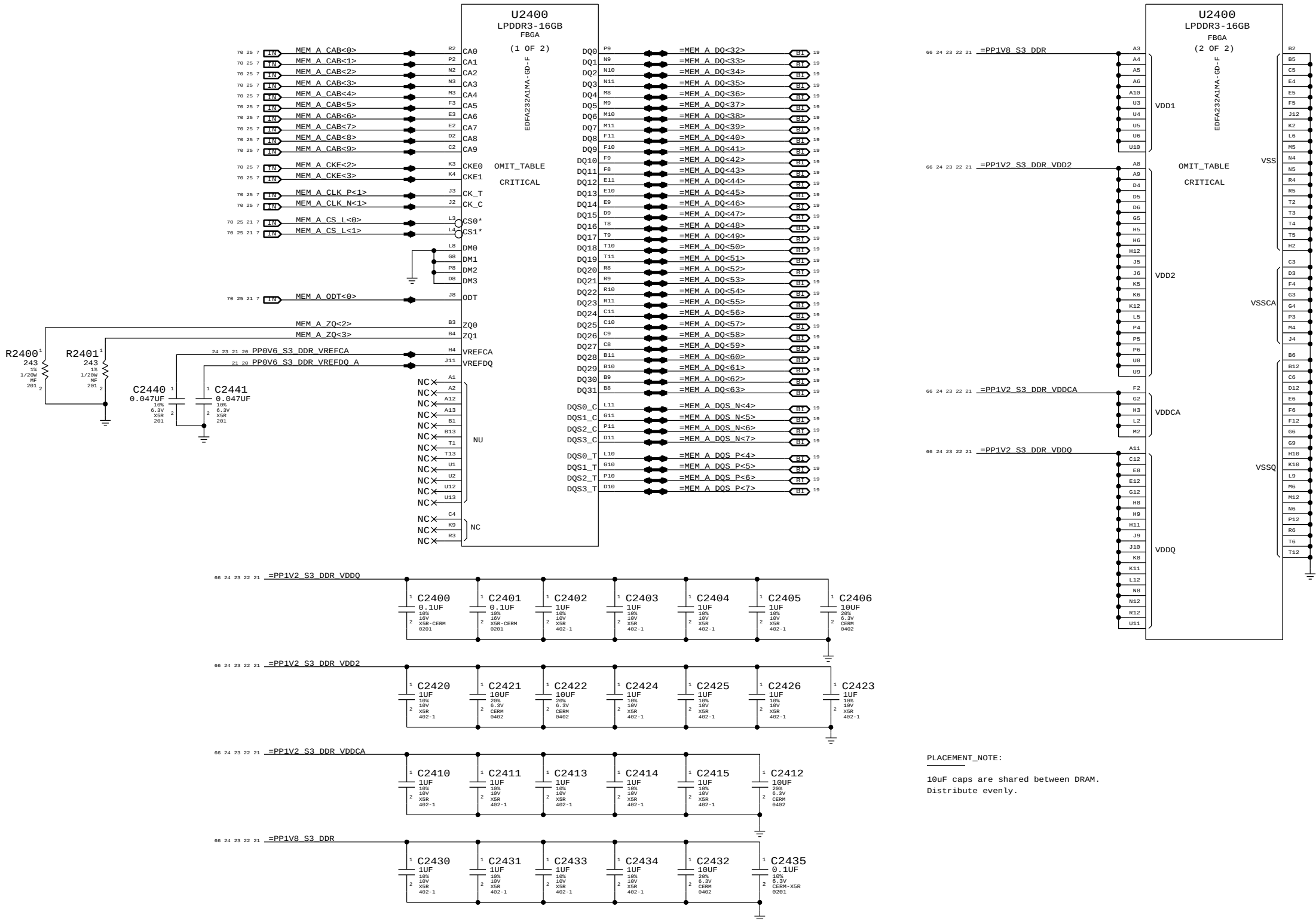


PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

SYNC MASTER=J41 MLB		SYNC DATE=09/03/2013	
PAGE TITLE		LPDDR3 DRAM Channel A (0-31)	
DRAWING NUMBER		051-1407	SIZE D
REVISION		A.0.0	BRANCH
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LPDDR3 CHANNEL A (32-63)

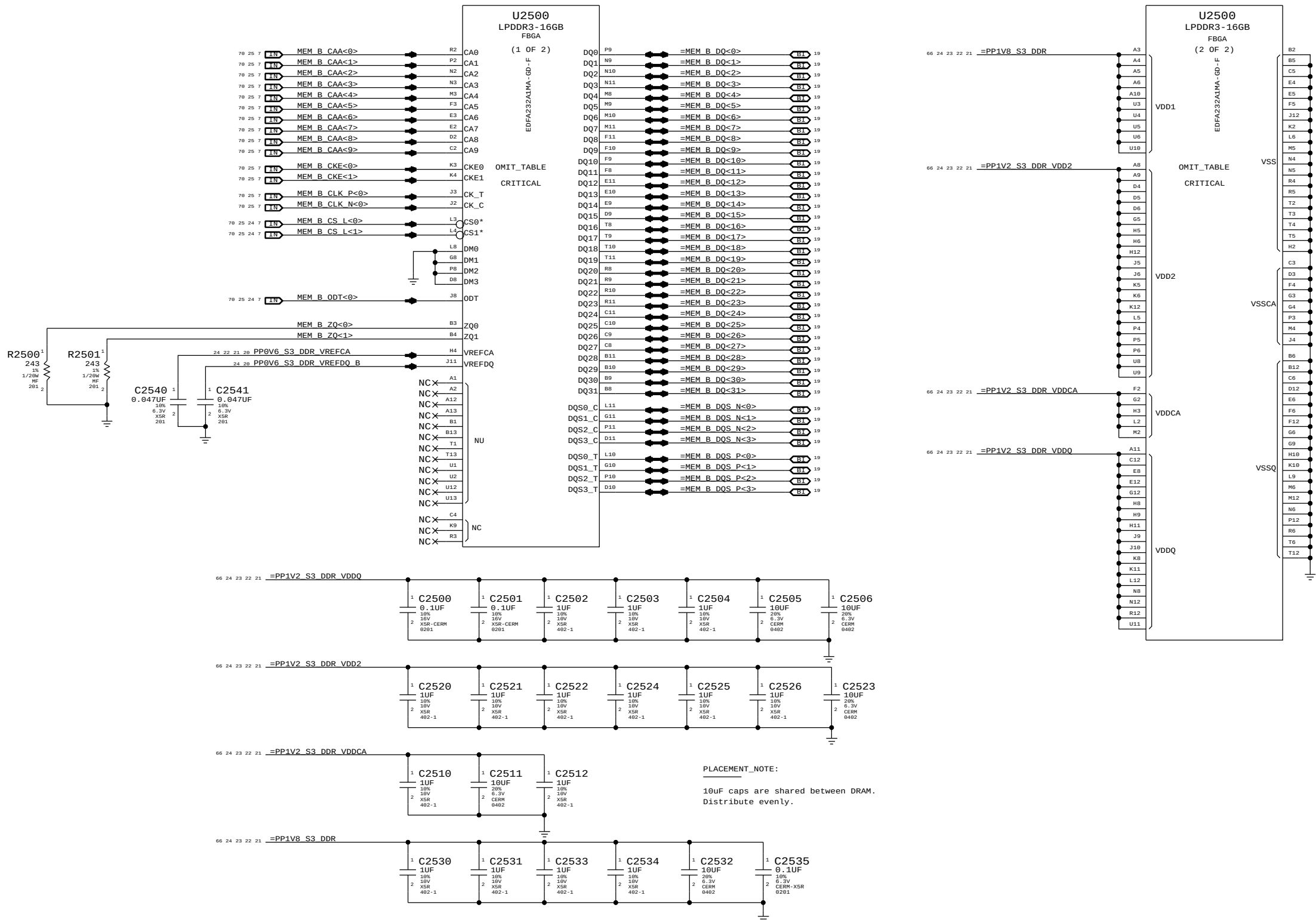


PLACEMENT_NOTE:

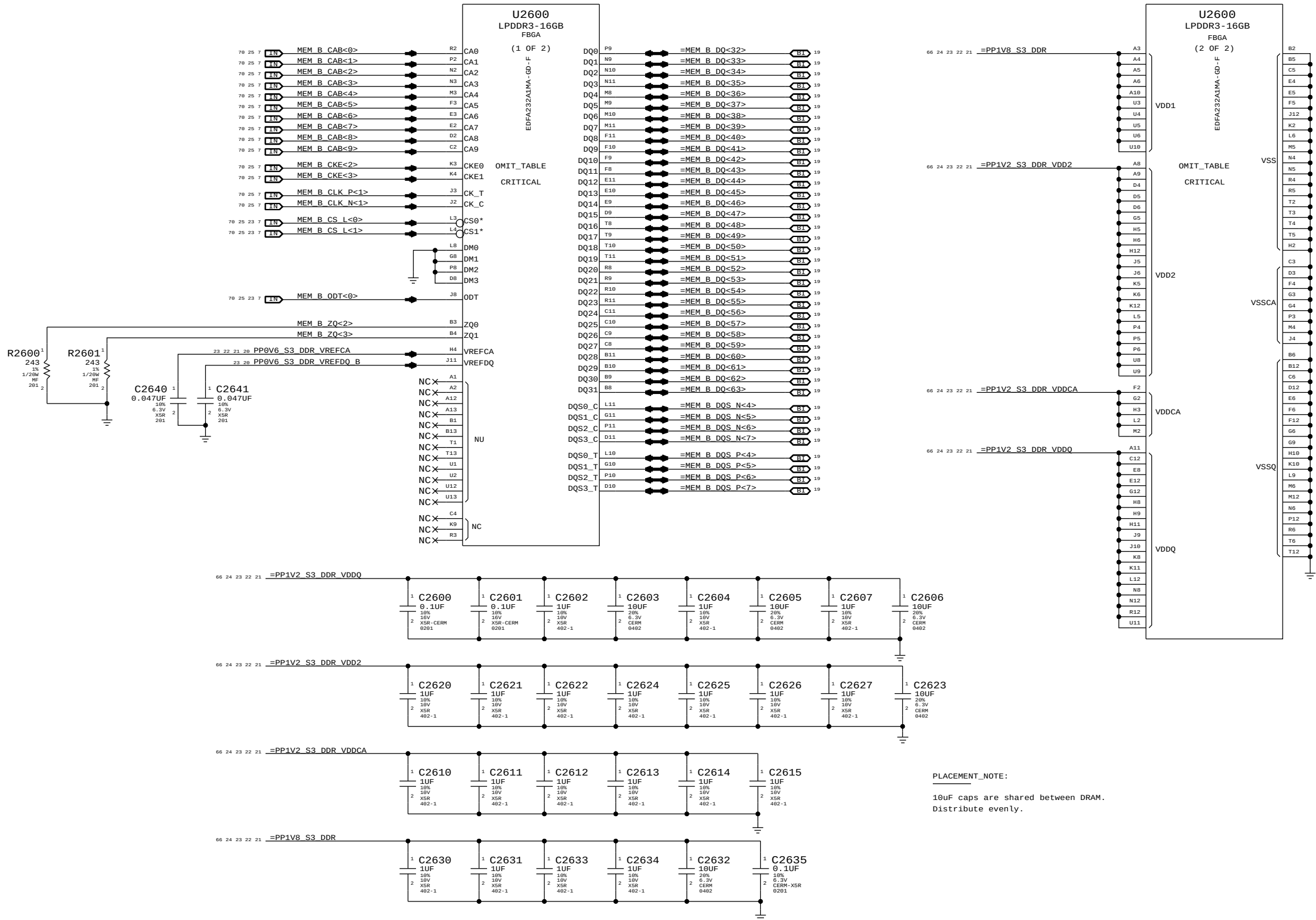
10uF caps are shared between DRAM.
Distribute evenly.

SYNC MASTER=J41 MLB		SYNC DATE=09/03/2013	
PAGE TITLE		LPDDR3 DRAM Channel A (32-63)	
DRAWING NUMBER		051-1407	SIZE D
REVISION		A.0.0	BRANCH
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LPDDR3 CHANNEL B (0-31)



LPDDR3 CHANNEL B (32-63)



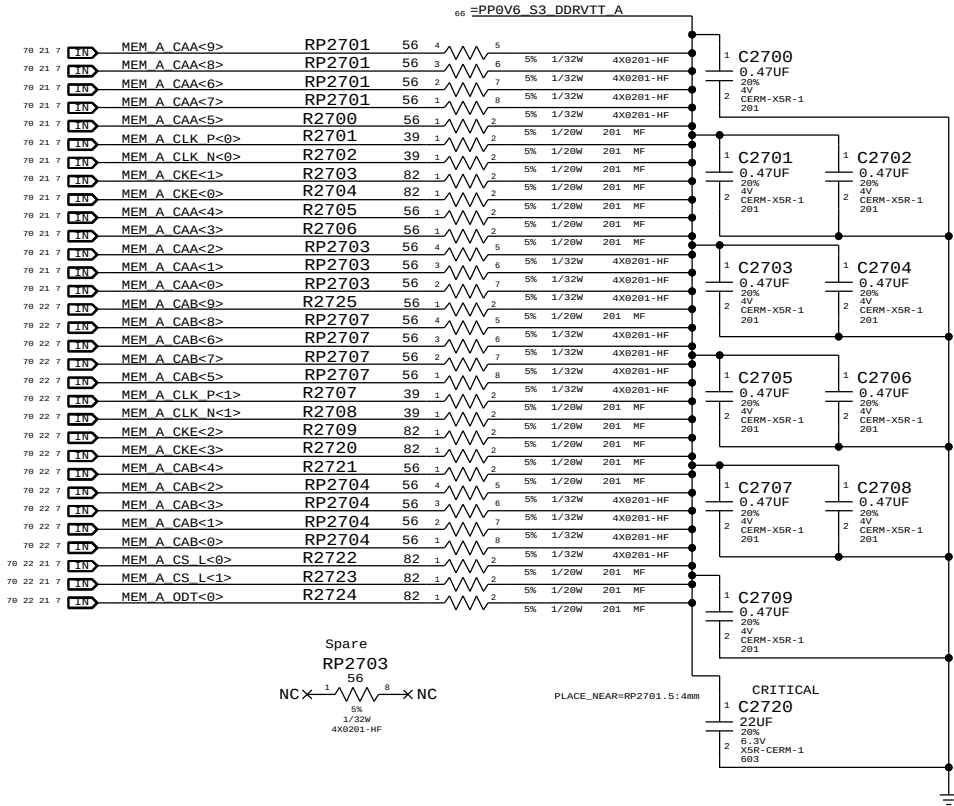
D

C

B

A

Intel recommends 55 Ohm for CMD/ADDR, 80 Ohm for CTRL/CKE, 38 Ohm for CLK

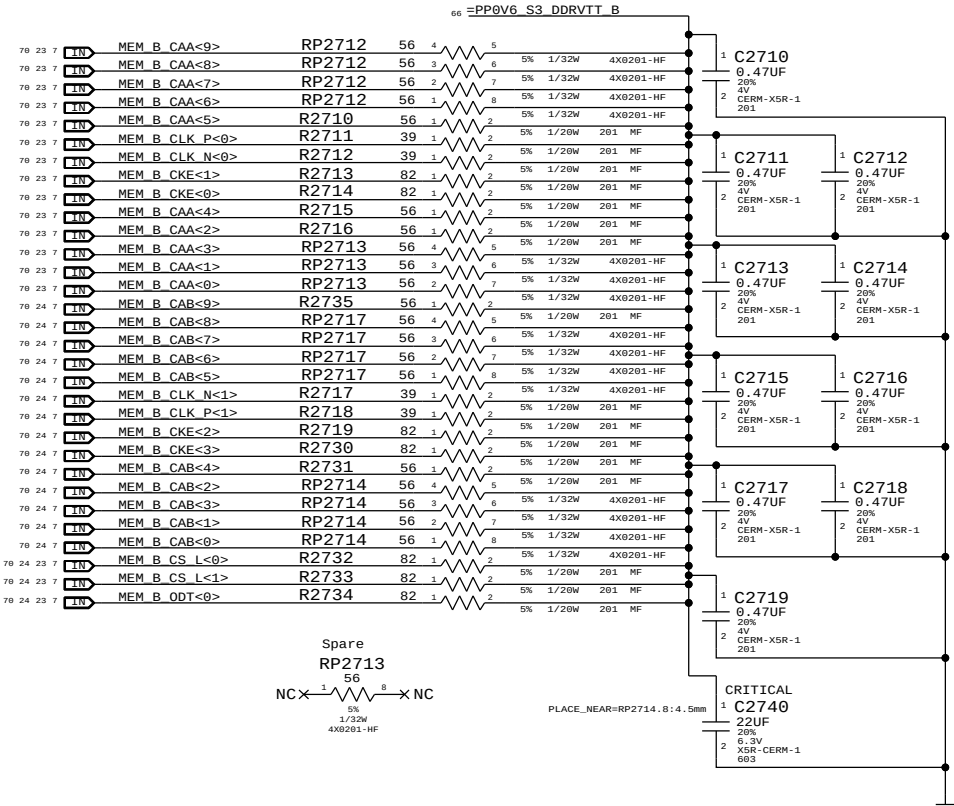


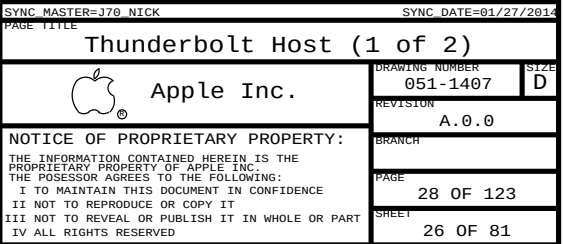
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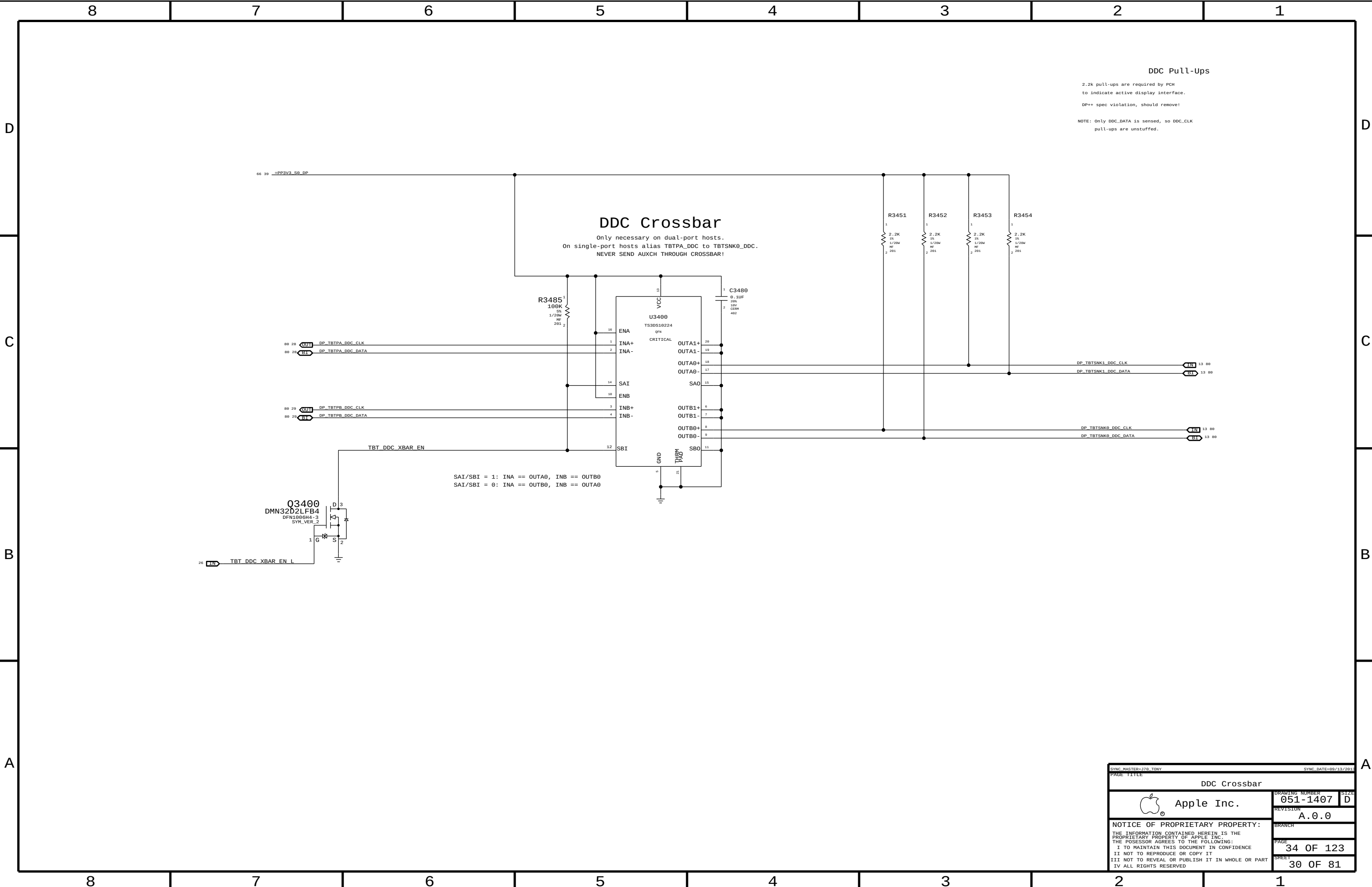
C

B

A





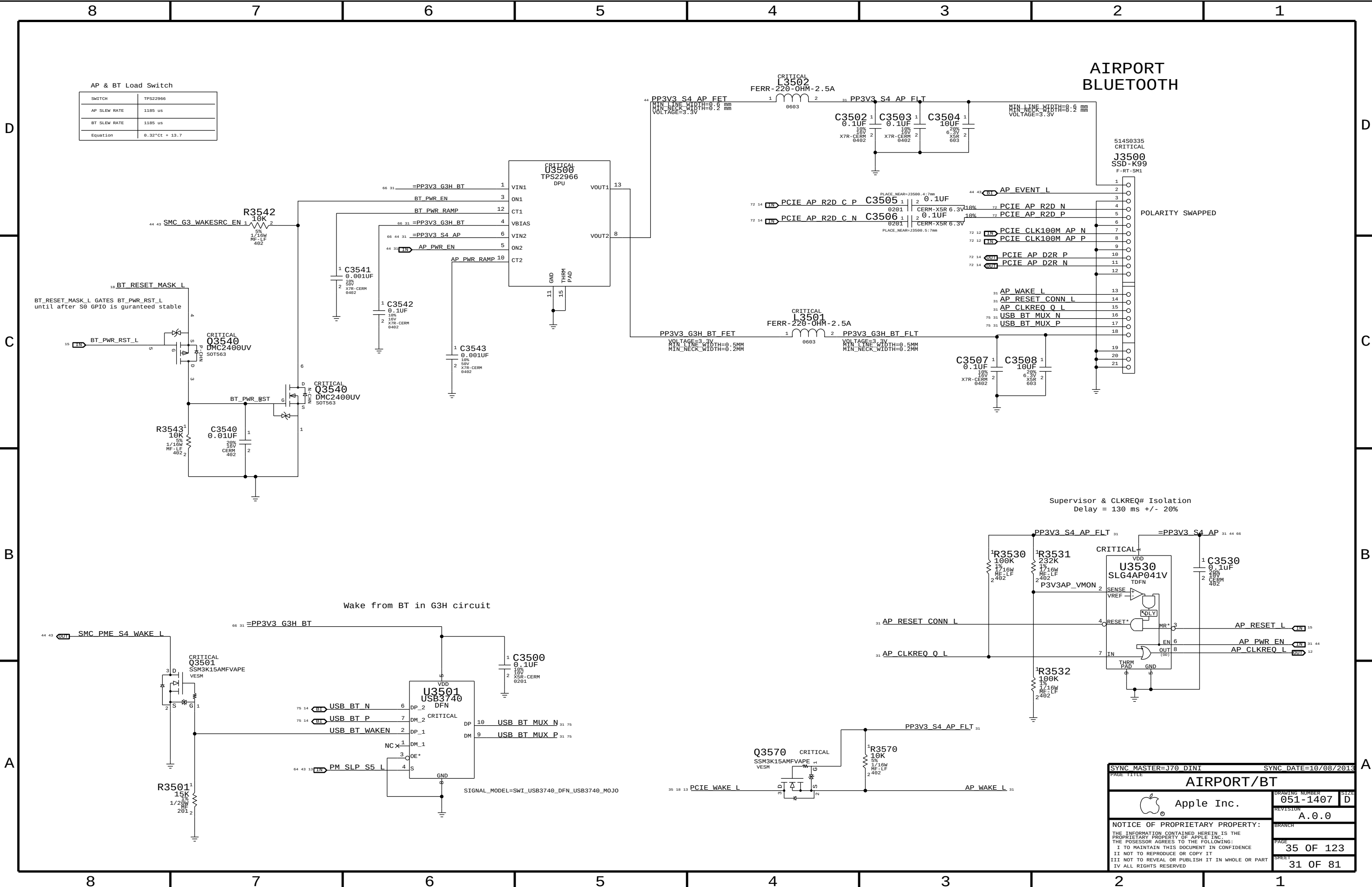


DDC Pull-Ups

2.2k pull-ups are required by PCH to indicate active display interface.

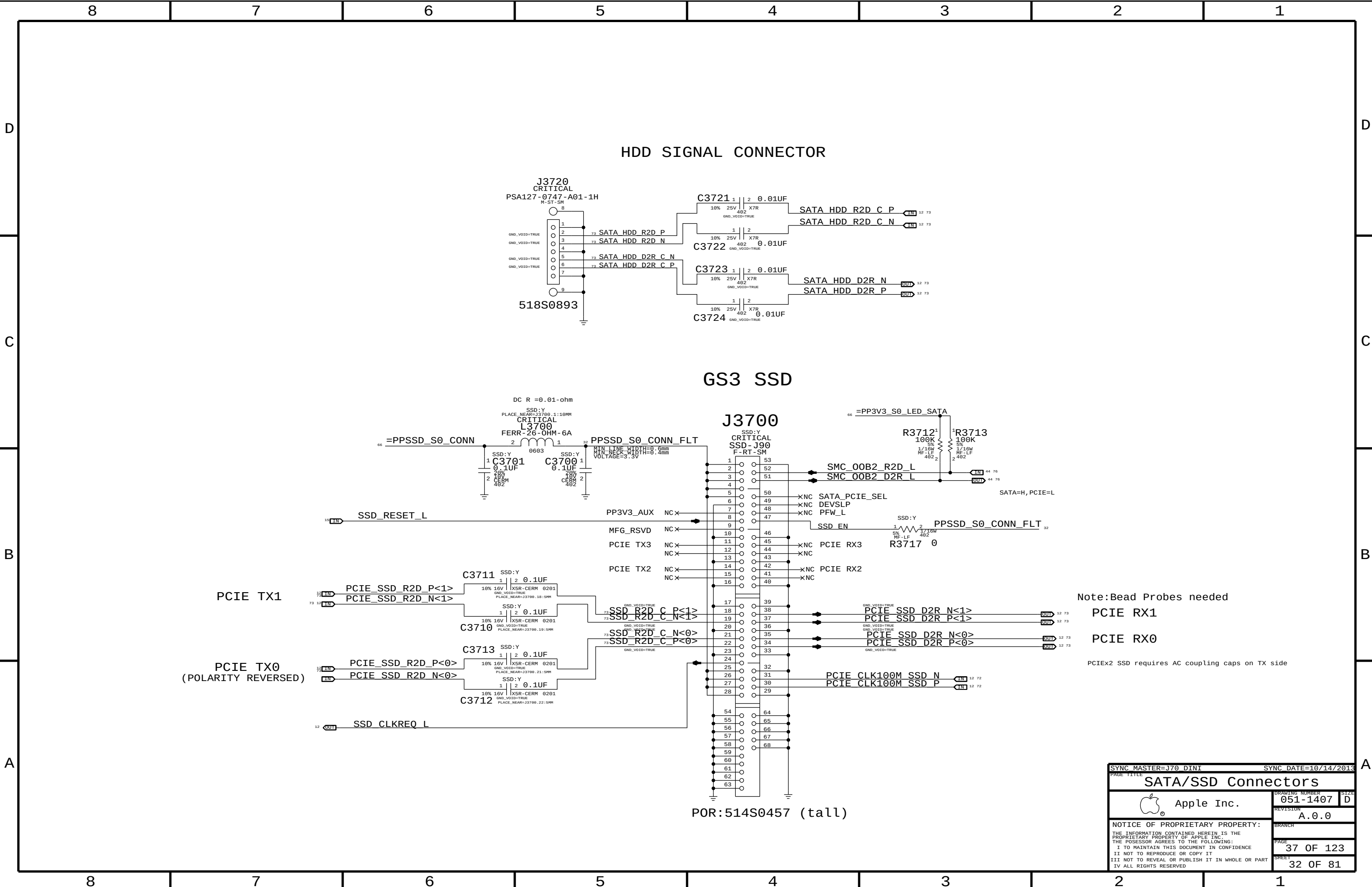
DP++ spec violation, should remove!

NOTE: Only DDC_DATA is sensed, so DDC_CLK pull-ups are unstuffed.



AIRPORT BLUETOOTH

SYNC MASTER=J70 DINI		SYNC DATE=10/08/2013	
PAGE TITLE		AIRPORT/BT	
Apple Inc.		DRAWING NUMBER	051-1407
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Note: Bead Probes needed
PCIE RX1
PCIE RX0
PCIex2 SSD requires AC coupling caps on TX side

SYNC MASTER=J70 DINI		SYNC DATE=10/14/2013	
PAGE TITLE			
SATA/SSD Connectors			
 Apple Inc.		DRAWING NUMBER	051-1407
		REVISION	A.0.0
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		PAGE	37 OF 123
		SHEET	32 OF 81

8	7	6	5	4	3	2	1
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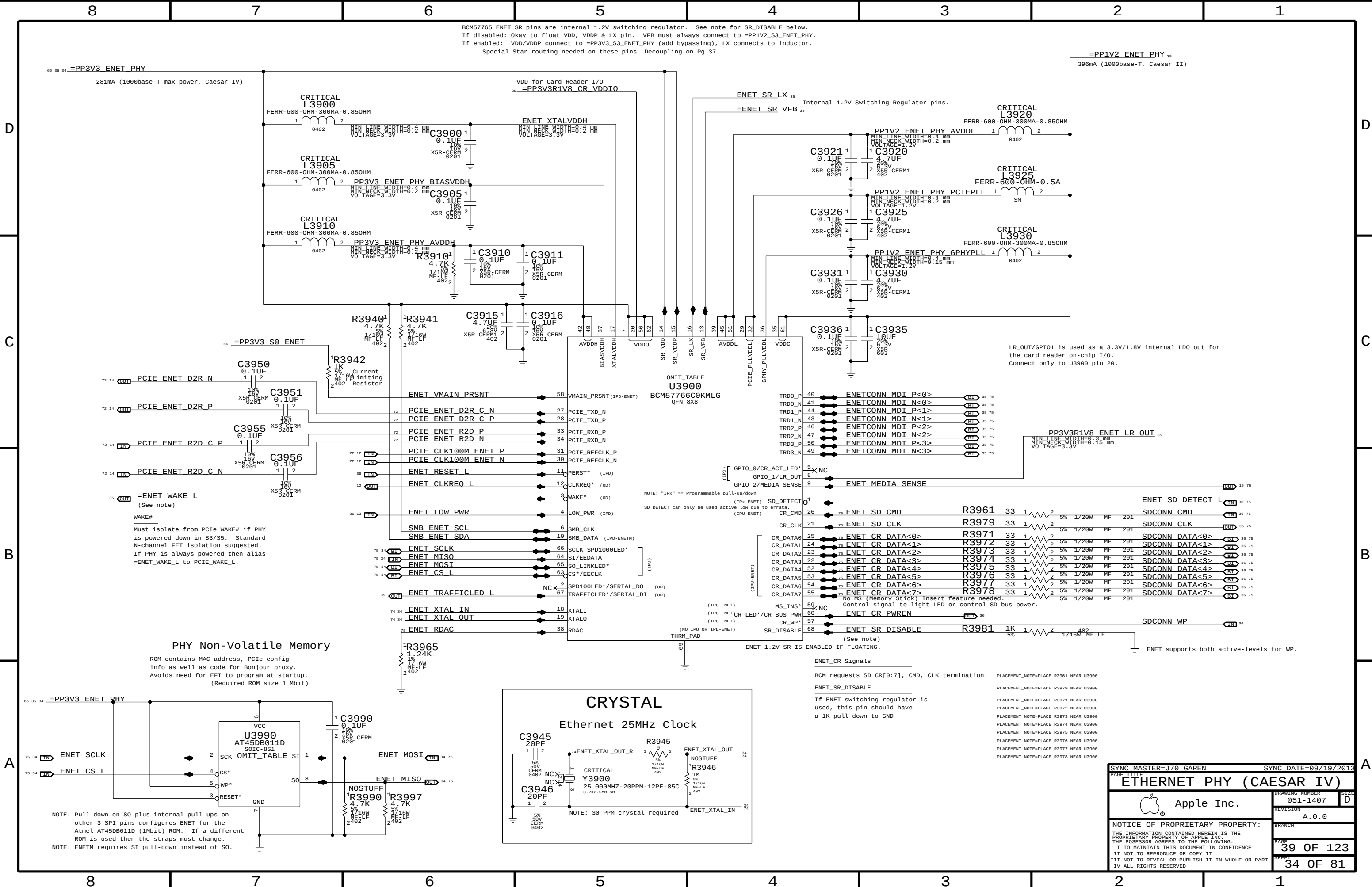
D | D

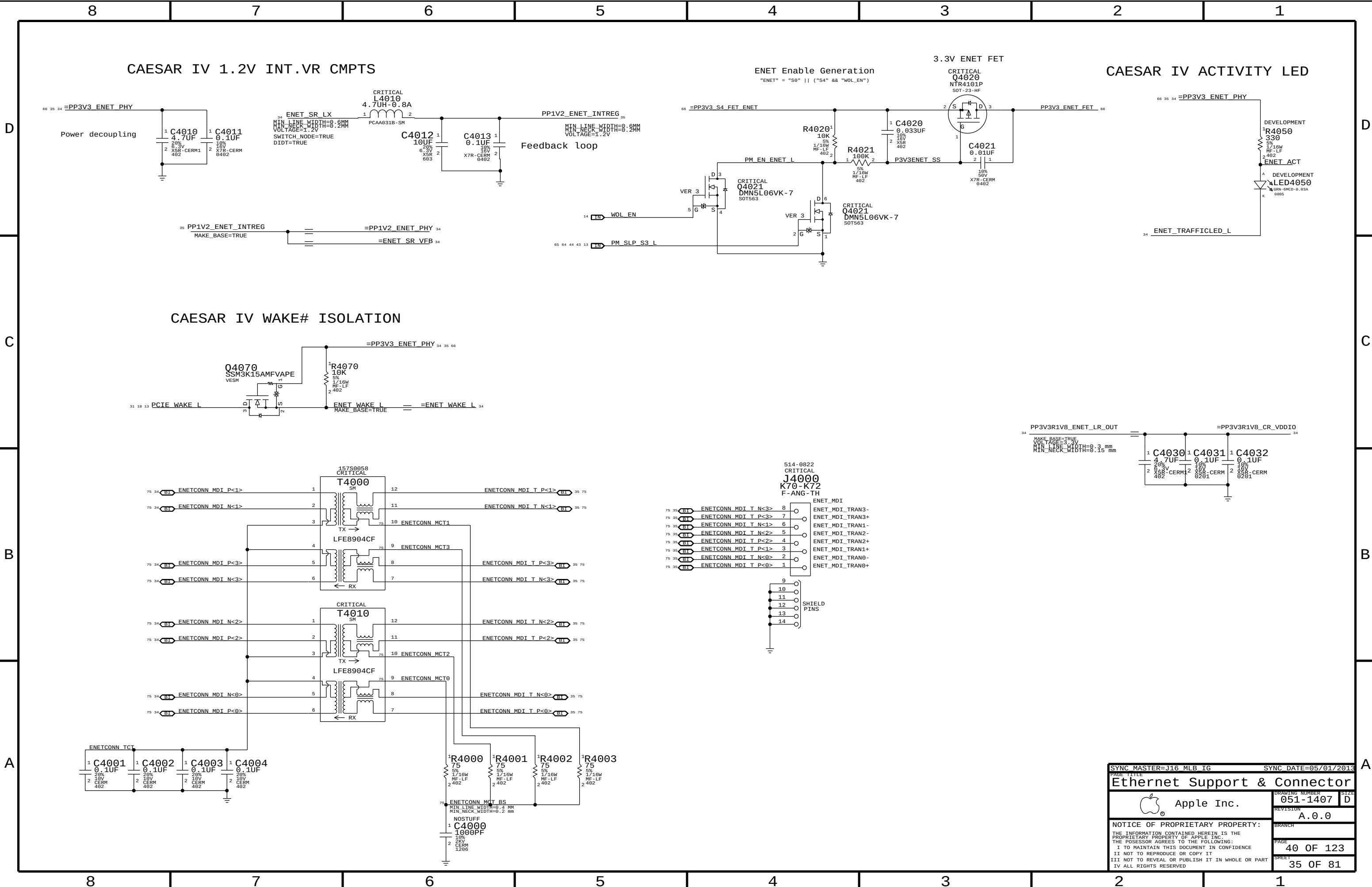


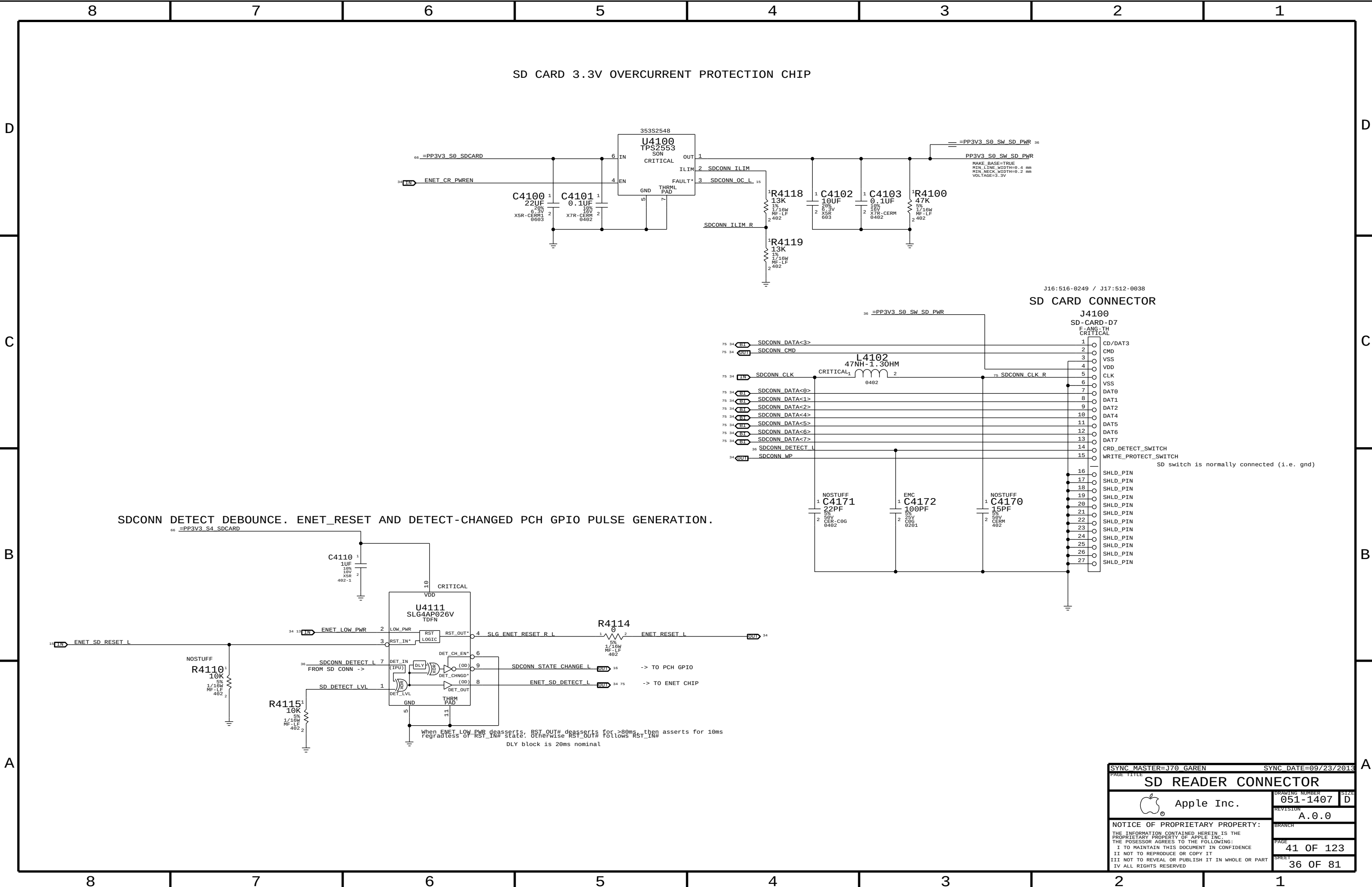
A
A

 SYNC MASTER=J16 MLB IG SYNC DATE=08/27/2013

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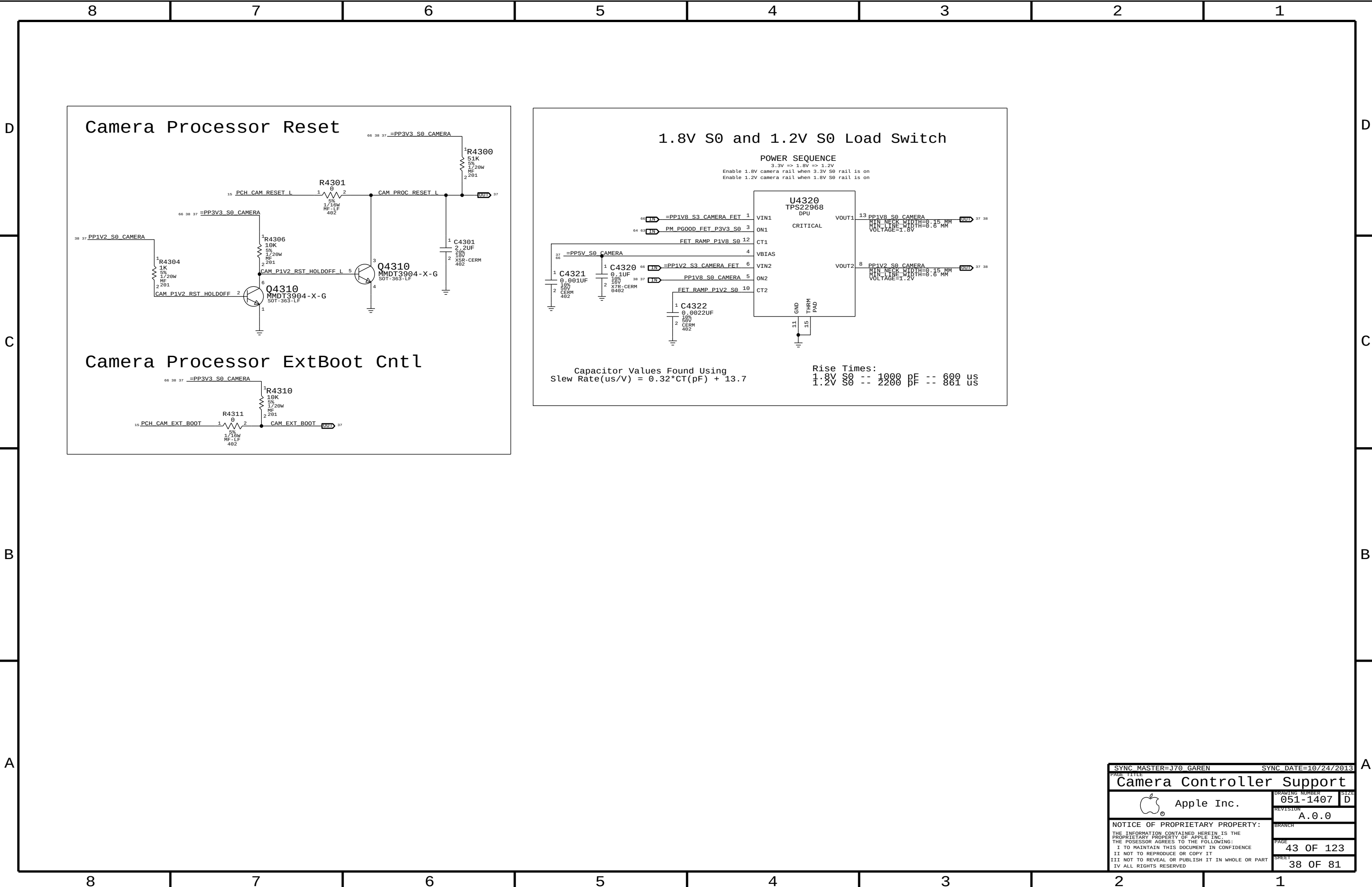




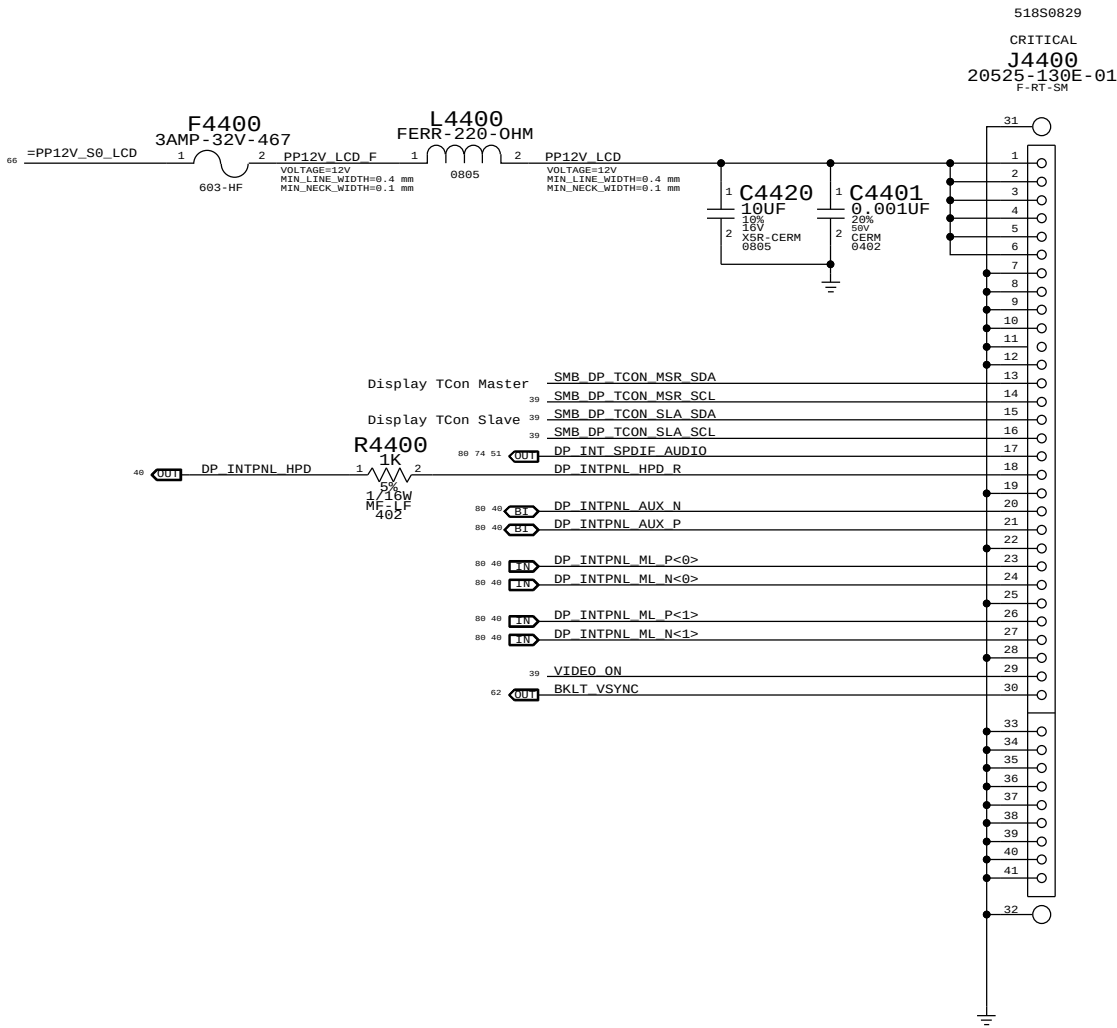


D

A

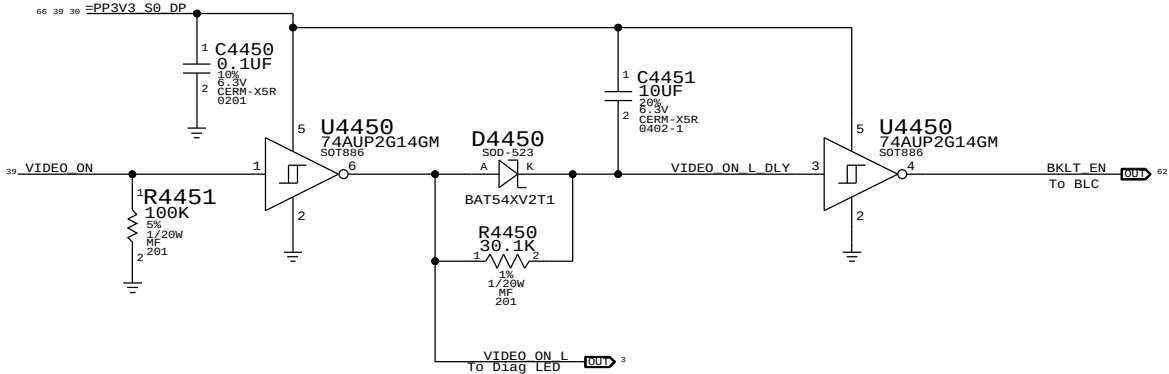


Internal DP Connector

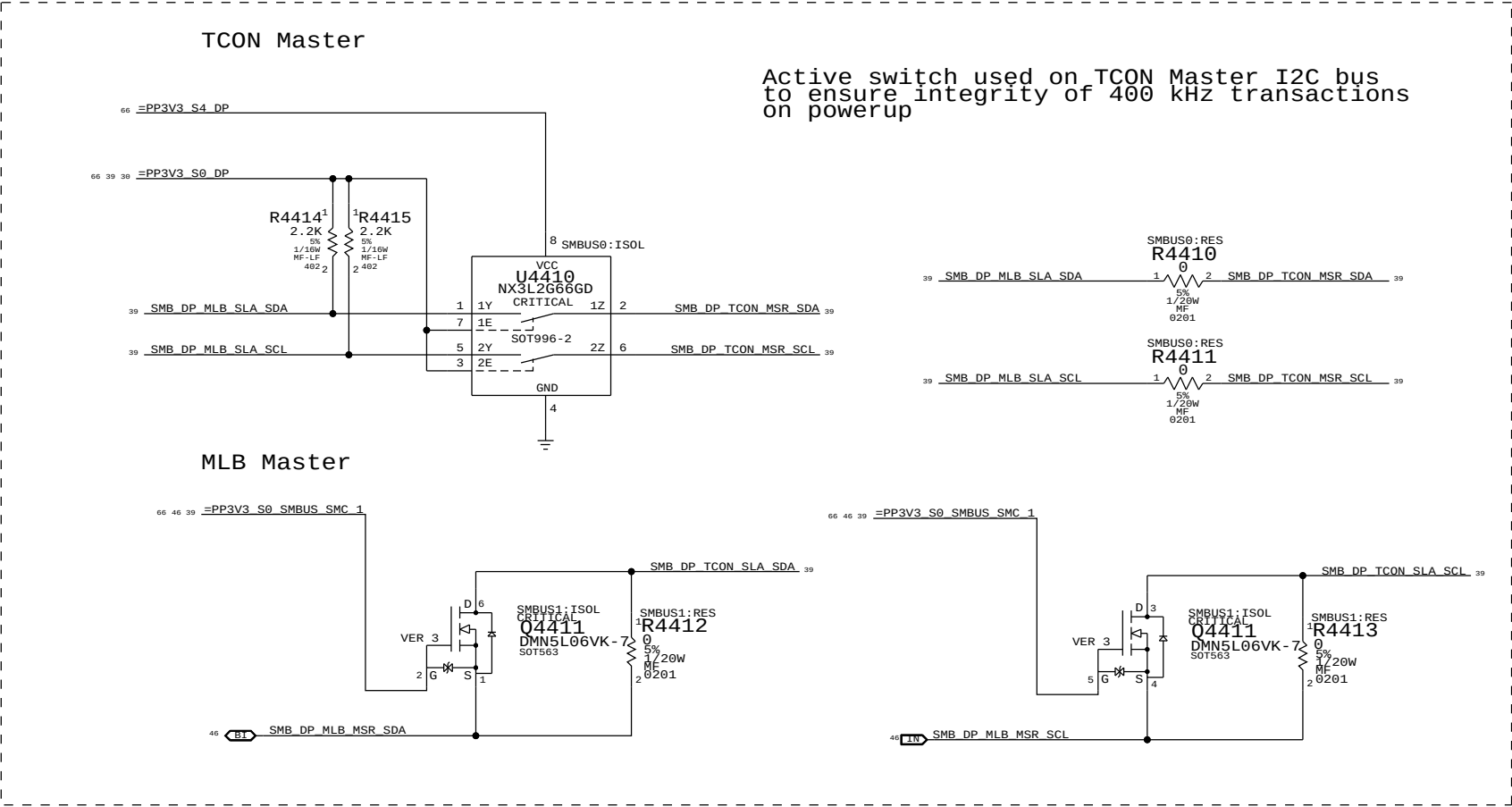


Backlight Control

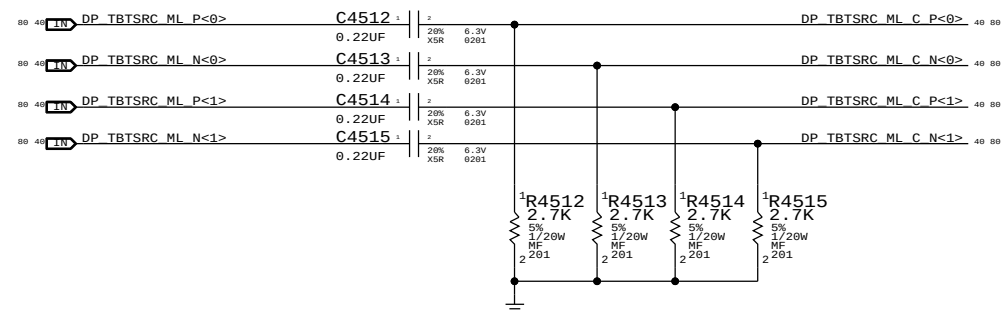
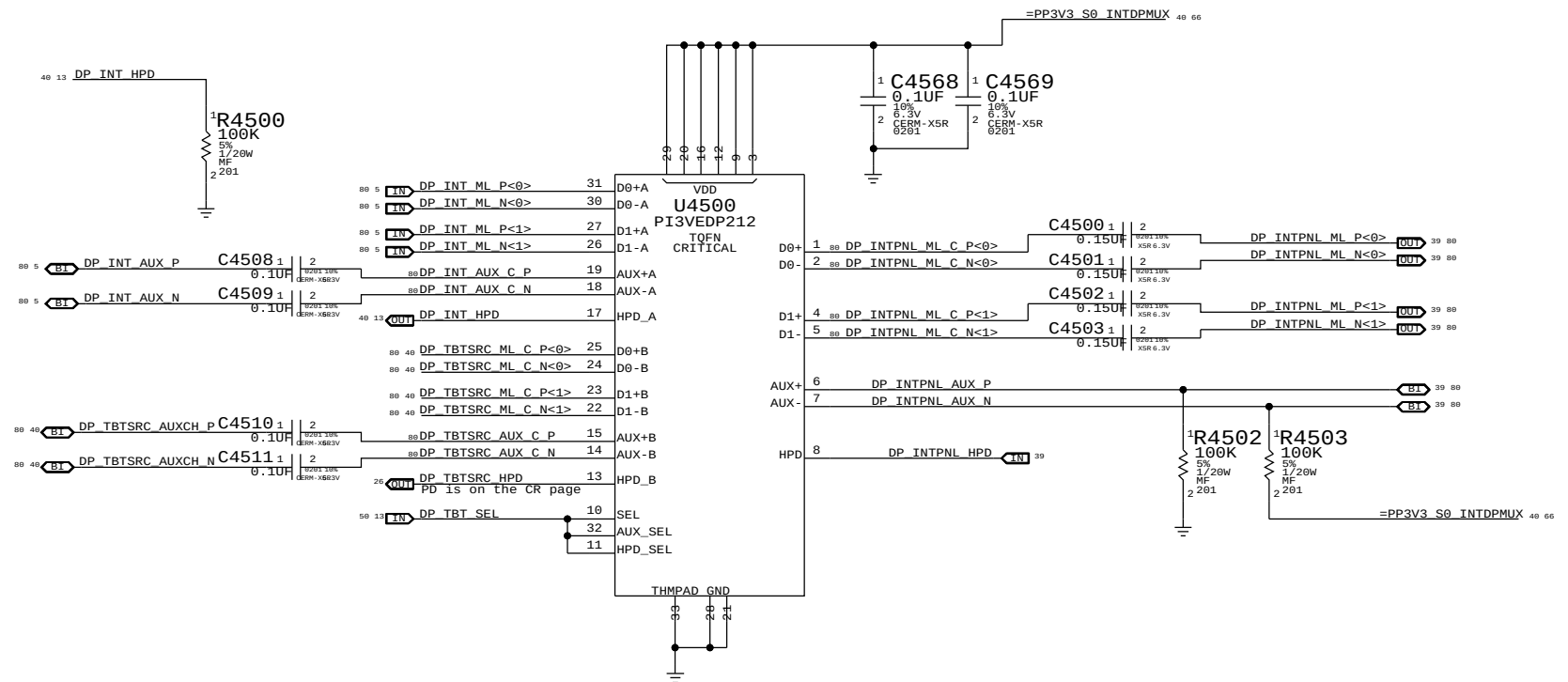
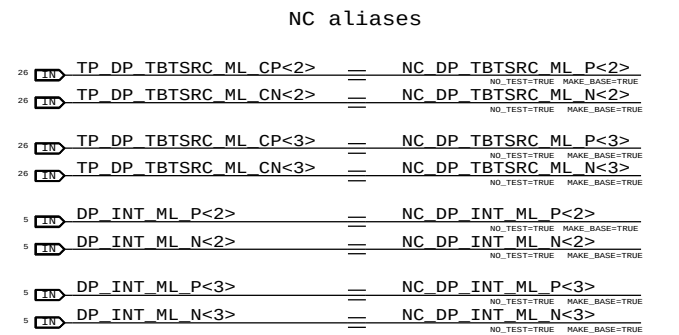
Delay applies only on a L->H transition on VIDEO_ON. This guarantees video is valid before the backlight is enabled.
On a H->L transition, output follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video



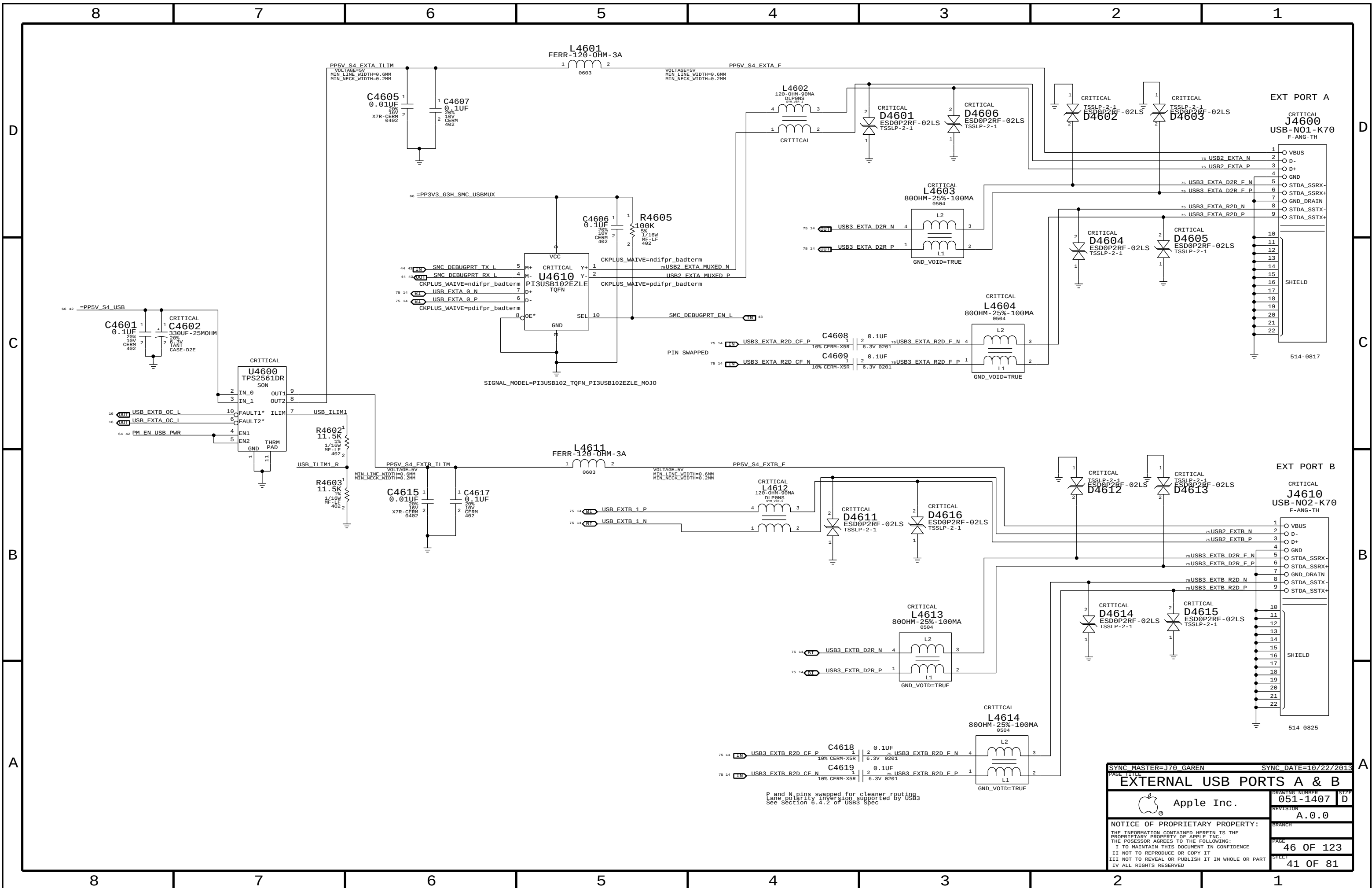
SMBus Isolation



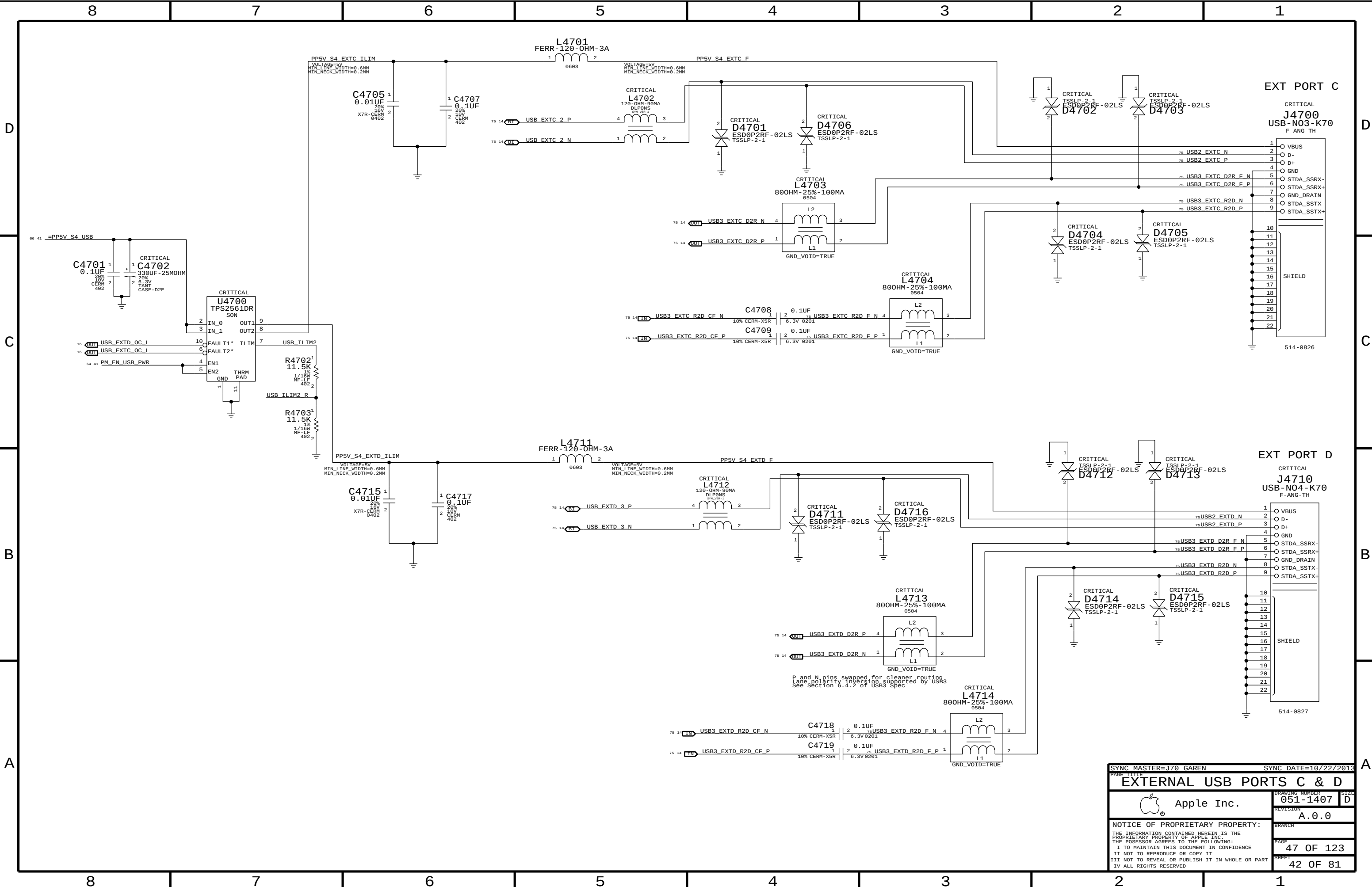
Active switch used on TCON Master I2C bus to ensure integrity of 400 kHz transactions on powerup



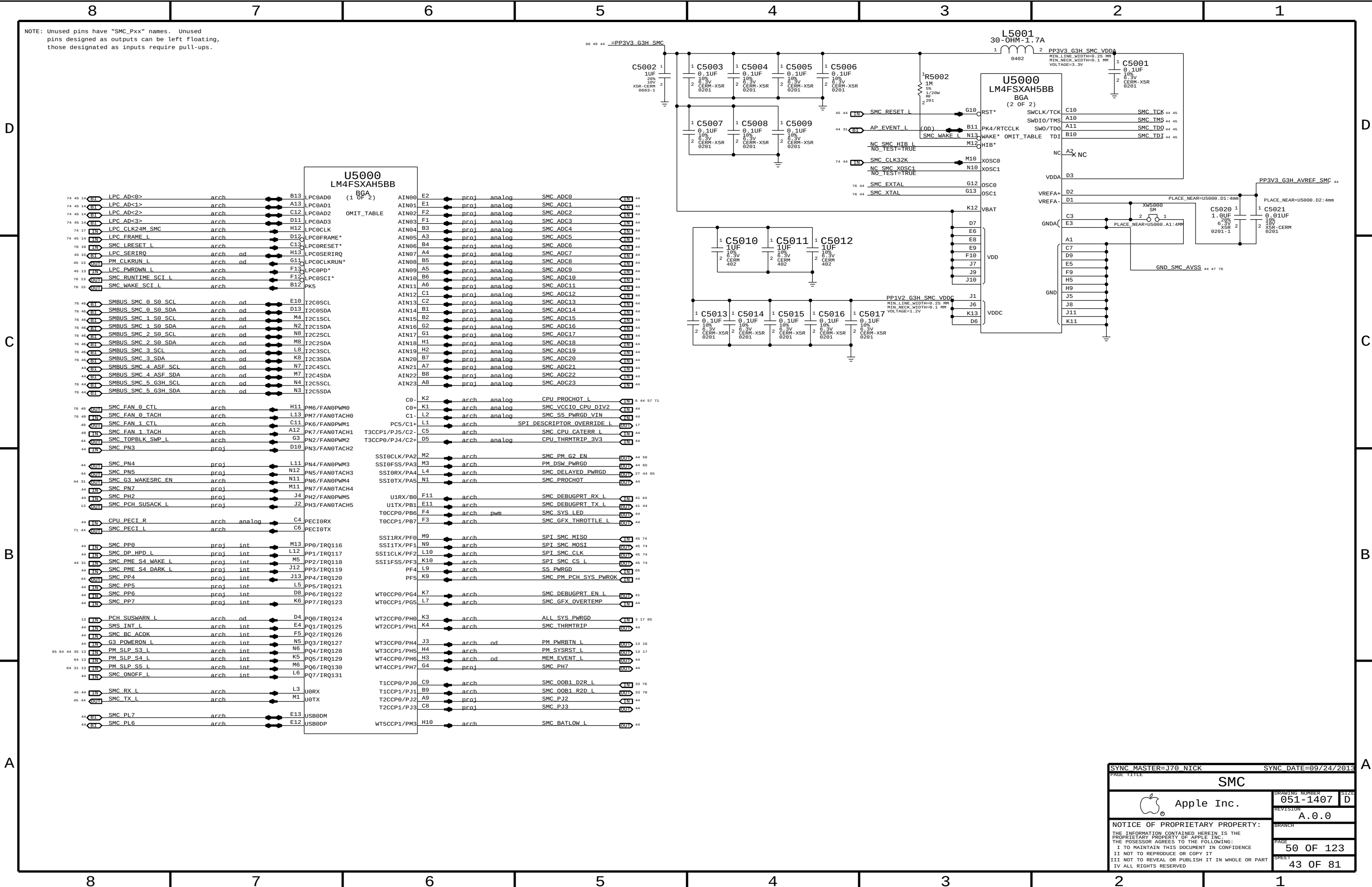
SYNC MASTER=J70 TONY		SYNC DATE=09/05/2013	
PAGE TITLE		Internal DP MUXing	
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		SIZE D	
		REVISION A.0.0	
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		SHEET 40 OF 81	

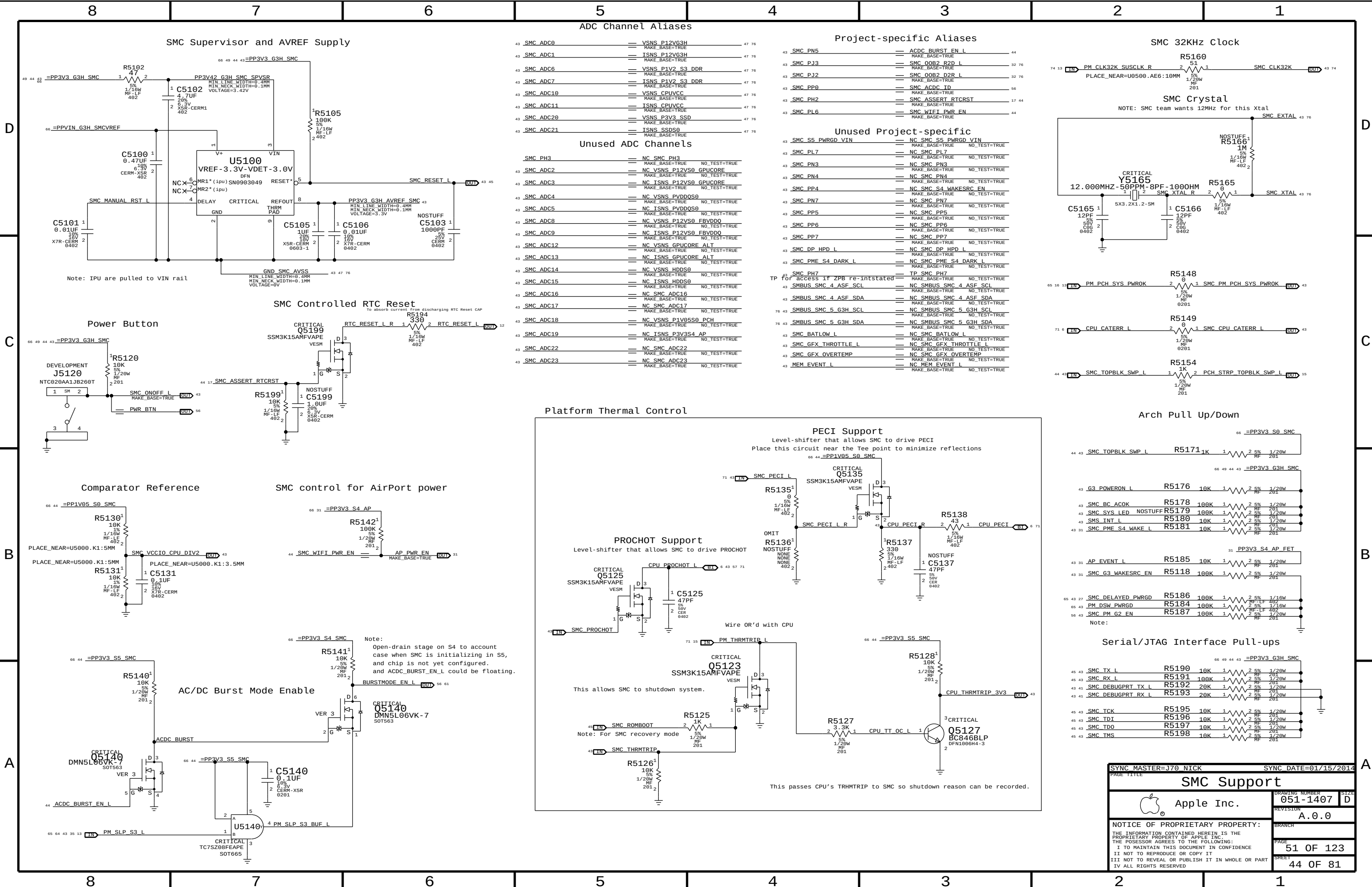


SYNC MASTER=J70 GAREN		SYNC DATE=10/22/2013	
PAGE TITLE		EXTERNAL USB PORTS A & B	
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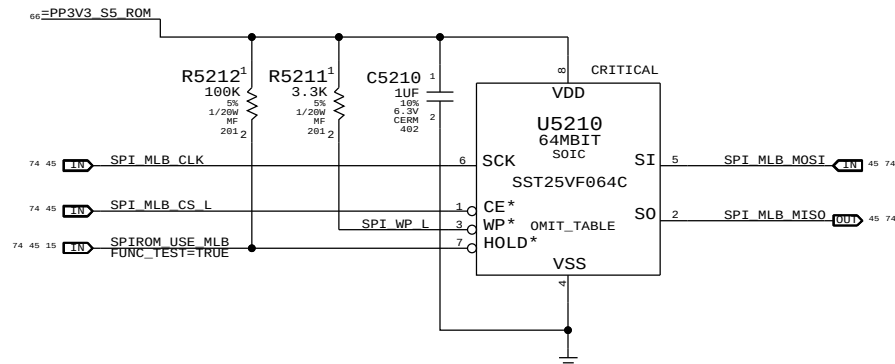


SYNC MASTER=J70 GAREN		SYNC DATE=10/22/2013	
PAGE TITLE		EXTERNAL USB PORTS C & D	
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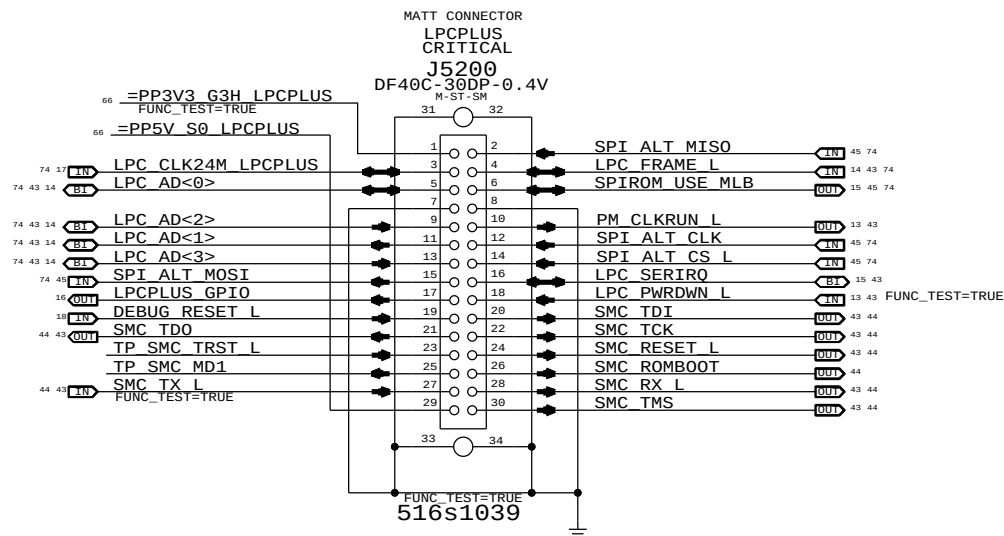




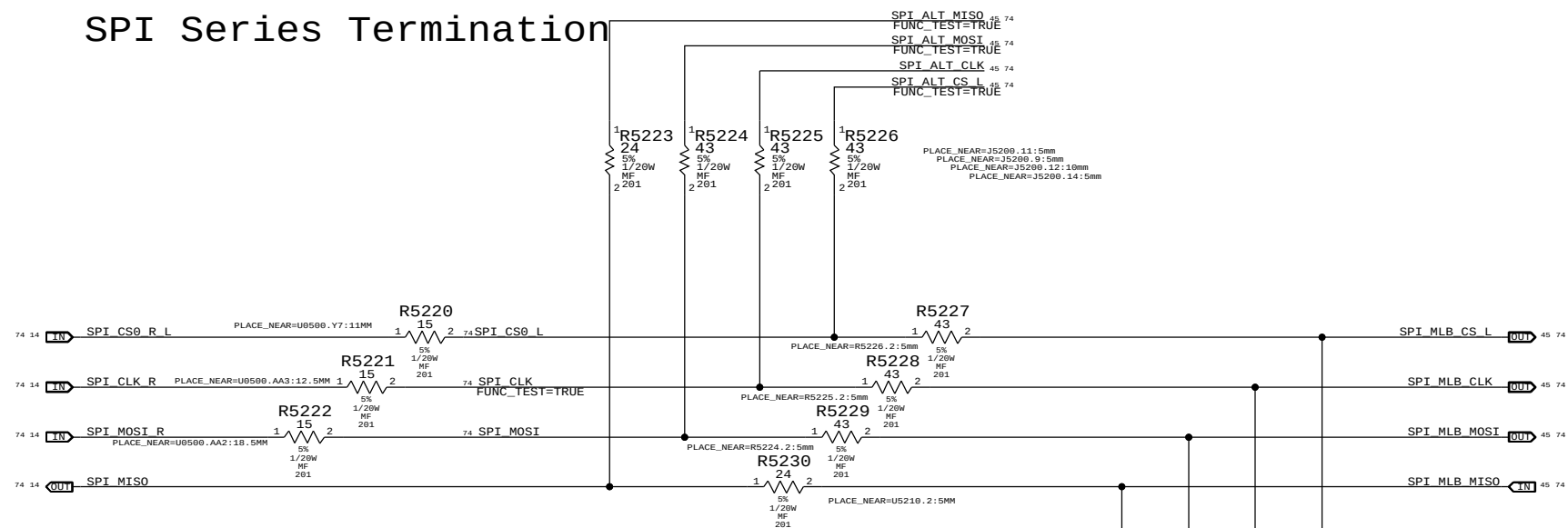
SPI BootROM



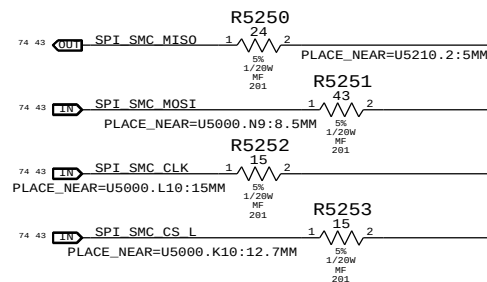
LPC+SPI Connector



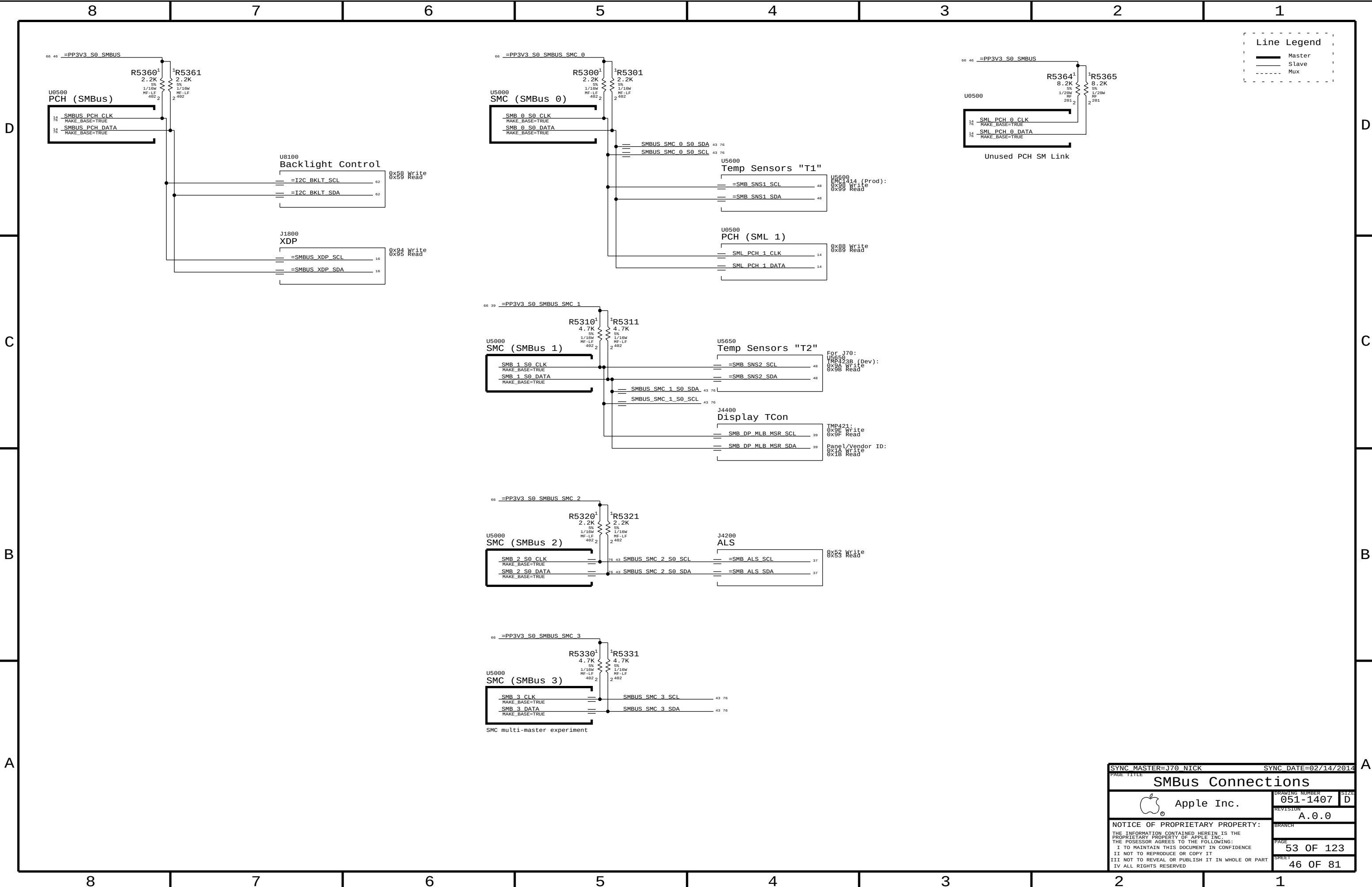
SPI Series Termination

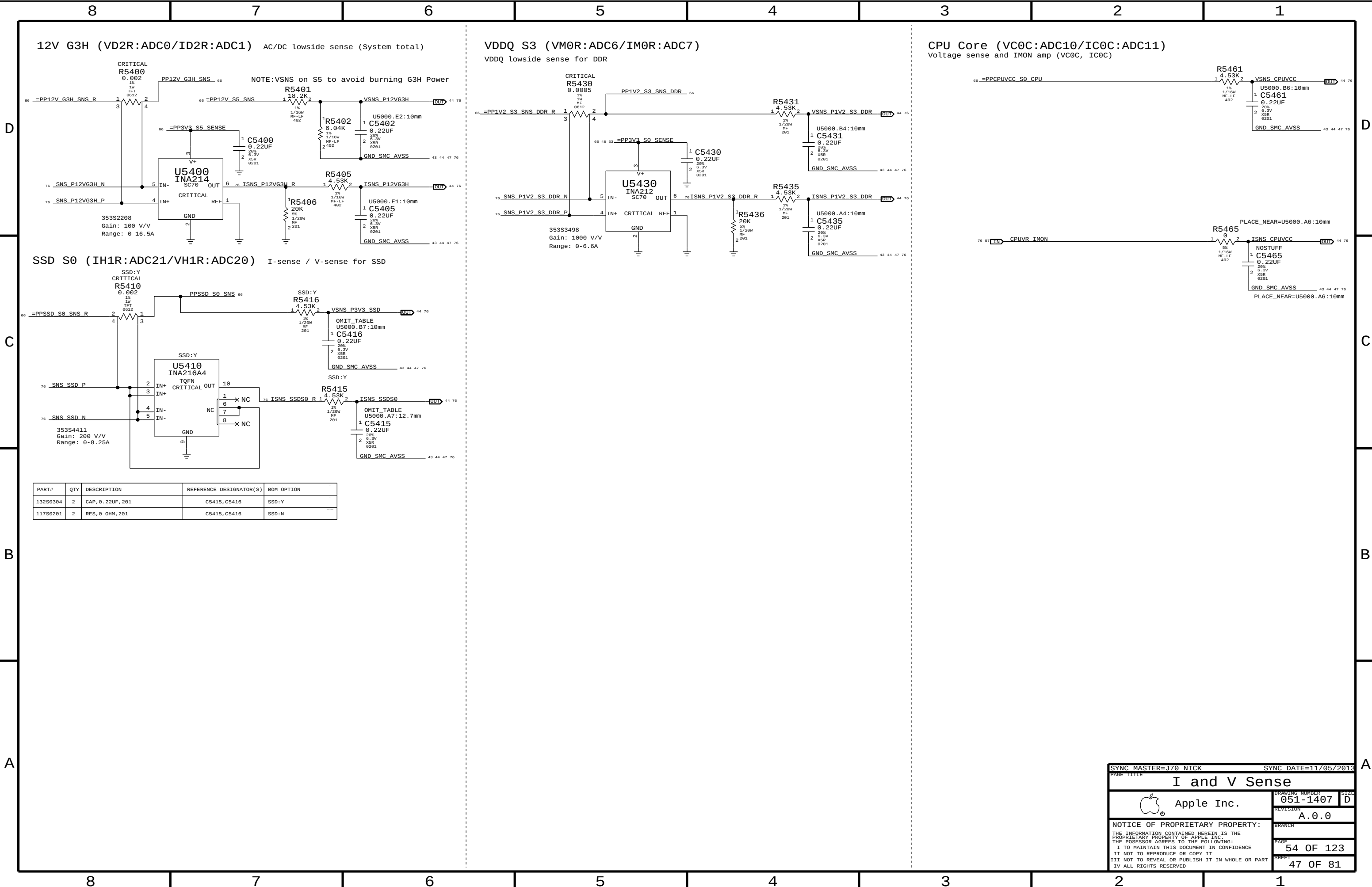


SMC SPI Support

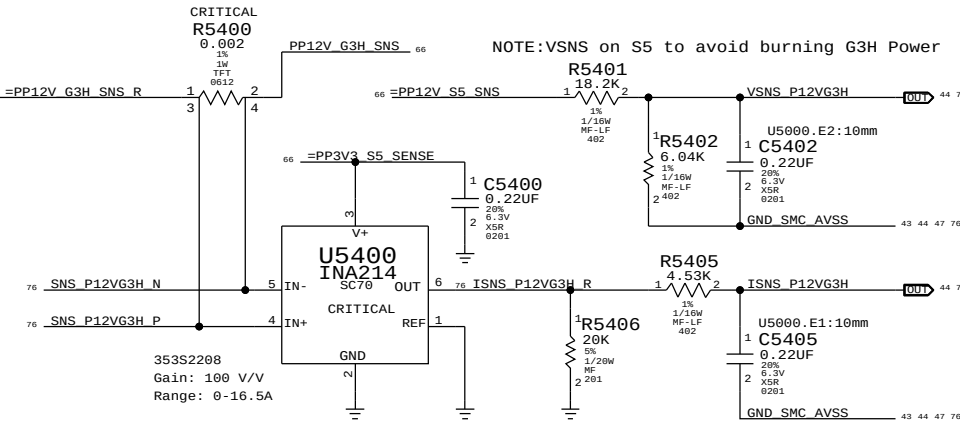


SYNC MASTER=J70 TONY		SYNC DATE=09/24/2013	
PAGE TITLE		SIZE	
SPI and Debug Connector		051-1407 D	
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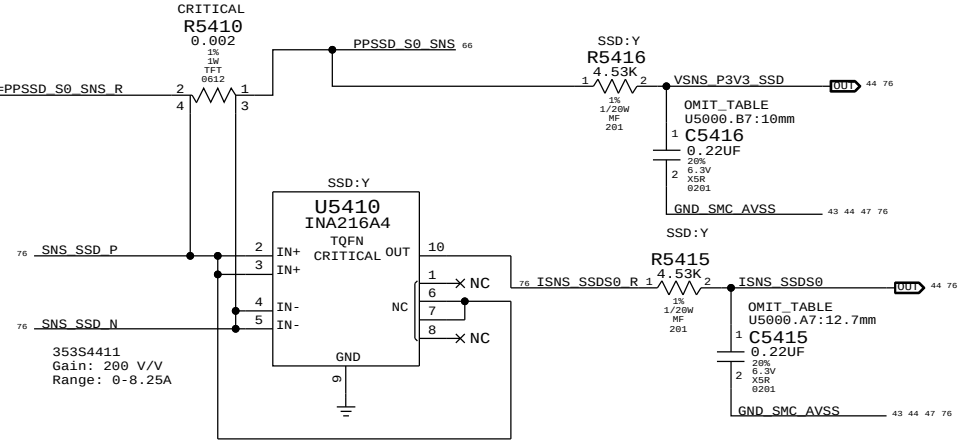




12V G3H (VD2R:ADC0/ID2R:ADC1) AC/DC lowside sense (System total)



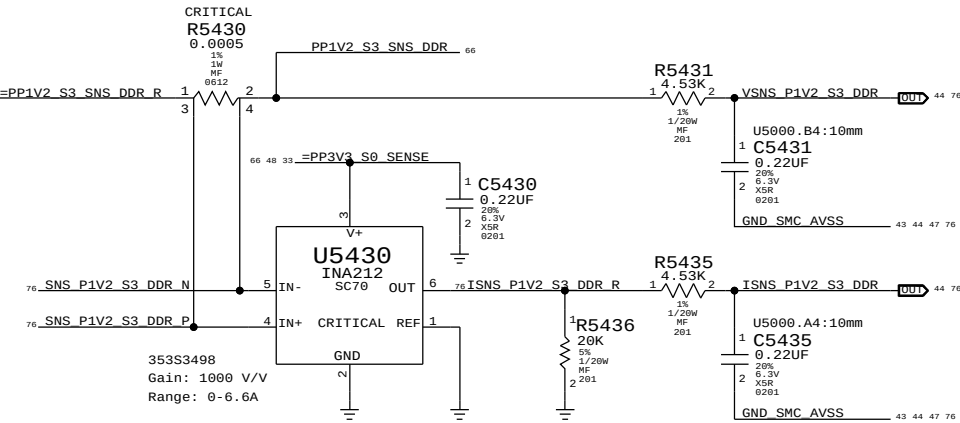
SSD S0 (IH1R:ADC21/VH1R:ADC20) I-sense / V-sense for SSD



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP, 0.22UF, 201	C5415, C5416	SSD:Y
117S0201	2	RES, 0 OHM, 201	C5415, C5416	SSD:N

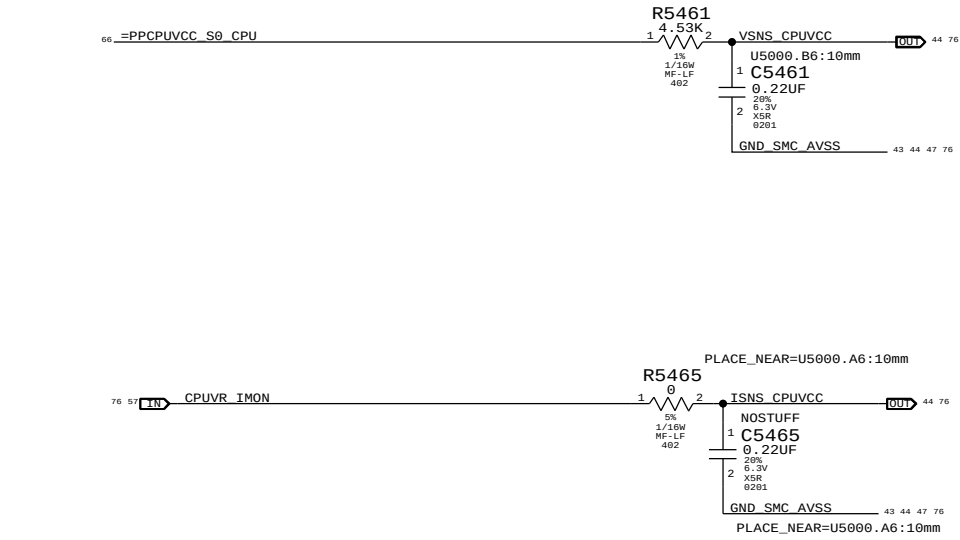
VDDQ S3 (VM0R:ADC6/IM0R:ADC7)

VDDQ lowside sense for DDR



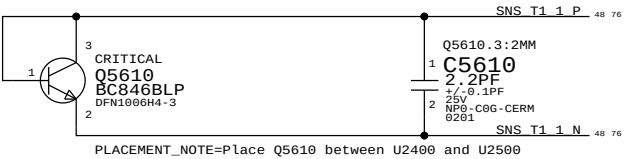
CPU Core (VC0C:ADC10/IC0C:ADC11)

Voltage sense and IMON amp (VC0C, IC0C)

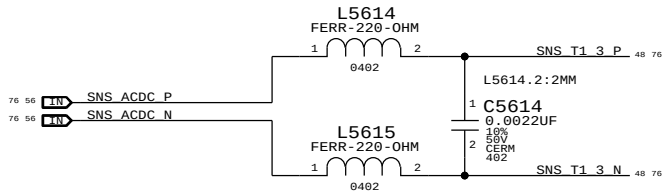


Temperature Sensor T1

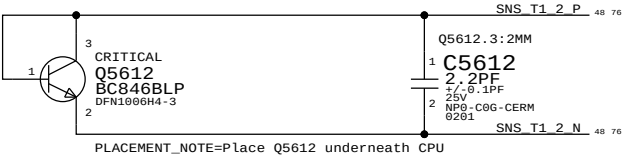
LPDDR3 Proximity



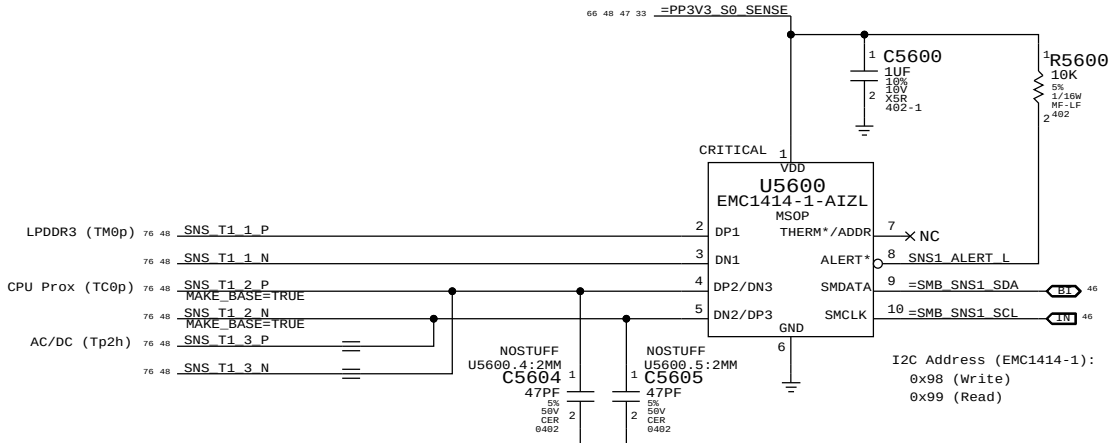
AC/DC Diode on supply



CPU Proximity

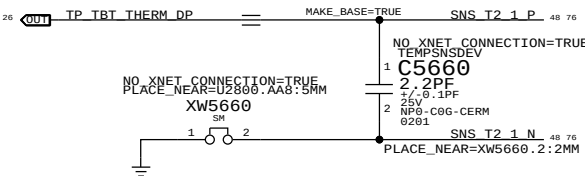


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

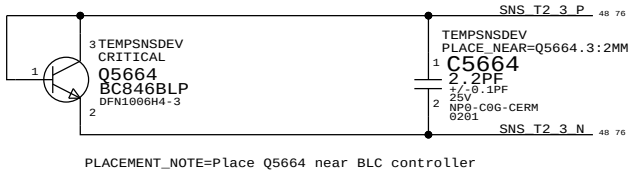


Temperature Sensor T2

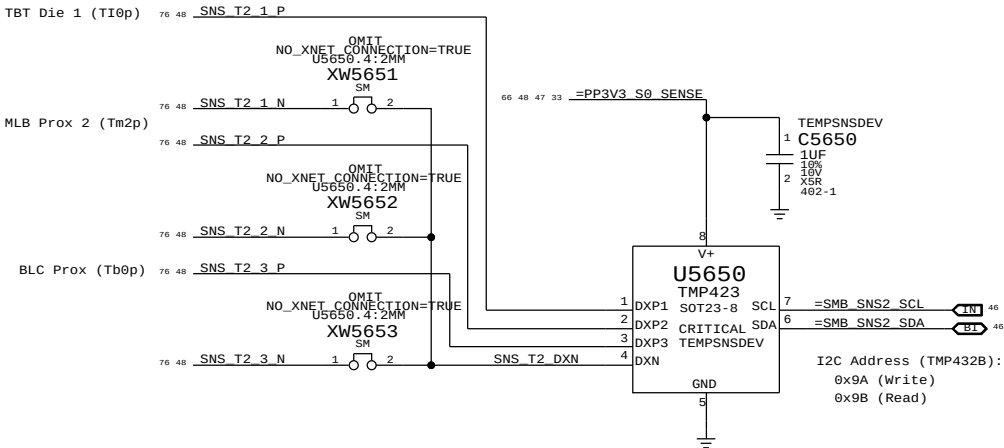
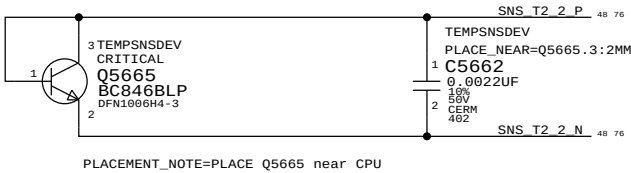
TBT On Die



BLC Proximity



MLB Misc 0



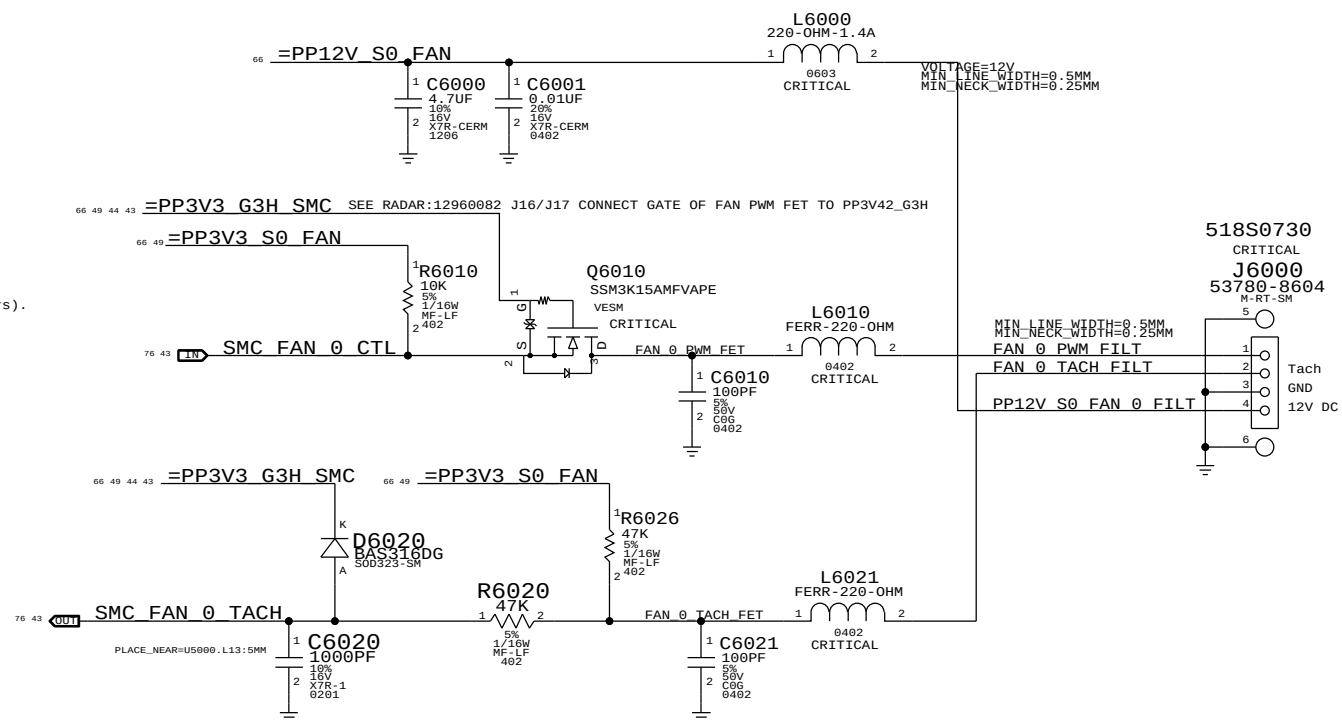
SYNC MASTER=J70 NICK		SYNC DATE=11/05/2013	
PAGE TITLE			
Temperature Sensors			
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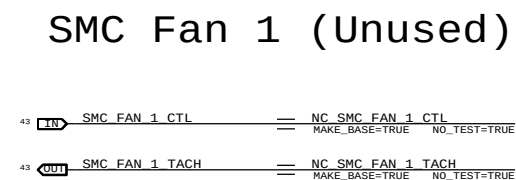
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

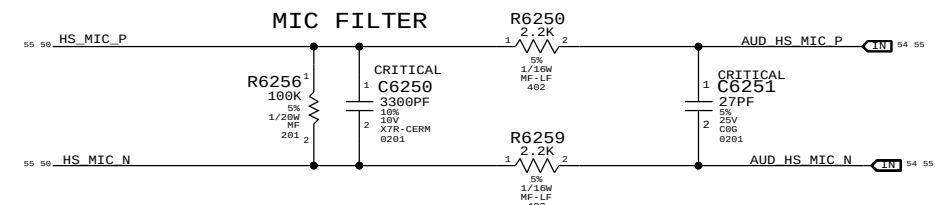
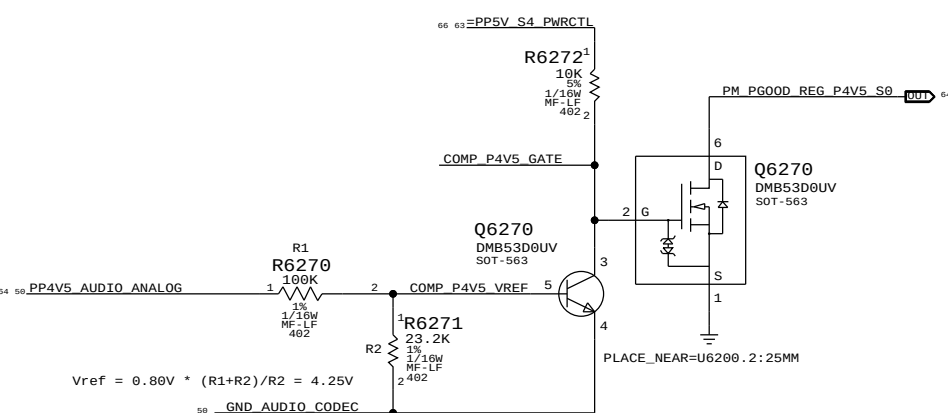
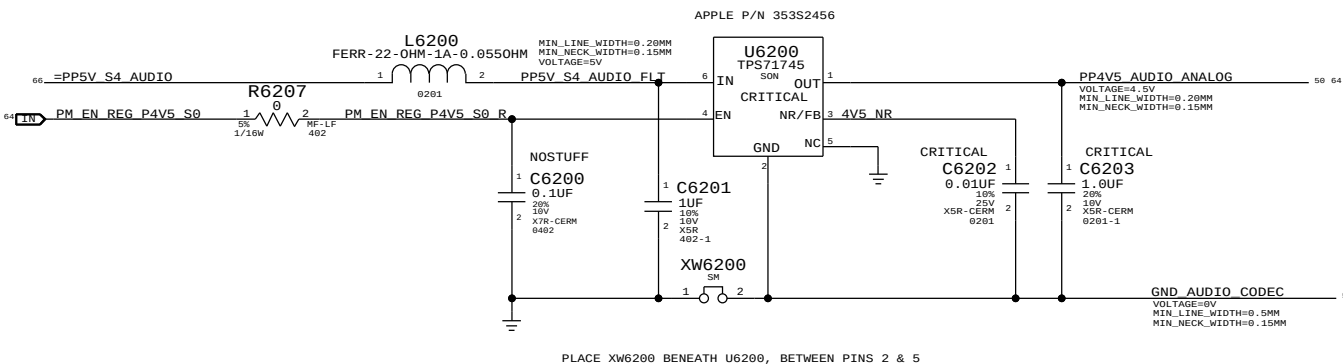
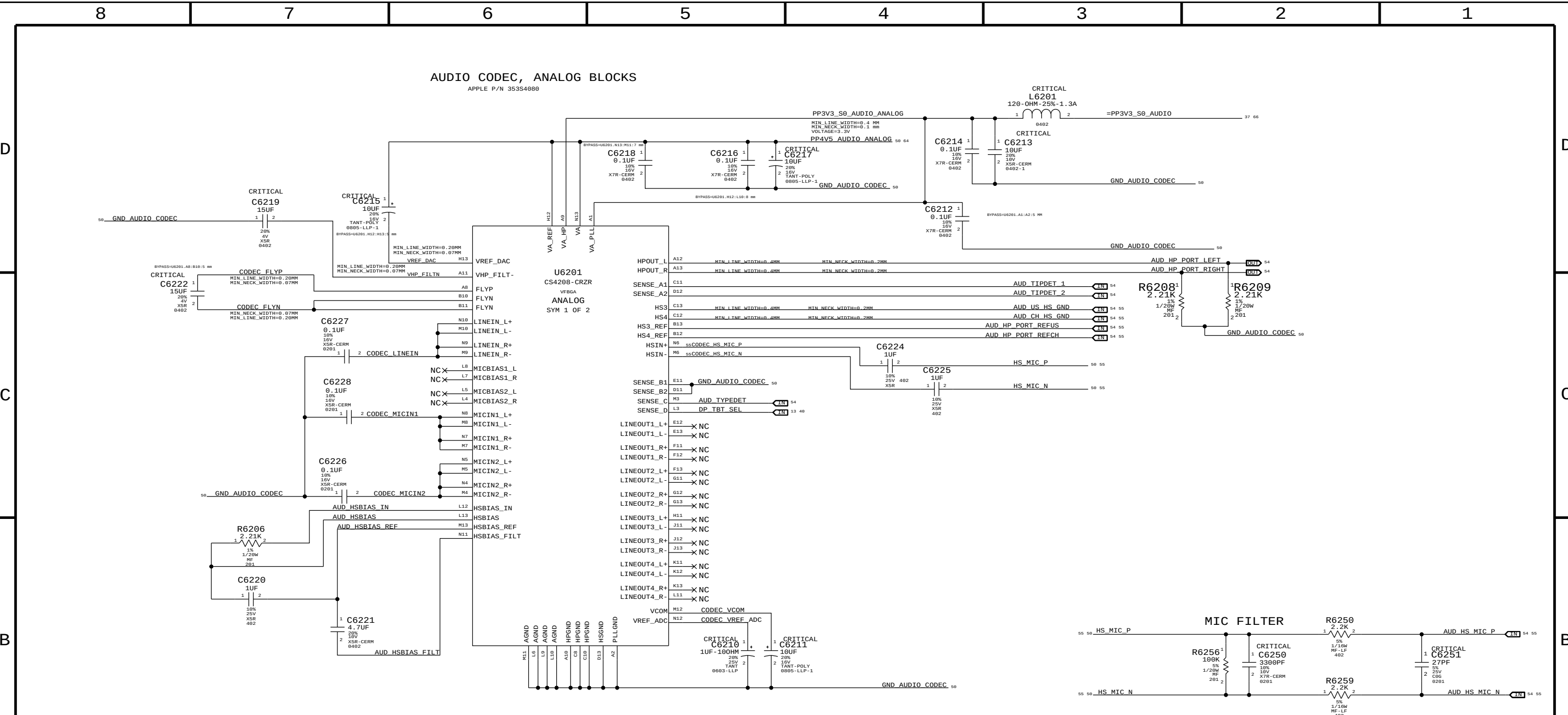
Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



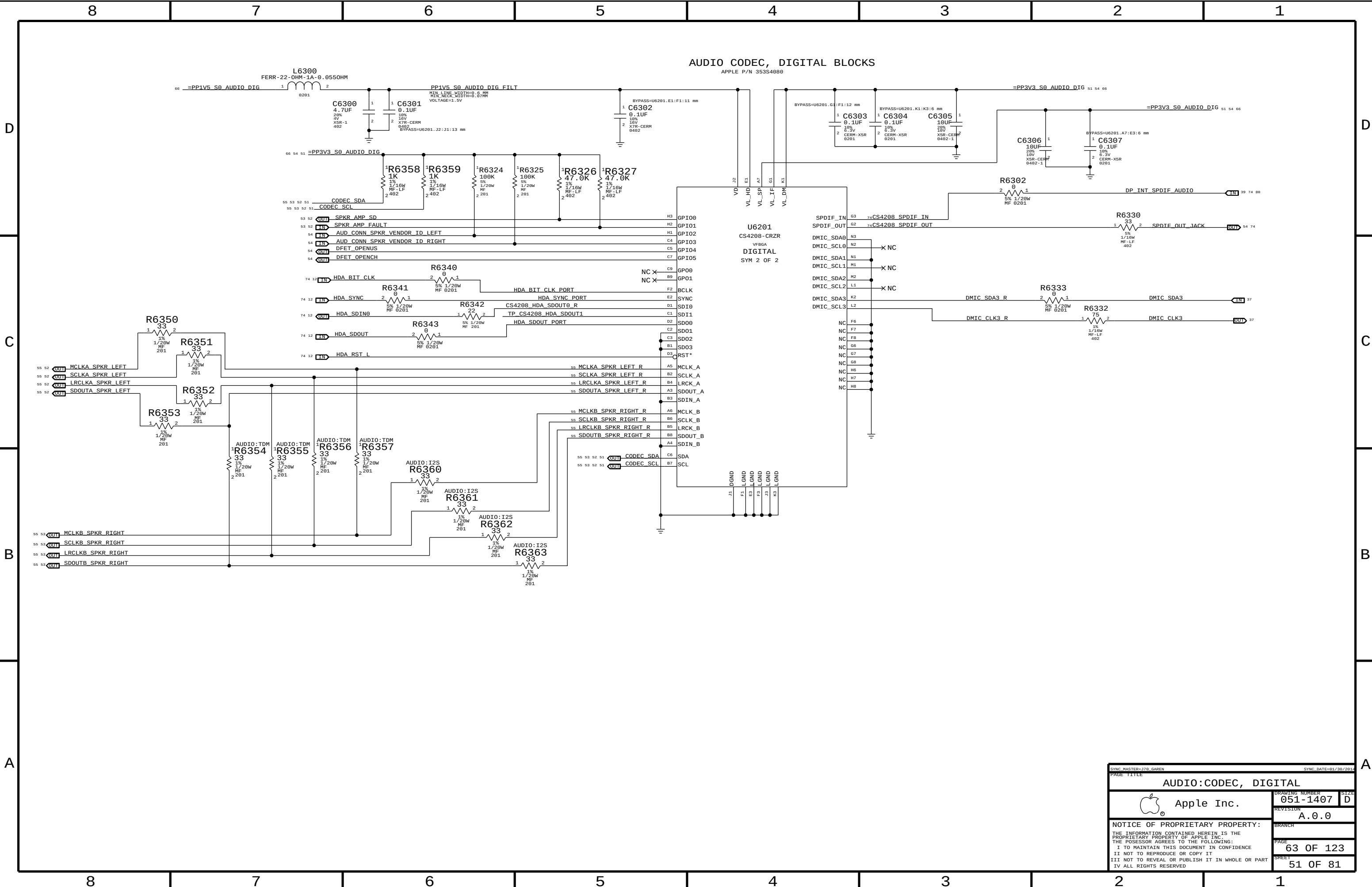
Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.



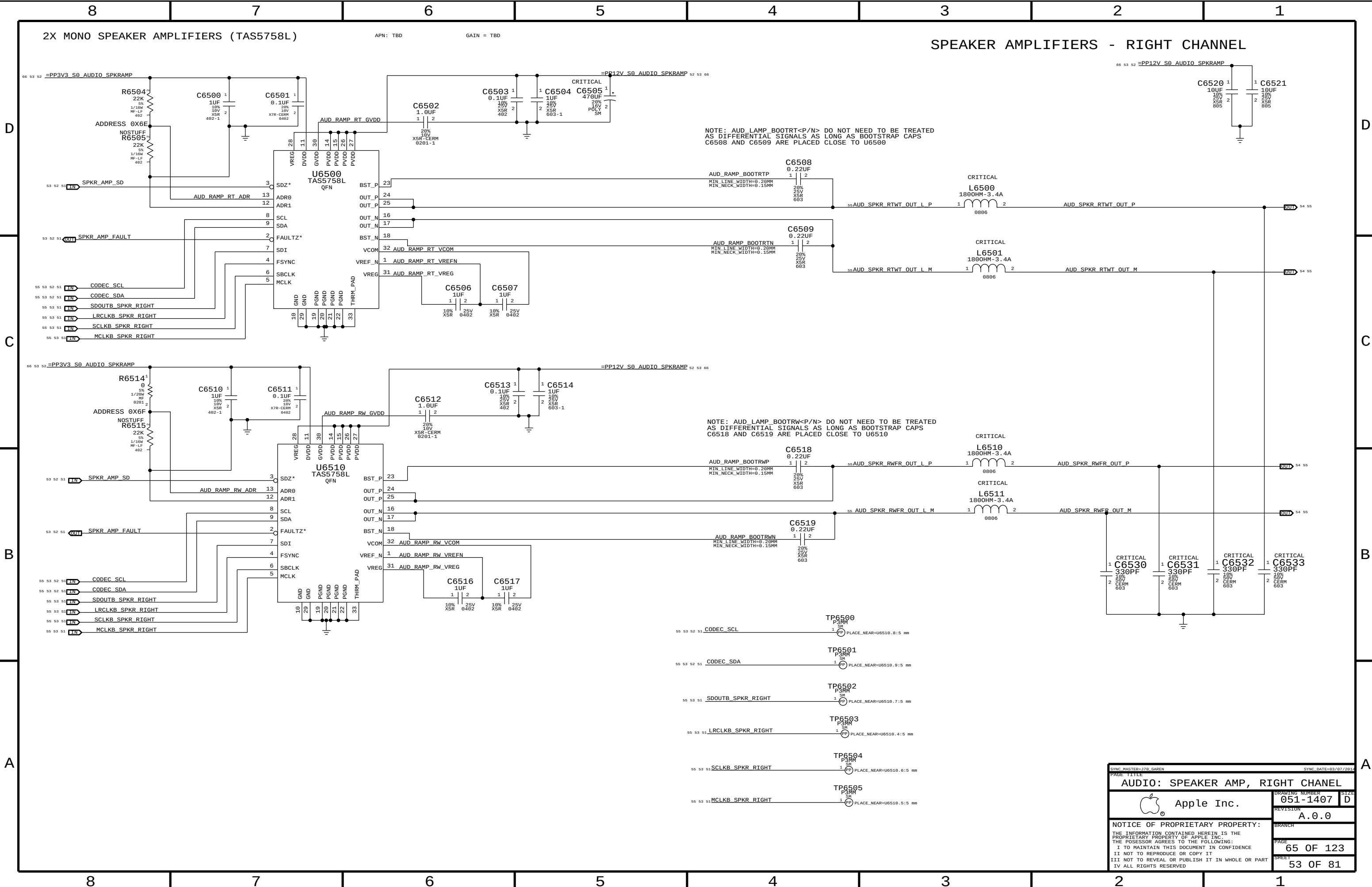
SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
PAGE TITLE			
System Fan			
		Apple Inc.	
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		SIZE	
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		BRANCH	
		PAGE	
		60 OF 123	
		SHEET	
		49 OF 81	

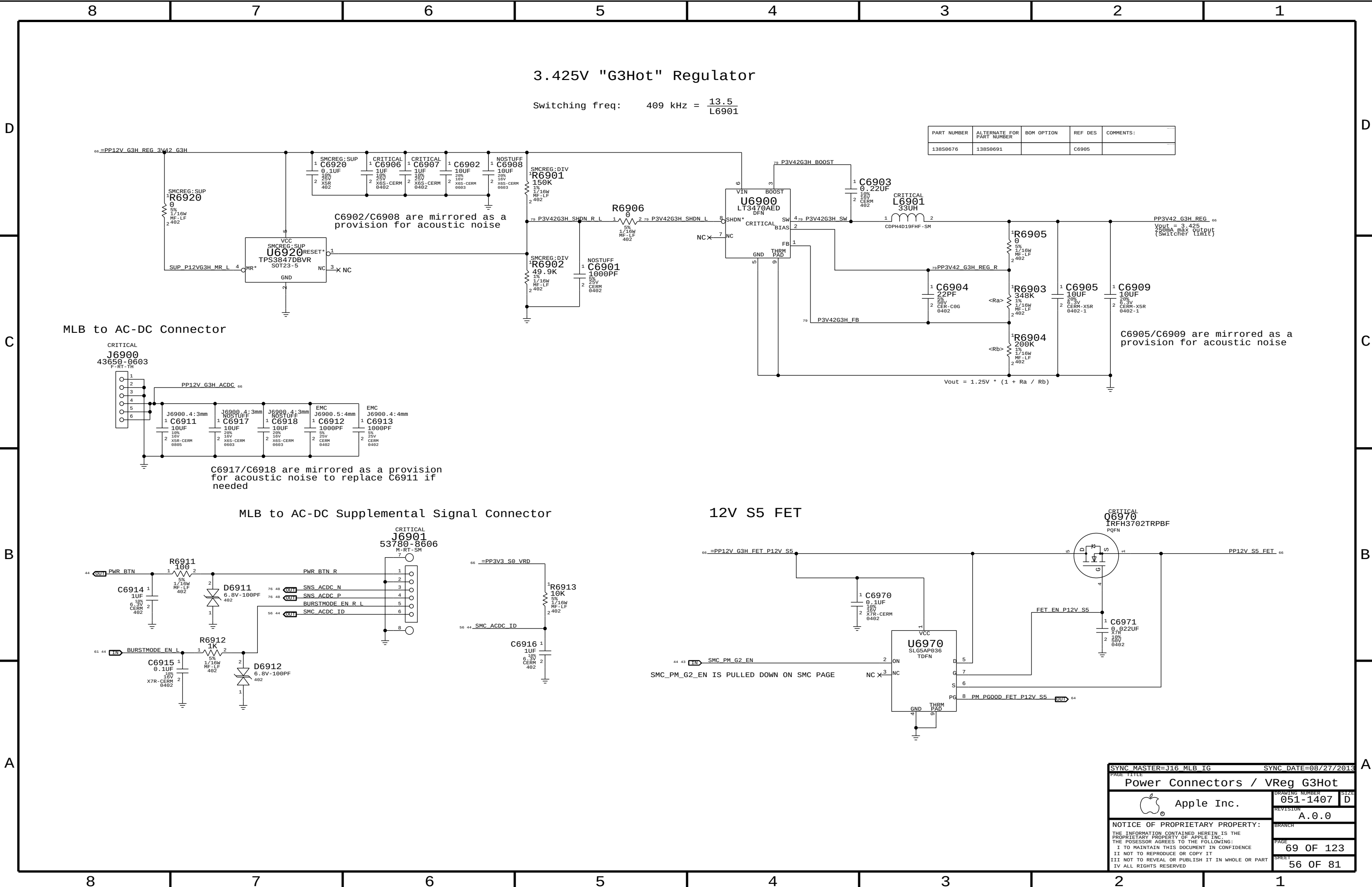


SYNCH MASTER=J70 GAREN		SYNCH DATE=12/17/2013	
PAGE 1 TITLE			
AUDIO: CODEC ANALOG			
 Apple Inc.		DRAWING NUMBER 051-1407	
		SIZE D	
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		SHEET 50 OF 81	



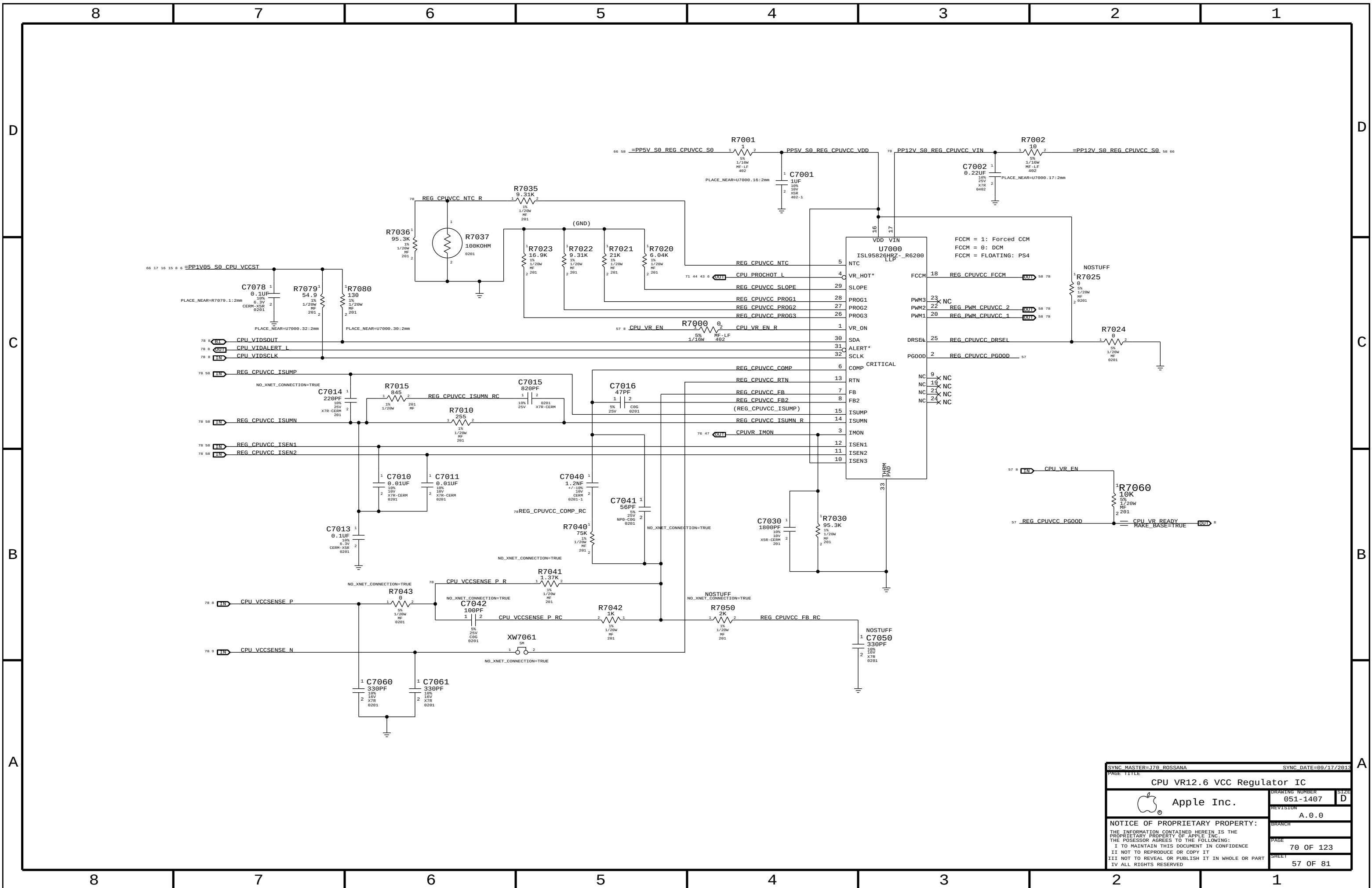
SYNC MASTER=JTD, GAREN		SYNC DATE=01/30/2011	
PAGE TITLE			
AUDIO:CODEC, DIGITAL			
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		SHEET	51 OF 81



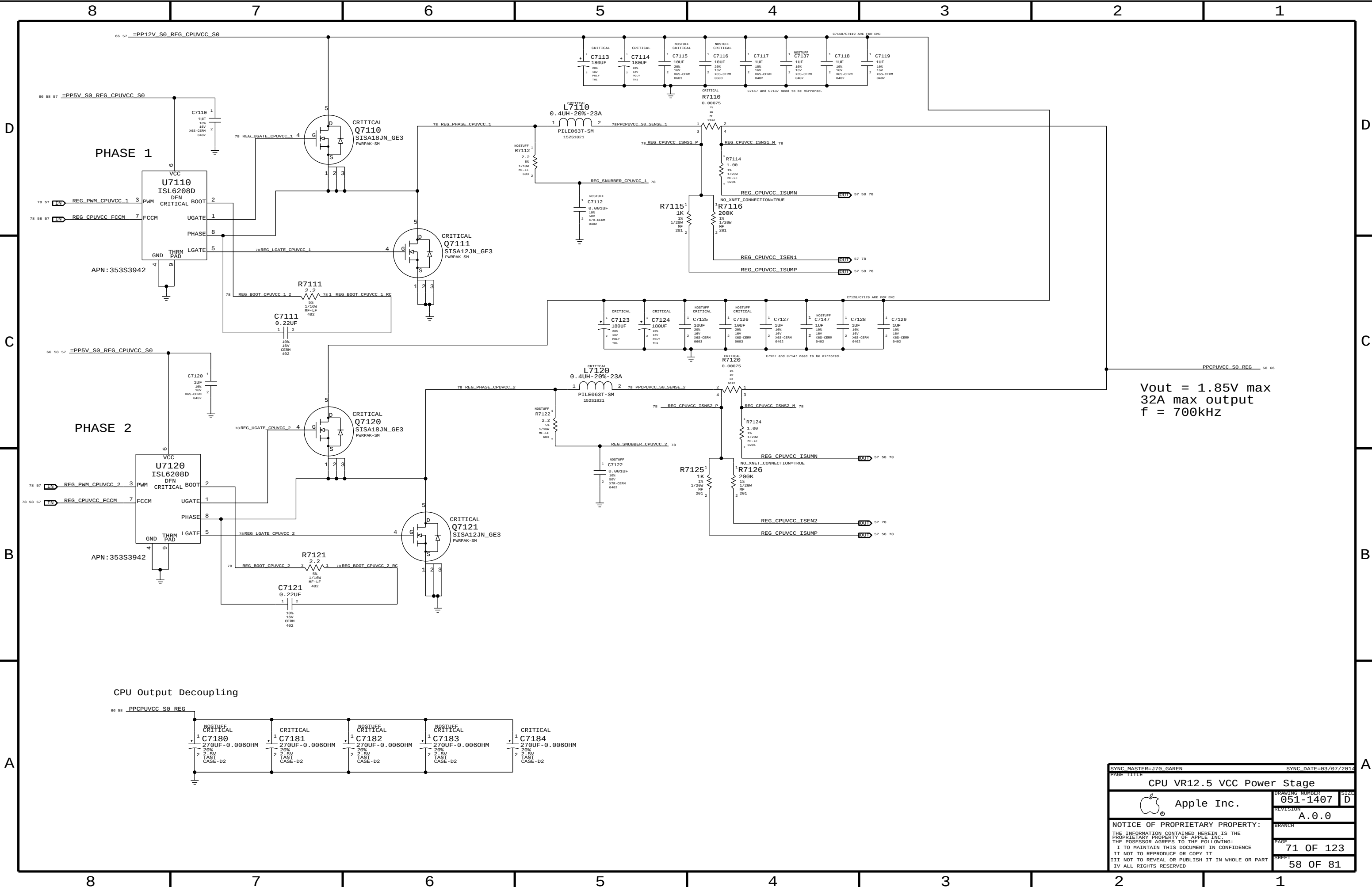


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6905	

SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
PAGE TITLE		Power Connectors / VReg G3Hot	
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SYNC MASTER=J78 ROSSANA		SYNC DATE=09/17/2013	
PAGE TITLE			
CPU VR12.6 VCC Regulator IC			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-1407	D
		REVISION	
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		PAGE	70 OF 123
		SHEET	57 OF 81



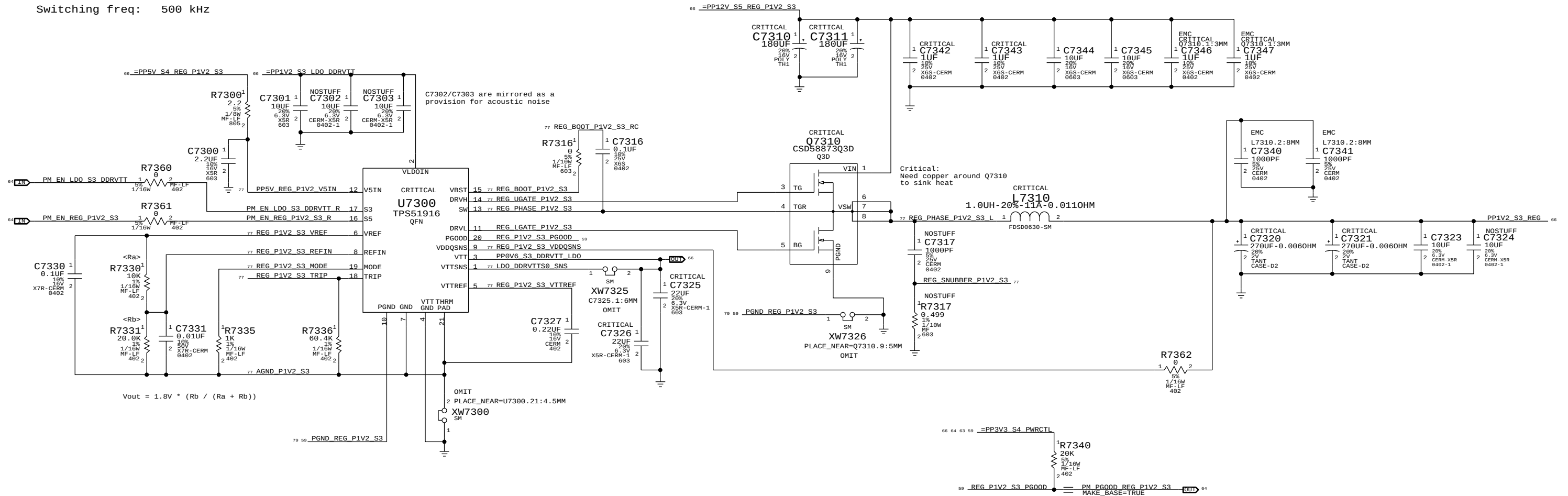
SYNC MASTER=J78 GAREN		SYNC DATE=83/07/2814	
PAGE TITLE			
CPU VR12.5 VCC Power Stage			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1407		D
	REVISION		
		A.0.0	
BRANCH			
PAGE			
		71 OF 123	
SHEET			
		58 OF 81	

VDDQ 1.2V S3 Regulator

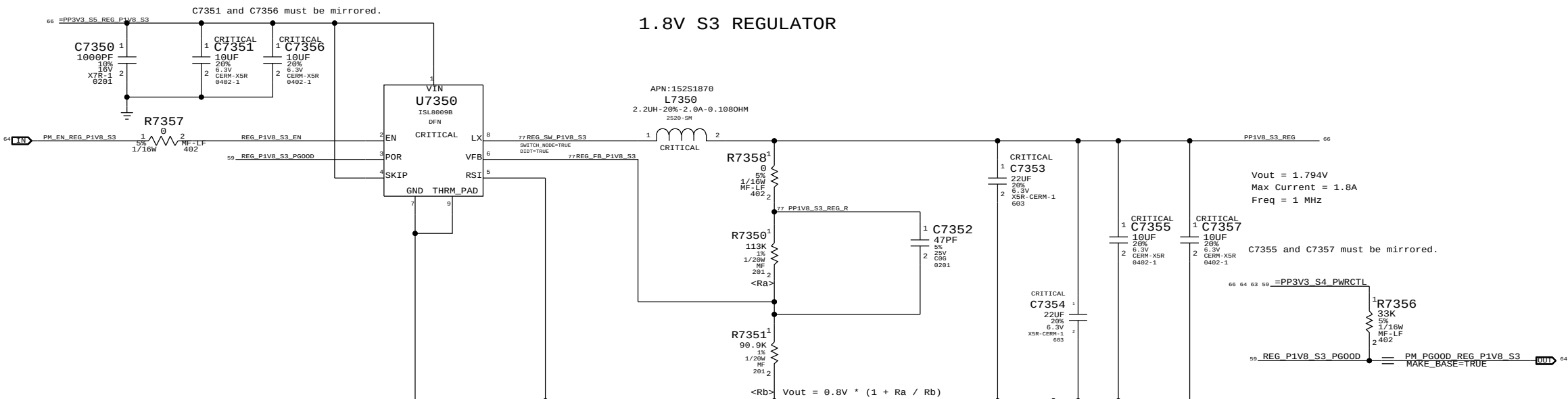
OC trip point: $30.4 \text{ A VDDQ} = \frac{R7336}{8 \text{ E5} * \text{Rds}(Q7310)} + \frac{0.65625}{L7310 * f(\text{switch})}$

3 A VTT (FIXED)
10 mA VTTREF (FIXED)

Switching freq: 500 KHz



1.8V S3 REGULATOR



SYNC MASTER=J70 ROSSANA		SYNC DATE=09/12/2013	
PAGE TITLE		VReg VDDQ S3 / 1.8V S3	
Apple Inc.		DRAWING NUMBER	051-1407
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3.3V S5 Regulator

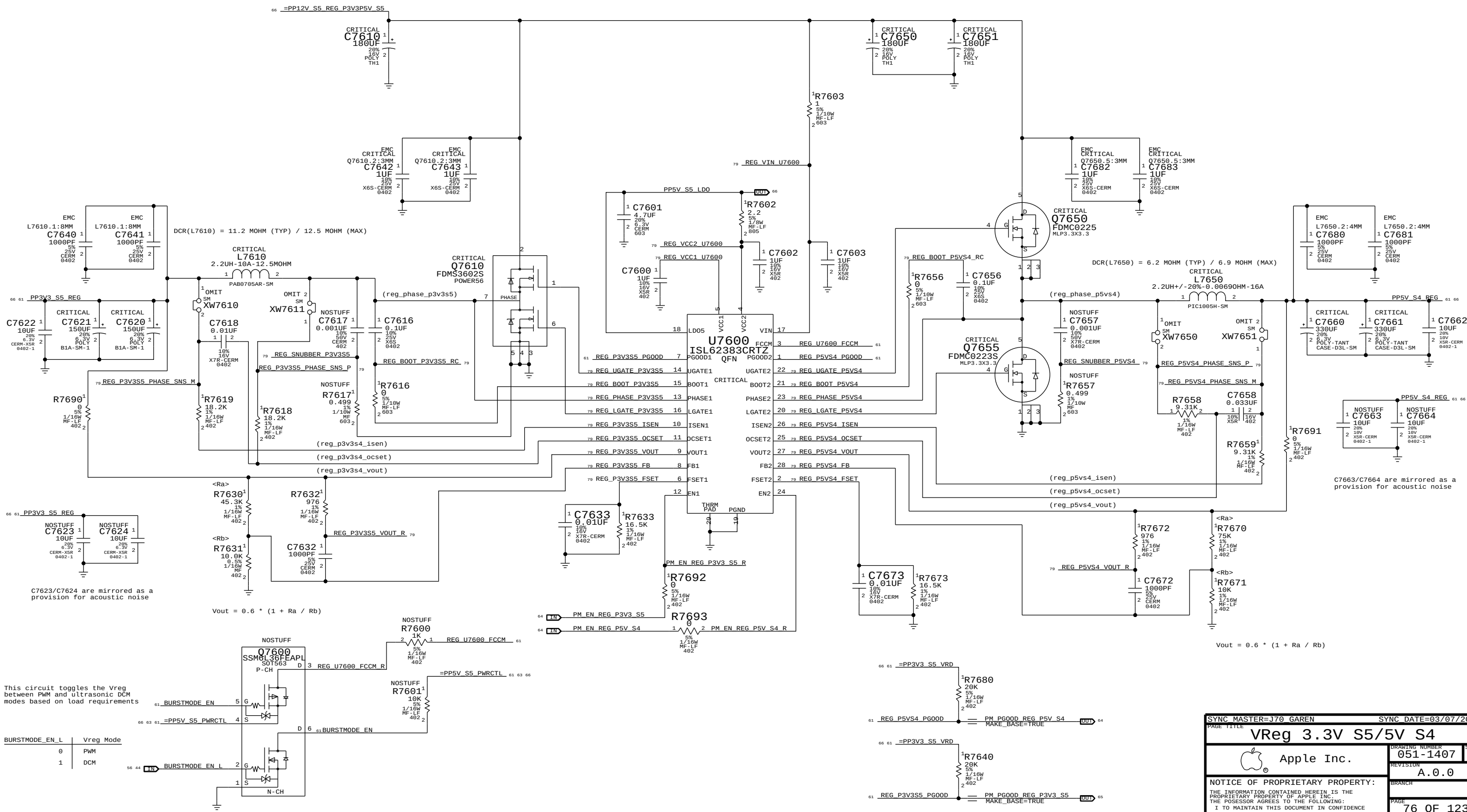
OC trip point: $12.5 \text{ A} = \frac{R7618 * 10 \text{ E-6}}{\text{DCR}(\text{L7610})}$

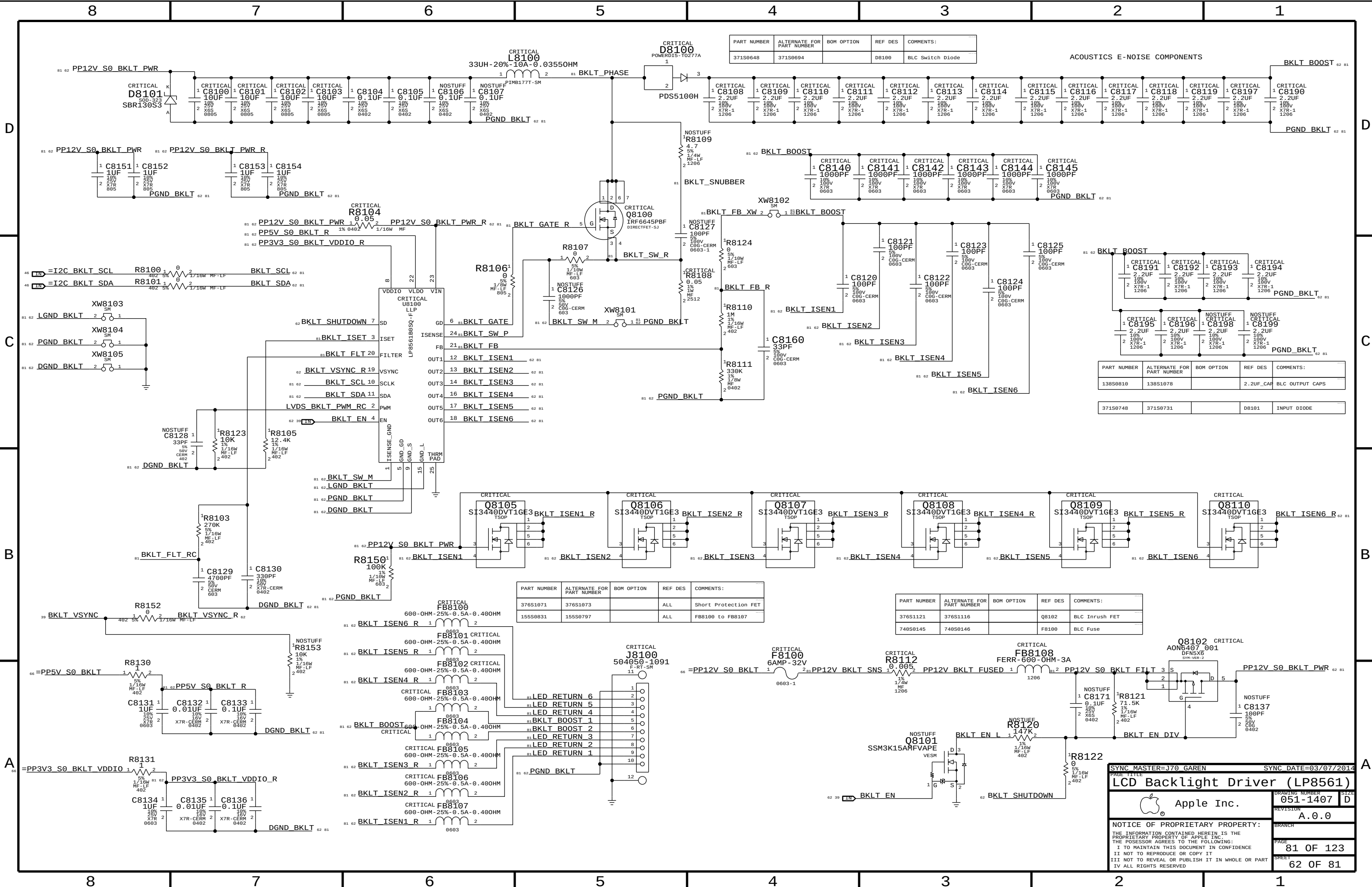
Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7633}$

5V S4 Regulator

OC trip point: $14.1 \text{ A} = \frac{R7658 * 10 \text{ E-6}}{\text{DCR}(\text{L7650})}$

Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7673}$





PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0648	371S0694		D8100	BLC Switch Diode

ACOUSTICS E-NOISE COMPONENTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0810	138S1078		2.2UF_CAF	BLC OUTPUT CAPS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0748	371S0731		D8101	INPUT DIODE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1071	376S1073		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8100 to FB8107

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1121	376S1116		Q8102	BLC Inrush FET
740S0145	740S0146		F8100	BLC Fuse

SYNC MASTER=J70 GAREN

SYNC DATE=03/07/2014

LCD Backlight Driver (LP8561)

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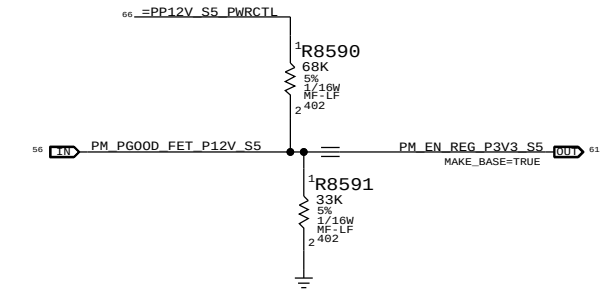
PAGE

81 OF 123

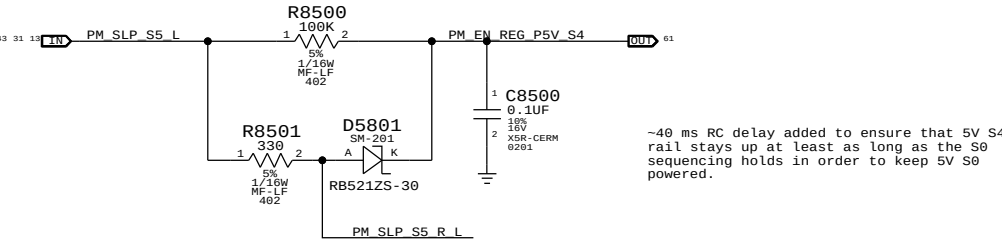
SHEET

62 OF 81

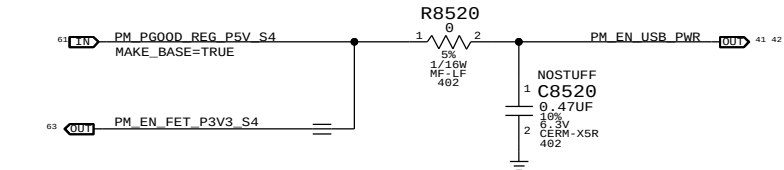
S5 Enable



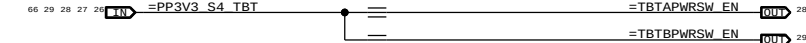
S4 Enables



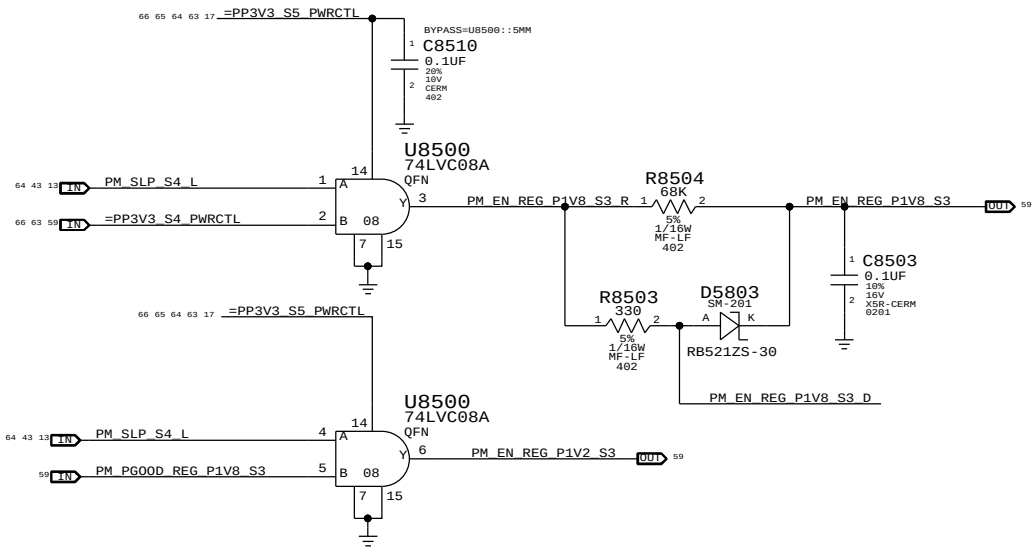
S4 USB Enable



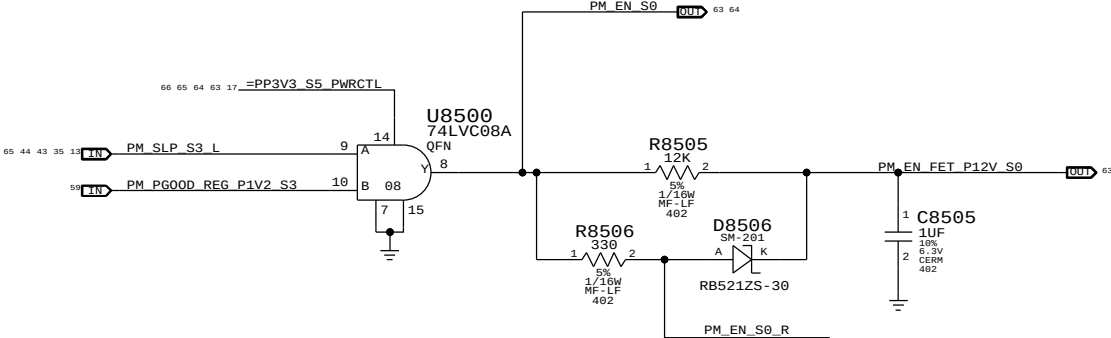
S4 TBT Port Enable



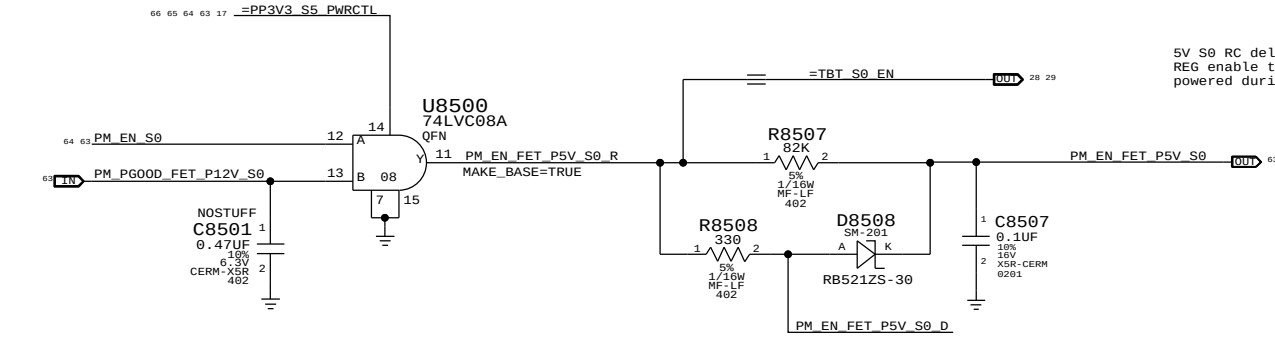
S3 Enables



S0 Enables

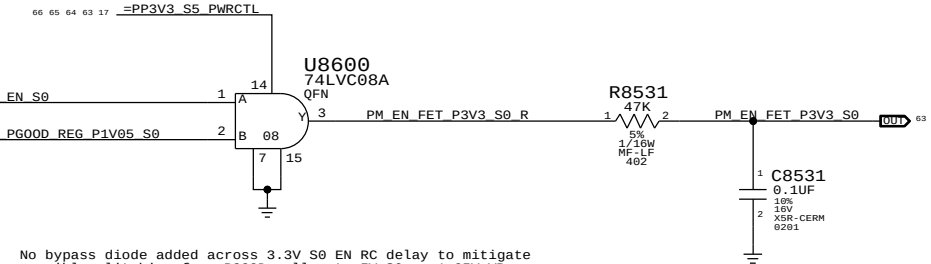
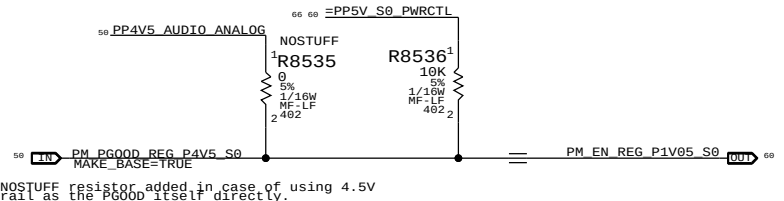
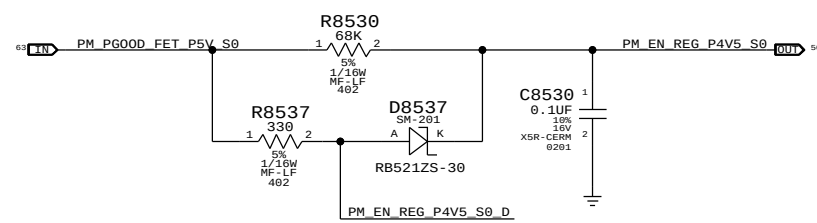


12V S0 EN RC delay must be >= downstream delay on 4.5V REG enable which in turn enables 1.05V S0. This allows for 12V S0 to hold as long as 1.05V S0 regulator is powered.



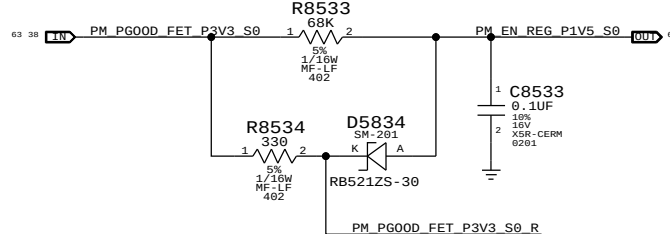
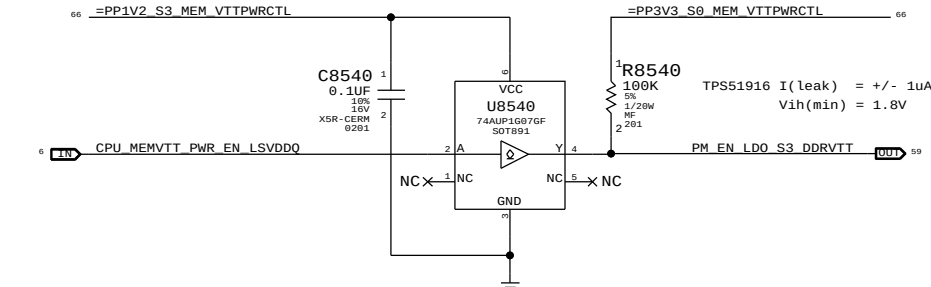
5V S0 RC delay must be >= downstream delay on 4.5V REG enable to allow for 4.5V regulator to remain powered during power down sequence.

Audio + PCH Sequencing Requirements:
4.5V -> 1.05V -> 3.3V -> 1.5V -> ALL SYS GOOD



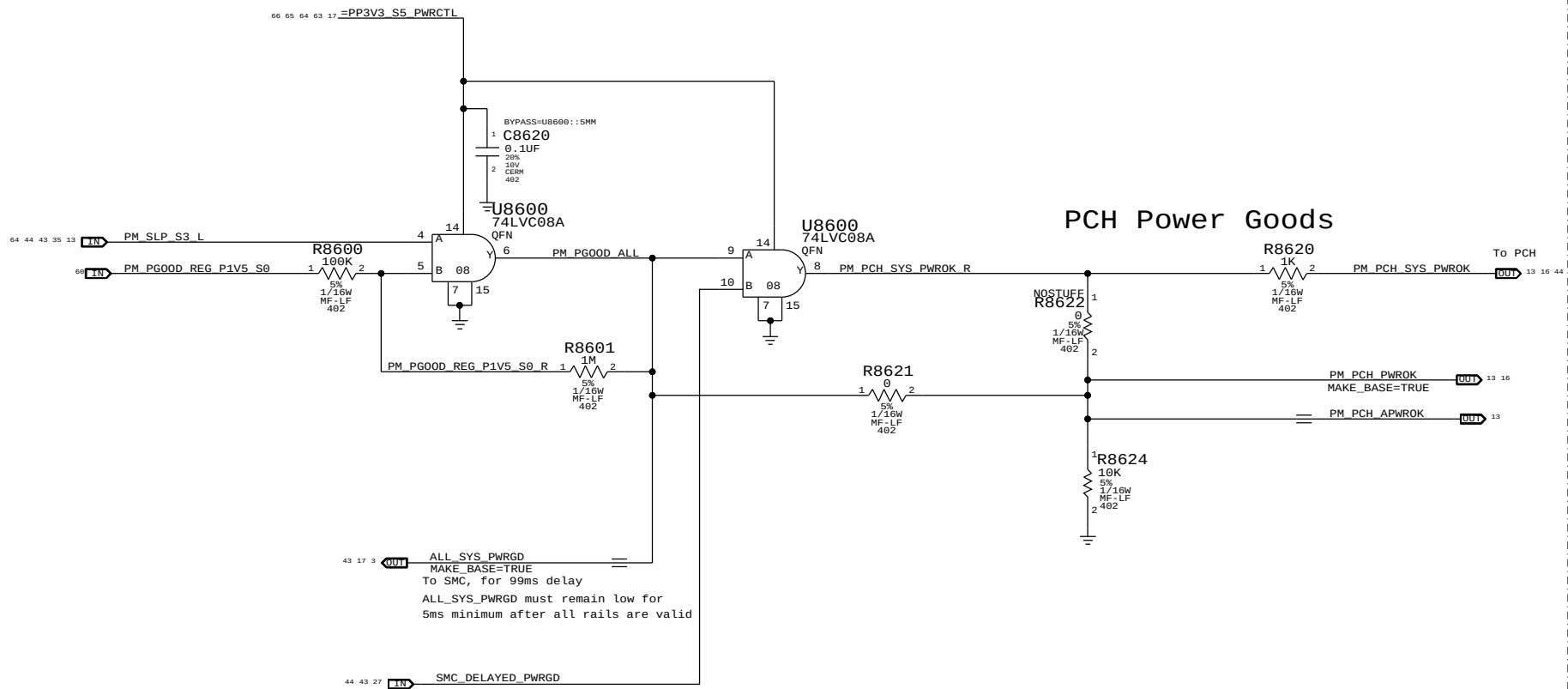
Memory VTT Enable Level-Shifter

CPU output is on VDDQ rail (1.2V), TPS51916 has 1.8V Vih(min).



SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
PAGE TITLE			
PM Regulator Enables			
 Apple Inc.		DRAWING NUMBER	051-1407
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		BRANCH	
		PAGE	85 OF 123
		SHEET	64 OF 81

ALL_SYS_PWRGD, PCH_PWROK & SYS_PWROK Generation



Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

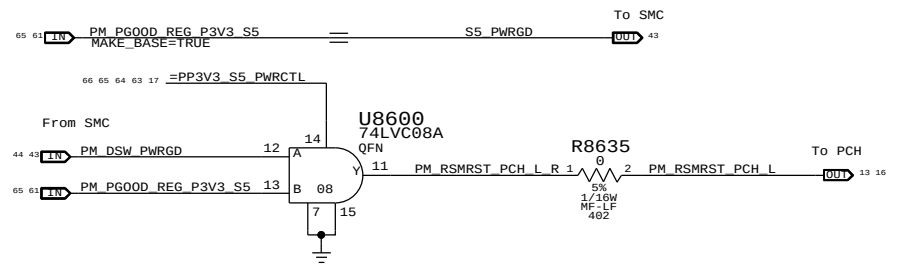
Note:
The iMac J70 design does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:

- Power on:
 - Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
 - Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
 - to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



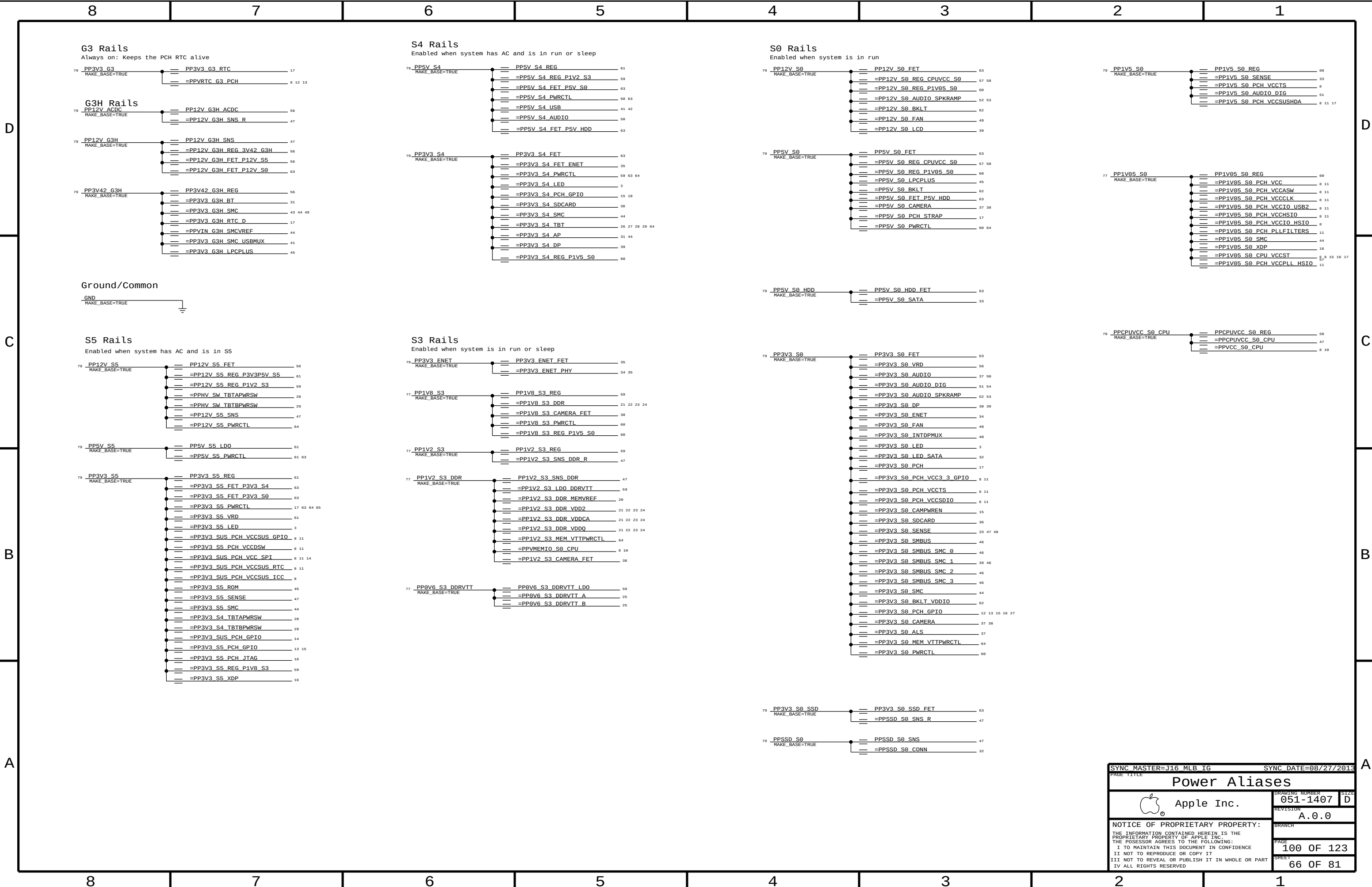
Rail definitions

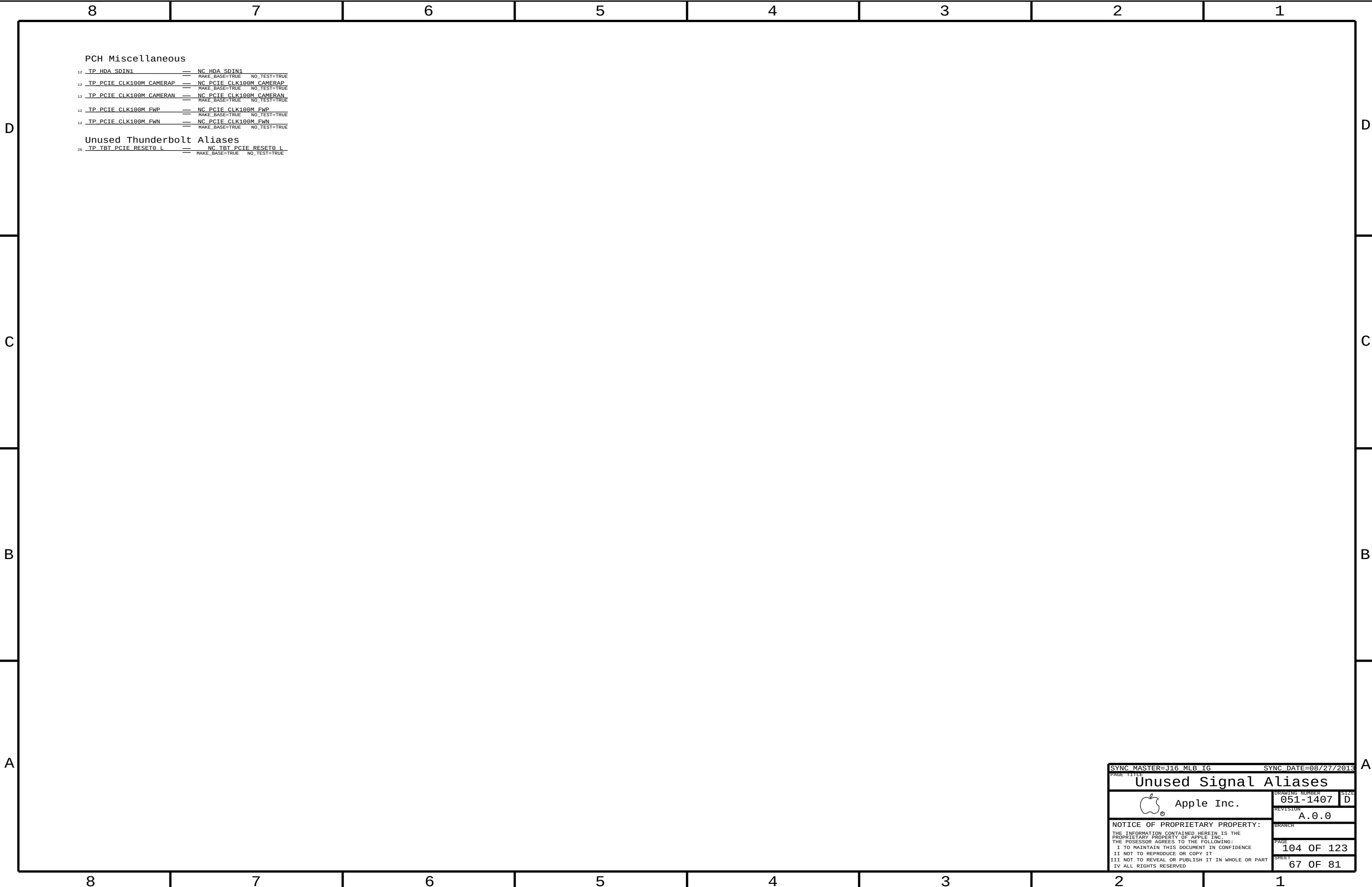
Platform: All processor non-Core and non-Graphics (5V, 3.3V, 1.5V, 1.05V for PCH/TBT/GPU)
Uncore: 1.8V and 1.2V for DDR3

Notes on sequencing requirements

- Intel:
- No hard specification on platform rails
 - SMC guarantees timing on PCH DPWROK and PWROK
 - VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
 - VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
 - VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
 - VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

SYNC MASTER=J70 TONY		SYNC DATE=10/10/2013	
PAGE TITLE			
PM Power Good			
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		BRANCH	
		PAGE	86 OF 123
		SHEET	65 OF 81





SYNC MASTER=J16 MLB IG

SYNC DATE=08/27/2013

PAGE TITLE

Unused Signal Aliases



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PAGE

104 OF 123

SHEET

67 OF 81

8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

SYNC MASTER=J16 MLB IG

SYNC DATE=08/27/2013

PAGE TITLE

Functional / ICT Test

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PAGE

105 OF 123

SHEET

68 OF 81

D

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PAGE TITLE			
J70 RULE DEFINITIONS			
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		BRANCH	
		PAGE	110 OF 123
		SHEET	69 OF 81

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D

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH_ISO	*	=4:1_SPACING	?
COMP_PCH_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH	*	*	COMP_PCH_ISO

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC_55S	LPC	LPC_AD<3..0> 14 43 45
	LPC_55S	LPC	LPC_AD_R<3..0> 14
	LPC_55S	LPC	LPC_FRAME_L 14 43 45
	LPC_55S	LPC	LPC_FRAME_R_L 14
LPC Clocks			
	CLK_LPC_55S	CLK_LPC	LPC_CLK24M_LPCPLUS 17 45
	CLK_LPC_55S	CLK_LPC	LPC_CLK24M_LPCPLUS_R 12 17
	CLK_LPC_55S	CLK_LPC	LPC_CLK24M_SMC 17 43
	CLK_LPC_55S	CLK_LPC	LPC_CLK24M_SMC_R 12 17

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
1653P	CLK_XTAL	XTAL	PCH_CLK24M_XTALIN 12 17
1653P	CLK_XTAL	XTAL	PCH_CLK24M_XTALOUT 12 17
1654E	CLK_XTAL	XTAL	PCH_CLK24M_XTALOUT_R 17
PCH RTC 32K			
1653P	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 12 17
1653P	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 12 17
1653P	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 17
SMC 32K			
1653E	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R 13 44
1653P	CLK_PCH_55S	CLK_PCH	SMC_CLK32K 43 44

25 MHz XTALS

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
	CLK_XTAL	XTAL	TBT_CLK25M_IN 26
	CLK_XTAL	XTAL	TBT_CLK25M_OUT 26
	CLK_XTAL	XTAL	TBT_CLK25M_OUT_R 26
	CLK_XTAL	XTAL	ENET_XTAL_IN 34
	CLK_XTAL	XTAL	ENET_XTAL_OUT 34
	CLK_XTAL	XTAL	ENET_XTAL_OUT_R 34

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC_ISO	*	=1.5:1_SPACING	?
CLK_LPC_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC	*	*	LPC_ISO
CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA_ISO	*	=2x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL_ISO	*	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI_ISO	*	=2:1_SPACING	?

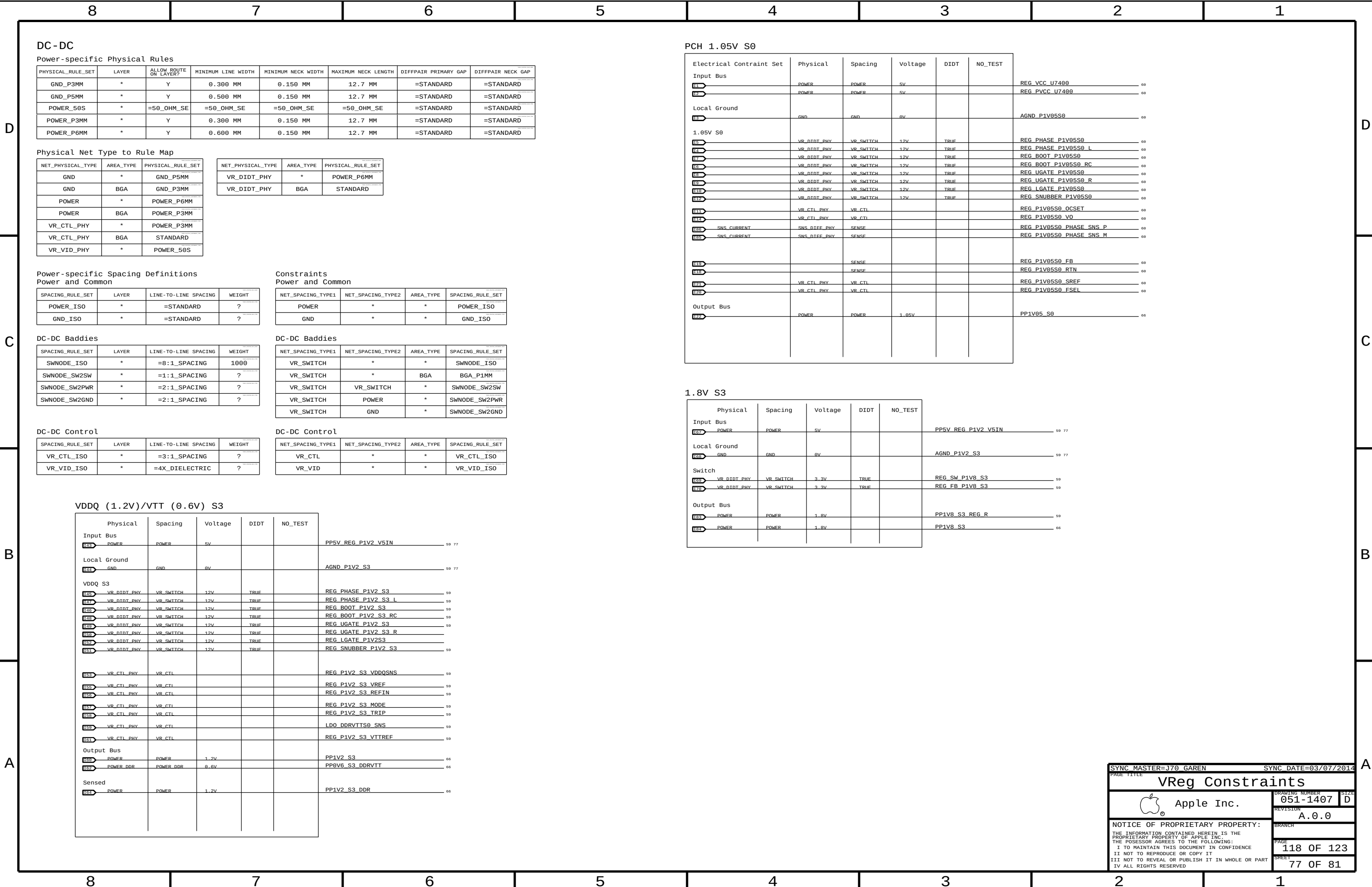
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI	*	*	SPI_ISO

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
HDB0 HDA_CLK	HDA_55S	HDA	HDA_BIT_CLK 12 51
HDB1	HDA_55S	HDA	HDA_BIT_CLK_R 12
HDB2 HDA_RST	HDA_55S	HDA	HDA_RST_L 12 51
HDB3	HDA_55S	HDA	HDA_RST_R_L 12
HDB4 HDA_OUIT	HDA_55S	HDA	HDA_SDOUT 12 51
HDB5	HDA_55S	HDA	HDA_SDOUT_R 12 17
HDB6 HDA_SYNC	HDA_55S	HDA	HDA_SYNC 12 51
HDB7	HDA_55S	HDA	HDA_SYNC_R 12
HDB8 HDA_TN	HDA_55S	HDA	HDA_SPDIO 12 51
HDB9	HDA_55S	HDA	AUD_SDI_R
SPDIF			
HDB10	HDA_55S	HDA	DP_INT SPDIF_AUDIO 39 51 89
HDB11	HDA_55S	HDA	SPDIF_OUT_JACK 51 54
HDB12	HDA_55S	HDA	CS4208 SPDIF_IN 51
HDB13	HDA_55S	HDA	CS4208 SPDIF_OUT 51

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
	SPT 50S	SPT	SPI CLK R 14 45
	SPT 50S	SPT	SPI CLK 45
	SPT 50S	SPT	SPI ALT CLK 45
	SPT 50S	SPT	SPI SMC CLK 43 45
	SPT 50S	SPT	SPI MLB CLK 45
	SPT 50S	SPT	SPI CS0 R L 14 45
	SPT 50S	SPT	SPI CS0 L 45
	SPT 50S	SPT	SPI ALT CS L 45
	SPT 50S	SPT	SPI SMC CS L 43 45
	SPT 50S	SPT	SPI MLB CS L 45
	SPT 50S	SPT	SPI MOSI R 14 45
	SPT 50S	SPT	SPI MOSI 45
	SPT 50S	SPT	SPI ALT MOSI 45
	SPT 50S	SPT	SPI SMC MOSI 43 45
	SPT 50S	SPT	SPI MLB MOSI 45
	SPT 50S	SPT	SPI MISO 14 45
	SPT 50S	SPT	SPI ALT MISO 45
	SPT 50S	SPT	SPI SMC MISO 43 45
	SPT 50S	SPT	SPI MLB MISO 45
	SPT 50S	SPT	SPIROM USE MLB 15 45



[illegible]

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTD_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C_ISO	*	=2X_DIELECTRIC	?
TBT_SPI_ISO	*	=2X_DIELECTRIC	?
TBTD_ISO	*	=5X_DIELECTRIC	?
TBTD_ISO	TOP,BOTTOM	=7X_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_ISO	*	=3:1_SPACING	?	DISPLAYPORT	*	*	DP_ISO

Pairs should be within 100 mils of clock length.

Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.

DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

Electrical Constraint Set	Physical	Spacing			
EE900	DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_ML_C P<3...0>	5 26	
EE901		DP_85D	DP TBTSNK0_ML_C N<3...0>	5 26	
EE902	DP TBTSNK0_ML	DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_ML P<3...0>	26
EE903	DP TBTSNK0_ML	DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_ML N<3...0>	26
EE904		DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_AUXCH_C P	13 26
EE905		DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_AUXCH_C N	13 26
EE906	DP TBTSNK0_AUX	DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_AUXCH P	26
EE907	DP TBTSNK0_AUX	DP_85D	DTSPDISPLAYPORT	DP TBTSNK0_AUXCH N	26
EE908		DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_ML_C P<3...0>	5 26
EE909		DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_ML_C N<3...0>	5 26
EE910	DP TBTSNK1_ML	DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_ML P<3...0>	26
EE911	DP TBTSNK1_ML	DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_ML N<3...0>	26
EE912		DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_AUXCH_C P	13 26
EE913		DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_AUXCH_C N	13 26
EE914	DP TBTSNK1_AUX	DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_AUXCH P	26
EE915	DP TBTSNK1_AUX	DP_85D	DTSPDISPLAYPORT	DP TBTSNK1_AUXCH N	26
EE916					
EE917					
EE918	DP TINTPN1_TBT_ML_MUX	DP_85D	DTSPDISPLAYPORT	DP TBTSRC_ML P<3...0>	40
EE919	DP TINTPN1_TBT_ML_MUX	DP_85D	DTSPDISPLAYPORT	DP TBTSRC_ML N<3...0>	40
EE920	DP TINTPN1_TBT_ML_MUX	DP_85D	DTSPDISPLAYPORT	DP TBTSRC_ML_C P<3...0>	40
EE921	DP TINTPN1_TBT_ML_MUX	DP_85D	DTSPDISPLAYPORT	DP TBTSRC_ML_C N<3...0>	40
EE922	DP TINTPN1_TBT_AUX_MUX	DP_85D	DTSPDISPLAYPORT	DP TBTSRC_AUXCH P	40
EE923	DP TINTPN1_TBT_AUX_MUX	DP_85D	DTSPDISPLAYPORT	DP TBTSRC_AUXCH N	40
EE924		DP_85D	DTSPDISPLAYPORT	DP TBTSRC_AUX_C P	40
EE925		DP_85D	DTSPDISPLAYPORT	DP TBTSRC_AUX_C N	40
EE926					
EE927					
EE928	TBT_SPT_CLK	TBT_SPT_55S	TBT_SPT	TBT_SPT_CLK	26
EE929	TBT_SPT_MOST	TBT_SPT_55S	TBT_SPT	TBT_SPT_MOST	26
EE930	TBT_SPT_MISO	TBT_SPT_55S	TBT_SPT	TBT_SPT_MISO	26
EE931	TBT_SPT_CS_I	TBT_SPT_55S	TBT_SPT	TBT_SPT_CS_L	26

*: Only used on hosts supporting T29 video-in

DisplayPort

Electrical Constraint Set		Physical	Spacing	
Graphics Source				
LE95P	DP INTPNL TG MI MUX	DP 85D	DISPLAYPORT	DP INT ML P<1..0>
LE95P	DP INTPNL TG MI MUX	DP 85D	DISPLAYPORT	DP INT ML N<1..0>
LE95P	DP INTPNL TG AUX MUX	DP 85D	DISPLAYPORT	DP INT AUX P
LE95P	DP INTPNL TG AUX MUX	DP 85D	DISPLAYPORT	DP INT AUX N
LE95P		DP 85D	DISPLAYPORT	DP INT AUX C P
LE95P		DP 85D	DISPLAYPORT	DP INT AUX C N
Internal Panel				
LE95P		DP 85D	DISPLAYPORT	DP INTPNL ML C P<3..0>
LE95P		DP 85D	DISPLAYPORT	DP INTPNL ML C N<3..0>
LE95P	DP INTPNL MI CONN	DP 85D	DISPLAYPORT	DP INTPNL ML P<3..0>
LE95P	DP INTPNL MI CONN	DP 85D	DISPLAYPORT	DP INTPNL ML N<3..0>
LE95P	DP INTPNL AUX CONN	DP 85D	DISPLAYPORT	DP INTPNL AUX P
LE95P	DP INTPNL AUX CONN	DP 85D	DISPLAYPORT	DP INTPNL AUX N
Internal DP SPDIF				
LE95P		HDA		DP INT SPDIF_AUDIO
DDC				
LE95P	TBT I2C 55S	TBT I2C		DP TBT5NK0 DDC CLK
LE95P	TBT I2C 55S	TBT I2C		DP TBT5NK0 DDC DATA
LE95P	TBT I2C 55S	TBT I2C		DP TBT5NK1 DDC CLK
LE95P	TBT I2C 55S	TBT I2C		DP TBT5NK1 DDC DATA
LE95P	TBT I2C 55S	TBT I2C		DP TBT6P DDC CLK
LE95P	TBT I2C 55S	TBT I2C		DP TBT6P DDC DATA
LE95P	TBT I2C 55S	TBT I2C		DP TBT6P DDC CLK
LE95P	TBT I2C 55S	TBT I2C		DP TBT6P DDC DATA

TBT/DP Net Properties

Electrical Constraint Set		Physical	Spacing	
Port A				
R649	TBT_R2D_RVSD	T8TDP_980	T8TDP	TBT A R2D C P<1>
R649	TBT_R2D_RVSD	T8TDP_980	T8TDP	TBT A R2D C N<1>
R649		T8TDP_980	T8TDP	TBT A R2D P<1>
R649		T8TDP_980	T8TDP	TBT A R2D N<1>
R649	TBT_R2D	T8TDP_980	T8TDP	TBT A R2D C P<0>
R649	TBT_R2D	T8TDP_980	T8TDP	TBT A R2D C N<0>
R649		T8TDP_980	T8TDP	TBT A R2D P<0>
R649		T8TDP_980	T8TDP	TBT A R2D N<0>
R650	DP_ML1	DP_850	DISPLAYPORT	DP T8TPA ML C P<1>
R650	DP_ML1	DP_850	DISPLAYPORT	DP T8TPA ML C N<1>
R649		DP_850	DISPLAYPORT	DP T8TPA ML P<1>
R649		DP_850	DISPLAYPORT	DP T8TPA ML N<1>
R650	DP_ML3	DP_850	DISPLAYPORT	DP T8TPA ML C P<3>
R650	DP_ML3	DP_850	DISPLAYPORT	DP T8TPA ML C N<3>
R649		DP_850	DISPLAYPORT	DP T8TPA ML P<3>
R649		DP_850	DISPLAYPORT	DP T8TPA ML N<3>
R649	DP_LSX	DP_850	DISPLAYPORT	DP A LSX ML P<1>
R649	DP_LSX	DP_850	DISPLAYPORT	DP A LSX ML N<1>
R650	TBT_D2R1_RVSD	T8TDP_980	T8TDP	TBT A D2R P<1>
R650	TBT_D2R1_RVSD	T8TDP_980	T8TDP	TBT A D2R N<1>
R649		T8TDP_980	T8TDP	TBT A D2R C P<1>
R649		T8TDP_980	T8TDP	TBT A D2R C N<1>
R649	TBT_D2R8_RVSD	T8TDP_980	T8TDP	TBT A D2R P<0>
R649	TBT_D2R8_RVSD	T8TDP_980	T8TDP	TBT A D2R N<0>
R649		T8TDP_980	T8TDP	TBT A D2R C P<0>
R649		T8TDP_980	T8TDP	TBT A D2R C N<0>
R650	TBT_AUXDDC	T8TDP_980	T8TDP	TBT A D2R1 AUXDDC P
R650	TBT_AUXDDC	T8TDP_980	T8TDP	TBT A D2R1 AUXDDC N
R649	TBT_AUXCH	DP_850	DISPLAYPORT	DP T8TPA AUXCH C P
R649	TBT_AUXCH	DP_850	DISPLAYPORT	DP T8TPA AUXCH C N
R649		DP_850	DISPLAYPORT	DP T8TPA AUXCH P
R649		DP_850	DISPLAYPORT	DP T8TPA AUXCH N
Port B				
R649	TBT_R2D_RVSD	T8TDP_980	T8TDP	TBT B R2D C P<1>
R649	TBT_R2D_RVSD	T8TDP_980	T8TDP	TBT B R2D C N<1>
R649		T8TDP_980	T8TDP	TBT B R2D P<1>
R649		T8TDP_980	T8TDP	TBT B R2D N<1>
R649	TBT_R2D	T8TDP_980	T8TDP	TBT B R2D C P<0>
R649	TBT_R2D	T8TDP_980	T8TDP	TBT B R2D C N<0>
R649		T8TDP_980	T8TDP	TBT B R2D P<0>
R649		T8TDP_980	T8TDP	TBT B R2D N<0>
R649	DP_ML1	DP_850	DISPLAYPORT	DP T8TPB ML C P<1>
R649	DP_ML1	DP_850	DISPLAYPORT	DP T8TPB ML C N<1>
R649		DP_850	DISPLAYPORT	DP T8TPB ML P<1>
R649		DP_850	DISPLAYPORT	DP T8TPB ML N<1>
R649	DP_ML3_RVSD	DP_850	DISPLAYPORT	DP T8TPB ML C P<3>
R649	DP_ML3_RVSD	DP_850	DISPLAYPORT	DP T8TPB ML C N<3>
R649		DP_850	DISPLAYPORT	DP T8TPB ML P<3>
R649		DP_850	DISPLAYPORT	DP T8TPB ML N<3>
R649	DP_LSX	DP_850	DISPLAYPORT	DP B LSX ML P<1>
R649	DP_LSX	DP_850	DISPLAYPORT	DP B LSX ML N<1>
R650	TBT_D2R1_RVSD	T8TDP_980	T8TDP	TBT B D2R P<1>
R650	TBT_D2R1_RVSD	T8TDP_980	T8TDP	TBT B D2R N<1>
R649		T8TDP_980	T8TDP	TBT B D2R C P<1>
R649		T8TDP_980	T8TDP	TBT B D2R C N<1>
R649	TBT_D2R8_RVSD	T8TDP_980	T8TDP	TBT B D2R P<0>
R649	TBT_D2R8_RVSD	T8TDP_980	T8TDP	TBT B D2R N<0>
R649		T8TDP_980	T8TDP	TBT B D2R C P<0>
R649		T8TDP_980	T8TDP	TBT B D2R C N<0>
R650	TBT_AUXDDC	T8TDP_980	T8TDP	TBT B D2R1 AUXDDC P
R650	TBT_AUXDDC	T8TDP_980	T8TDP	TBT B D2R1 AUXDDC N
R649	TBT_AUXCH	DP_850	DISPLAYPORT	DP T8TPB AUXCH C P
R649	TBT_AUXCH	DP_850	DISPLAYPORT	DP T8TPB AUXCH C N
R649		DP_850	DISPLAYPORT	DP T8TPB AUXCH P
R649		DP_850	DISPLAYPORT	DP T8TPB AUXCH N

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		PAGE	121 OF 123
		SHEET	80 OF 81

